

Nonparametric inference for ratios of densities via uniformly valid and powerful permutation tests

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Abstract

We propose the density ratio permutation test, a hypothesis test that assesses whether the ratio between two densities is proportional to a known function based on independent samples from each distribution. The test uses an efficient Markov Chain Monte Carlo scheme to draw weighted permutations of the pooled data, yielding exchangeable samples and finite sample validity. For power, if the statistic is an integral probability metric, our procedure is consistent under mild assumptions on the defining function class; specializing to a reproducing kernel Hilbert space, we introduce the shifted maximum mean discrepancy and prove minimax optimality of our test when a normalized difference between the densities lies in a Sobolev ball. We extend to the case of an unknown density ratio by estimating it on an independent training sample and derive type I error bounds in terms of the estimation error as well as power results. This allows adapting our method to conditional two sample testing, making it a versatile tool for assessing covariate-shift and related assumptions, which frequently arise in transfer learning and causal inference. Finally, we validate our theoretical findings through experiments on both simulated and real-world datasets.

Some key words: Conditional two-sample testing; Density ratio; Distributional shift; Minimax optimality; Permutation test.

1 Introduction

Density ratios are foundational objects in statistics. Commonly known as the importance sampling literature (Tokdar and Kass, 2010), they can be used to implement importance weighting techniques (e.g. Kahn and Harris, 1951; Horvitz and Thompson, 1952; Owen and Zhou, 2000), which assign varying weights to data points to correct for biases and emphasize relevant observations. These quantities remain central to contemporary statistics for several reasons. For instance, many modern analyses involve data drawn from different sources or environments (Koh et al., 2020). When distributions differ across datasets, principled procedures are needed to transport information so that inferences target the population of interest. Density ratio tools provide exactly this mechanism, quantifying how one distribution departs from another and enabling methods that combine heterogeneous datasets for prediction and inference (e.g. Storkey, 2009; Weiss et al., 2016). In medical applications, practitioners might be interested in making predictions in a

particular experimental setting or using specific equipment, while also utilizing data collected under different conditions or with alternative tools (e.g. Guan and Liu, 2022). Likewise, in the context of large language models, thoughtfully incorporating human-authored data alongside model-generated content during training helps maintain diversity and enhance overall performance (e.g. Ji et al., 2025). Beyond transportability, density ratios underpin a wider range of statistical problems that admit a classification formulation, where the Bayes optimal decision rule amounts to thresholding a class conditional ratio. A prominent example is simulation based inference, which refers to methodologies that leverage simulators to perform Bayesian analysis when the likelihood is intractable or unavailable (e.g. Deistler et al., 2025). In this context, many pipelines estimate density ratios, e.g. via neural ratio estimators, to target posterior-to-prior or likelihood-to-evidence ratios (Hermans et al., 2020; Thomas et al., 2022; Miller et al., 2022; Delaunoy et al., 2022; Leonte et al., 2025), thereby enabling downstream tasks such as posterior approximation. Additional applications include nonparametric regression (Ma et al., 2023), efficient estimation with additional incomplete data (Berrett, 2024), testing under distributional shifts (Thams et al., 2023), quantile function estimation (Chen and Liu, 2013), reinforcement learning (e.g. Sutton and Barto, 2018), and causal inference (Robins et al., 2000).

Although the statistical literature on density ratio estimation is well developed (e.g. Sugiyama et al., 2010), a persistent practical challenge is not only how to estimate the ratio but also whether a given estimate is adequate for the downstream task. Rather than plugging estimated ratios directly into importance weighting or transportability analyses, practitioners in many settings expect diagnostics before trusting the estimate. For example, common approaches to the estimation of causal treatment effects rely on the estimation of propensity scores, often assuming a parametric model, in order to balance covariate distributions between treatment and control groups. Best practice requires diagnostics to confirm the weights achieve balance. Austin and Stuart (2015, Section 4) recommend assessing covariate balance using reweighted moments, graphical distributional comparisons via boxplots or empirical cumulative distribution functions, and summary metrics such as the weighted Kolmogorov—Smirnov distance. Such diagnostics are also implemented in popular software; see, for example Zhou et al. (2022) and the references therein. In parallel, transportability analyses emphasize that suitability of weights must be evaluated, not assumed: as Westreich et al. (2017) write, “In fact, the transportability of observational studies to a particular target population of interest is not guaranteed and must be evaluated carefully.”

Another setting where statistical tests for density ratios are required is simulation based inference, where modern procedures can yield miscalibrated posteriors leading to invalid downstream conclusions (Hermans et al., 2022). Although formal statistical tests exist for methods that target the posterior directly, as summarized in Section 3 and Appendix A5 in Deistler et al. (2025), for neural ratio estimation the literature offers diagnostics rather than formal tests. The need for quantitative checks is evidenced both in the statistical literature, e.g. Hermans et al. (2020, Figure 3) and Miller et al. (2022, Figure 3), and in real scientific workflows, such as in high energy physics (ATLAS Collaboration, 2025). In particular, these works assess density ratio quality using the same diagnostics as listed above, alongside the receiver operating characteristic metric for a classifier that distinguishes the target distribution from a reweighted version of the reference distribution, trained on independent samples from each using a density-ratio-weighted empirical loss.

The foregoing discussion highlights the need to supplement density ratio estimation with inferential

procedures that quantify uncertainty. In this paper, we focus on the foundational problem of testing

$$H_0 : g \propto r f, \tag{1}$$

based on independent data $(X_1, \dots, X_n, Y_1, \dots, Y_m) \sim P_f^{\otimes n} \otimes P_g^{\otimes m}$, where P_f, P_g are distributions on a domain $\mathcal{X}$ with densities f, g with respect to a common dominating measure, and where r is a fixed function. This can be viewed as the most fundamental nonparametric hypothesis testing problem for an unknown density ratio g/f , where we test proportionality to a known function, analogous to goodness-of-fit testing for densities (e.g. [Ingster and Suslina, 2003](#)) where we would test the equality of an unknown density and a fixed function. While simple, this problem itself is of interest in applications. For example, in simulation based inference the quality of a ratio estimator can be assessed using independent or held-out data; see [Rogozhnikov \(2016\)](#); [ATLAS Collaboration \(2025\)](#) for examples from the physics literature. The testing problem (1) can also be equivalently reformulated as a two-sample testing problem given biased data (e.g. [Kang and Nelson, 2009](#)), for which asymptotic results are available; see also the closely related problem of independence testing with biased data ([Tenzer et al., 2022](#)). More generally, nonparametric inference under biased sampling is well studied, with known weight functions arising through size-biased sampling and many other practical mechanisms; see, for example, [Cox \(1969\)](#); [Qin \(1993\)](#); [Efremovich \(2004\)](#) and the references therein. It can also be seen that (1) is equivalent to conditional two-sample testing (e.g. [Lee et al., 2024](#)) when the ratio of covariate densities can be assumed to be known, as would be the case in a Model-X framework ([Candès et al., 2018](#)) or randomized trials with perfect compliance ([Lei and Candès, 2021](#)). Nonetheless, moving beyond this foundational problem, in Section 5 we extend our study to test hypotheses that g/f lies in a restricted function class, for example parametric or depending only on a proper subset of variables. This, in turn, enables assessing a range of modelling assumptions, including tests for distributional shifts such as covariate shift, a setting common in domain adaptation and transfer learning (e.g. [Koh et al., 2020](#); [Tibshirani et al., 2019](#); [Jin and Candès, 2025](#)).

Our assessment of (1) is based on the novel density ratio permutation test defined in Algorithm 2, which uses an appropriate resampling scheme to calibrate any chosen test statistic and provide a p -value. Permutation methods are widely used in hypothesis testing, due to their guaranteed validity and strong power properties in both classical asymptotic ([Hoeffding, 1952](#); [Lehmann and Romano, 2005](#)) and non-asymptotic ([Berrett et al., 2021](#); [Kim et al., 2022](#)) senses. These methods approximate the null distribution of the test statistic by uniformly relabelling observations, which in classical two-sample settings is sufficient to satisfy the randomization hypothesis, thus yielding exact type I error control ([Lehmann and Romano, 2005](#)). However, this conventional approach is not appropriate for (1) since the two samples are not exchangeable under the null hypothesis. We address this limitation by introducing a modified permutation sampling approach, adapting ideas from conditional independence testing ([Berrett et al., 2020](#)) to the problem (1), which produces permutations of $(X_1, \dots, X_n, Y_1, \dots, Y_m)$ according to a distribution dependent on the hypothesized density ratio r , restoring exchangeability of the true and resampled datasets under H_0 and thus ensuring finite-sample validity. When r is constant, our method naturally reduces to the classical permutation test.

This non-uniform permutation sampling approach represents a departure from standard methodology, but aligns with recent developments tackling various testing problems. The weight-dependent permutation

method of [Kang and Nelson \(2009\)](#) tests $f = g$ using samples from the densities $w_1 f$ and $w_2 g$, where w_1, w_2 are known sampling bias functions, and is proved to have asymptotic validity and power. The conditional permutation test of [Berrett et al. \(2020\)](#) for testing $X \perp\!\!\!\perp Y \mid Z$ draws permutations based on the knowledge of the conditional law of $X \mid Z$. [Ramdas et al. \(2022\)](#) studies a general framework for the use of weighted permutations to test data exchangeability. Recent advances in conformal prediction under distributional shifts modify procedures to maintain coverage with non-exchangeable data ([Tibshirani et al., 2019](#); [Hu and Lei, 2020](#); [Prinster et al., 2024](#)). Despite this great recent interest in non-uniform permutation methods, power results are rare. To the best of our knowledge, our work is the first to prove general consistency results and the minimax rate optimality of such a uniformly-valid testing procedure.

An alternative way to tackle (1) is to use knowledge of r to reweight the data and reduce the task to classical two-sample testing. For example, one can apply importance sampling to $(X_1, \dots, X_n)$ to create a sample from a distribution proportional to $r f$. This can then be compared with $(Y_1, \dots, Y_m)$ using standard two-sample tests. This broad strategy has been employed in works such as [Thams et al. \(2023\)](#) and [Chau et al. \(2025\)](#) for other statistical problems. We deliberately avoid this route because importance sampling typically yields a much smaller effective sample size, which in practice leads to less powerful tests. We validate this empirically in Section 6.1. Moreover, importance sampling requires density ratios to be bounded, whereas our methods do not require such an assumption to be valid and consistent.

Our contributions are as follows. We begin by introducing an appropriate permutation distribution and sampling algorithm, through which we define p -values that we prove in Theorem 3 are uniformly valid for any choice of test statistic; the development here follows [Berrett et al. \(2020\)](#) and our contribution is in adapting this non-uniform permutation methodology and type I error theory to our problem. After laying these foundations we turn to our main technical novelty, which is in the construction of appropriate test statistics and the development of power results. Our new insights into the behaviour of our sampling algorithm under the alternative hypothesis lead to the introduction of a class of integral probability metrics and their sample versions as test statistics. We prove in Theorem 6 that the associated hypothesis tests are consistent in wide generality, even when the sampling algorithm is run for a finite amount of time and r is unbounded. In order to provide computable test statistics we specialize our class of integral probability metrics to those based on reproducing kernel Hilbert spaces, thus generalizing the maximum mean discrepancy ([Gretton et al., 2012](#)). In the canonical special case of continuous data under Sobolev smoothness conditions, we establish the minimax optimal separation rates for our problem; this is based on a detailed power analysis of our new methodology and novel minimax lower bounds. We further discuss how our methodology and power results extend to testing larger null hypotheses where r is an unknown member of a model.

We conclude this section with some notation that is used throughout the paper. We denote by $\mathbb{R}_+$ the set of positive real numbers and by $\mathbb{R}_{\geq 1}$ the set of real numbers greater than or equal to one. We further set $\mathbb{N}_+ = \mathbb{N} \setminus \{0\}$. The symmetric group of all permutations over the set $[n] := \{1, \dots, n\}$ is denoted by $\mathcal{S}_n$. For a set A , we write $\#A$ to denote its cardinality. The maximum and minimum operators are sometimes denoted by $\vee$ and $\wedge$, respectively. We define the support of a function f defined on a domain $\mathcal{X}$ to be the closure of the set where f does not vanish, that is, $\text{supp } f := \overline{\{x \in \mathcal{X} : f(x) \neq 0\}}$. The set of bounded and continuous functions on $\mathcal{X}$ is denoted by $\mathcal{C}_b^0(\mathcal{X})$, and δ_x denotes the Dirac delta measure centred at x . Given $1 \leq p < \infty$ and $d \geq 1$, we define the L^p norm of a function f as $\|f\|_p := (\int_{\mathbb{R}^d} |f(x)|^p dx)^{1/p}$, and the corresponding $L^p(\mathbb{R}^d)$ space as the set of all measurable functions for which this norm is finite. Furthermore, $\|\cdot\|_\infty$ denotes

the essential supremum norm, that is, $\|f\|_\infty := \text{ess sup}_{x \in \mathbb{R}^d} |f(x)|$, and $L^\infty(\mathbb{R}^d)$ refers to the set of functions that are bounded almost everywhere. Finally, we use $C^\infty(\mathcal{X})$ to denote the space of infinitely differentiable functions on a domain $\mathcal{X}$, and $\bar{z}$ for the complex conjugate of $z \in \mathbb{C}$.

2 Permutation methodology and validity

In this section we detail our general permutation procedure and its validity guarantees, which are largely adapted to our setting from the conditional independence testing problem studied in [Berrett et al. \(2020\)](#). Suppose we observe $Z = (Z_1, \dots, Z_{n+m}) = (X_1, \dots, X_n, Y_1, \dots, Y_m) \in \mathcal{X}^{n+m}$, consisting of $n + m$ independent random variables, where $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P_f$ and $Y_1, \dots, Y_m \stackrel{\text{i.i.d.}}{\sim} P_g$. Assume further that P_f and P_g are absolutely continuous with respect to a common dominating measure μ with densities f, g . We propose a refined permutation test designed to detect deviations from the null hypothesis (1) where $r : \mathcal{X} \rightarrow \mathbb{R}_+$ is a fixed, unnormalized function. By the definition of $\mathbb{R}_+$ given above, we have $r(x) > 0$ for all $x \in \mathcal{X}$. For simplicity, we also assume that the supports of f and g are the same. If this is not the case, i.e. if $r(x) > 0$ for all $x \in \mathcal{X}$ but $\text{supp } g \neq \text{supp } f$, then the testing problem becomes easier, as we expect to eventually observe signals in the regions corresponding to the symmetric difference of the supports. Similarly, if $\text{supp } r \neq \mathcal{X}$ but $\text{supp } g \subseteq \text{supp } r$, we can restrict the sample space to $\text{supp } r$ and carry out the analysis as before. If this inclusion does not hold, the testing problem again becomes easier. Finally, we shall also require that $\int g/r d\mu < \infty$ and $\int r f d\mu < \infty$, which ensure that the null hypothesis is nontrivial.

In line with permutation testing practice, the methodology we propose is to create copies $Z^{(1)}, \dots, Z^{(H)}$, with $H \geq 1$, that are exchangeable with Z under H_0 , and define a p -value by

$$p = \frac{1 + \sum_{h=1}^H \mathbb{1}\{T(Z^{(h)}) \geq T(Z)\}}{1 + H}, \quad (2)$$

where $T : \mathcal{X}^{n+m} \rightarrow \mathbb{R}$ is an arbitrary test statistic. Classically, these copies are uniformly shuffled versions of Z , but this is not appropriate here because the two samples are not exchangeable under the null. Intuitively, under the null hypothesis $g \propto r f$, points with larger $r(x)$ are more likely to originate from g than from f , hence we should favour permutations that assign observations with large $r(Z_i)$ to the last m positions. Based on this, we set

$$Z^{(h)} = Z_{\sigma^{(h)}} \quad \text{with} \quad \mathbb{P}\{\sigma^{(h)} = \sigma \mid Z\} = \frac{\prod_{i \in \{n+1, \dots, n+m\}} r(Z_{\sigma(i)})}{\sum_{\tilde{\sigma} \in \mathcal{S}_{n+m}} \prod_{i \in \{n+1, \dots, n+m\}} r(Z_{\tilde{\sigma}(i)})}, \quad (3)$$

where, for any vector $x = (x_1, \dots, x_{n+m})$ and permutation $\sigma \in \mathcal{S}_{n+m}$, we define $x_\sigma := (x_{\sigma(1)}, \dots, x_{\sigma(n+m)})$. As we see in the proof of Theorem 1 below, if we condition on the unordered set of observed data values, then the distribution of Z is the same as the distribution of $Z^{(h)}$ under H_0 . This leads to the exchangeability of $(Z, Z^{(1)}, \dots, Z^{(H)})$ and the validity of our tests, as summarized in the following result.

Theorem 1. *Assume that $H_0 : g \propto r f$ is true. Suppose that $\sigma^{(1)}, \dots, \sigma^{(H)}$ are drawn i.i.d. from (3) and write $Z^{(h)} = Z_{\sigma^{(h)}}$ for all $h \in [H]$. Then the sequence $(Z, Z^{(1)}, \dots, Z^{(H)})$ is exchangeable. In particular, for any statistic $T : \mathcal{X}^{n+m} \rightarrow \mathbb{R}$, the p -value defined in (2) is valid, satisfying $\mathbb{P}\{p \leq \alpha\} \leq \alpha$ for all $\alpha \in [0, 1]$.*

In practice, to conduct a statistical test we need to be able to sample permutations $\sigma^{(1)}, \dots, \sigma^{(H)}$ from (3).

We now address the challenge of generating these samples efficiently through Algorithm 1.

Algorithm 1 Pairwise sampler

- 1: Initial permutation σ_0 , integer $S \geq 1$. Call $K = \min\{n, m\}$.
- 2: **for** $t \in [S]$ **do**
- 3: Sample a vector of couples $\tau_t = \{(i_1^t, j_1^t), \dots, (i_K^t, j_K^t)\}$ such that $(i_1^t, \dots, i_K^t)$ are sampled uniformly and without replacement from $[n]$, and $(j_1^t, \dots, j_K^t)$ are sampled uniformly and without replacement from $\{n+1, \dots, n+m\}$, and initialize σ_t to be a copy of σ_{t-1} .
- 4: **for** $k \in [K]$ **do**
- 5: Draw a Bernoulli random variable B_{i_k, j_k}^t with

$$\mathbb{P}\{B_{i_k, j_k}^t = 1\} = \frac{r(Z_{\sigma_{t-1}(i_k^t)})}{r(Z_{\sigma_{t-1}(i_k^t)}) + r(Z_{\sigma_{t-1}(j_k^t)})} := p_{i_k, j_k}^t, \quad (4)$$

- 6: and swap $\sigma_t(i_k^t)$ with $\sigma_t(j_k^t)$ if $B_{i_k, j_k}^t = 1$.
 - 7: **end for**
 - 8: **return** σ_S .
-

Algorithm 1 is easily parallelizable due to the disjoint structure of the pairs in τ_t , and, most importantly, it accurately targets the distribution in (3), guaranteeing that the resulting Markov chain converges to the desired stationary distribution, as established in the following proposition.

Proposition 2. *For any initial permutation σ_0 , the distribution (3) of the permutation σ conditional on Z is the unique stationary distribution of the Markov chain defined in Algorithm 1.*

Remark 1. *By examining the proof of Proposition 2 in Appendix A.1, we see that other proposal mechanisms can also target the invariant distribution (3). Specifically, Step 5 in Algorithm 1 switches the indices i_k^t and j_k^t with probability p_{i_k, j_k}^t as in (4), but the key property used in the proof is $\mathbb{P}\{B_{i_k, j_k}^t = 1\} / \mathbb{P}\{B_{i_k, j_k}^t = 0\} = r(Z_{\sigma_{t-1}(i_k^t)}) / r(Z_{\sigma_{t-1}(j_k^t)})$. Thus, any alternative distribution on B_{i_k, j_k}^t satisfying this ratio would work equally well. For example, for the later power analysis of the methodology, it is more convenient to consider*

$$\tilde{p}_{i_k, j_k}^t := \frac{\hat{\lambda} m n r(Z_{\sigma_{t-1}(i_k^t)})}{\{n + \hat{\lambda} m r(Z_{\sigma_{t-1}(i_k^t)})\} \{n + \hat{\lambda} m r(Z_{\sigma_{t-1}(j_k^t)})\}} \quad (5)$$

with normalizing constant $\hat{\lambda}$ such that $\sum_{i=1}^{n+m} \{n + \hat{\lambda} m r(Z_i)\}^{-1} = 1$.

Remark 2. *Testing $H_0 : g \propto r f$ is equivalent to testing $H'_0 : f \propto \frac{1}{r} g$. We now show that our method is similarly invariant under taking reciprocals of r and relabelling the samples, using either p_{i_k, j_k}^t or $\tilde{p}_{i_k, j_k}^t$. For p_{i_k, j_k}^t , observe that the appropriate quantity after relabelling is*

$$(p_{i_k, j_k}^t)' := \frac{1/r(Z_{\sigma_{t-1}(j_k^t)})}{1/r(Z_{\sigma_{t-1}(i_k^t)}) + 1/r(Z_{\sigma_{t-1}(j_k^t)})} = \frac{r(Z_{\sigma_{t-1}(i_k^t)})}{r(Z_{\sigma_{t-1}(i_k^t)}) + r(Z_{\sigma_{t-1}(j_k^t)})},$$

which matches p_{i_k, j_k}^t exactly. In the case of $\tilde{p}_{i_k, j_k}^t$, define

$$(\tilde{p}_{i_k, j_k}^t)' := \frac{\hat{\theta}nm/r(Z_{\sigma_{t-1}(j_k^t)})}{\{m + \hat{\theta}n/r(Z_{\sigma_{t-1}(i_k^t)})\}\{m + \hat{\theta}n/r(Z_{\sigma_{t-1}(j_k^t)})\}},$$

with $\hat{\theta}$ such that $\sum_{i=1}^{n+m} \{m + n\hat{\theta}/r(Z_i)\}^{-1} = 1$. Here, n and m are switched because the roles of the X 's and Y 's are interchanged. By algebraic manipulation, one finds that $\hat{\theta} = \hat{\lambda}^{-1}$. Substituting back into $(\tilde{p}_{i_k, j_k}^t)'$ shows $(\tilde{p}_{i_k, j_k}^t)' = \tilde{p}_{i_k, j_k}^t$, confirming the invariance of our algorithm under reciprocal transformations of r .

By Proposition 2, Algorithm 1, when executed for sufficiently many steps S , generates a copy Z_{σ_S} which acts as an appropriate control for Z in testing H_0 . In fact, we can make a much stronger statement. Under the null hypothesis, conditionally on the unordered set of observed data values, the distribution of Z coincides with that of Z_σ when σ is sampled from (3). This implies that Z_{σ_S} represents a draw from the exact target distribution for any value of S , thereby providing a valid control for Z regardless of the number of steps taken. However, exact type I error control relies on the stronger requirement of the exchangeability of $(Z, Z^{(1)}, \dots, Z^{(H)})$. This can be achieved through a star-shaped sampling scheme, following the methodology introduced by Besag and Clifford (1989) and subsequently applied in permutation-based approaches by Berrett et al. (2020) and Ramdas et al. (2022). This results in the general testing procedure of Algorithm 2, which is the basis of our later methodological and theoretical contributions.

Algorithm 2 Density ratio permutation test

- 1: Data $Z = (Z_1, \dots, Z_{n+m}) = (X_1, \dots, X_n, Y_1, \dots, Y_m)$, initial permutation σ_0 , integers $S \geq 1$, $H \geq 1$, and test statistic T .
 - 2: Let σ_* be the output of Algorithm 1 after S steps, initialized at σ_0 .
 - 3: **for** $h \in [H]$, independently **do**
 - 4: Let $\sigma^{(h)}$ be the output of Algorithm 1 after S steps, initialized at σ_* .
 - 5: **end for**
 - 6: Compute $Z^{(h)} := Z_{\sigma^{(h)}}$ for all $h \in [H]$.
 - 7: **return** $p := (1 + H)^{-1}(1 + \sum_{h=1}^H \mathbb{1}\{T(Z^{(h)}) \geq T(Z)\})$.
-

Algorithm 2, when initialized with $\sigma_0 = \text{id}_{n+m}$, provides an exchangeable sampling mechanism, since the permutation σ_* lies S steps away from each of the permutations $(\sigma_0, \sigma^{(1)}, \dots, \sigma^{(H)})$ and the Markov chain associated to Algorithm 1 is reversible, as shown in the proof of Proposition 2. The following result verifies exchangeability and ensures that the results of Theorem 1 remain satisfied when the permuted vectors $Z^{(1)}, \dots, Z^{(H)}$ are obtained via Algorithm 2, thereby showing that the density ratio permutation test is valid.

Theorem 3. *Let p be the output of Algorithm 2 when initialized at $\sigma_0 = \text{id}_{n+m}$. If $H_0 : g \propto r f$ is true, then $\mathbb{P}\{p \leq \alpha\} \leq \alpha$ for all $\alpha \in [0, 1]$.*

Finally, although the permutation distribution is generally complex and Algorithm 1 is required to generate a sample from it, in Appendix D we explicitly characterize it using Fisher's noncentral hypergeometric distribution in the case of discrete data with finite support, significantly reducing computational costs.

3 Integral probability metrics and consistency theory

3.1 Integral probability metrics

We begin by establishing the consistency of our test under mild assumptions. A sequence of tests is consistent against a given class of alternatives if, as the sample size tends to infinity, the test rejects the null with probability 1 under every fixed alternative. In the classical $r \equiv 1$ case, this follows from [Hoeffding \(1952\)](#): letting σ be uniformly distributed over the symmetric group and assuming for simplicity $n/m \rightarrow \tau > 0$, we have, for all $\varphi \in C_b^0(\mathcal{X})$,

$$n^{-1} \sum_{i=1}^n \varphi(Z_{\sigma(i)}) \xrightarrow{\mathbb{P}} \int \varphi h_1 d\mu \quad \text{and} \quad m^{-1} \sum_{j=n+1}^{n+m} \varphi(Z_{\sigma(j)}) \xrightarrow{\mathbb{P}} \int \varphi h_1 d\mu, \quad (6)$$

where $C_b^0(\mathcal{X})$ denotes the space of bounded continuous functions on $\mathcal{X}$, and $h_1 = \tau(1+\tau)^{-1}f + (1+\tau)^{-1}g$. This means that the empirical distributions of both the first n and the last m permuted samples converge to the same mixture of f and g , so that Z_σ asymptotically satisfies the null. As a result, if we base the p -value (2) on an appropriate non-negative test statistic that is approximately zero only under the null, we shall see that $p \approx 0$ under every alternative, thus yielding consistency.

We will now extend this for general density ratios r . To this end, we first show that there exists a unique density h which satisfies the null hypothesis in finite samples while preserving the distribution of the combined data.

Lemma 4. *There exists a unique density h such that*

$$\frac{n}{n+m}h + \frac{m}{n+m} \frac{rh}{\int rh d\mu} = \frac{n}{n+m}f + \frac{m}{n+m}g,$$

and it is of the form $h = (nf + mg)/(n + \lambda_0 mr)$ for a suitable constant $\lambda_0 > 0$ such that $\int h d\mu = 1$.

We will use λ_0 to refer to such a normalizing constant from now on. The previous result, together with the fact that $\lambda_0 = (\int rh d\mu)^{-1}$, derived in the proof of Lemma 4 in Appendix A.2, implies that H_0 is equivalent to $f = h$ almost surely. This motivates the introduction, at the population level, of an integral probability metric of the form

$$T_{\mathcal{F},r}(f, g) := \sup_{\varphi \in \mathcal{F}} \left| \int \frac{n+m}{n+\lambda_0 mr} (\lambda_0 r f - g) \varphi d\mu \right| = \frac{n+m}{m} \sup_{\varphi \in \mathcal{F}} \left| \int (f - h) \varphi d\mu \right|,$$

for a suitable function class $\mathcal{F}$. It follows from Lemma 4 that we have $T_{\mathcal{F},r}(f, g) = 0$ under H_0 , but in order to have the reverse implication for a full characterization of the null we need additional assumptions on $\mathcal{F}$. Following similar lines to [Gretton et al. \(2012\)](#), we prove the following lemma.

Lemma 5. *Let $\mathcal{F} = \{\varphi \in \mathcal{H} : \|\varphi\|_{\mathcal{H}} \leq 1\}$, where $\mathcal{H}$ is a dense subset of $C_b^0(\mathcal{X})$ with respect to $\|\cdot\|_{\infty}$, and $\|\cdot\|_{\mathcal{H}}$ is a norm on $\mathcal{H}$. Then $T_{\mathcal{F},r}$ characterizes H_0 , meaning that $T_{\mathcal{F},r} = 0$ if and only if H_0 is true.*

Although this population measure of discrepancy depends on the sample sizes, we show in Proposition 17 in Appendix B that $T_{\mathcal{F},r}(f, g) \rightarrow \sup_{\varphi \in \mathcal{F}} \left| \int \{f - (g/r)(\int g/r d\mu)^{-1}\} \varphi d\mu \right|$ when $n/m \rightarrow 0$ and $T_{\mathcal{F},r}(f, g) \rightarrow$

$\sup_{\varphi \in \mathcal{F}} |\int \{(rf)(\int rfd\mu)^{-1} - g\} \varphi d\mu|$ when $n/m \rightarrow \infty$. This means that $T_{\mathcal{F},r}(f, g)$ is of constant order at fixed alternatives, even with imbalanced n, m . While it would be possible to use either of these simple limiting values as a population measure of discrepancy, we find $T_{\mathcal{F},r}$ to be a natural choice due to its invariance as in Remark 2 to relabelling of the samples; in Proposition 19 in Appendix B we prove that $\psi_r := (n+m)m^{-1}(f-h)$ is unchanged under this transformation. Furthermore, from a technical perspective, it allows a more elegant and straightforward analysis of Algorithm 1, which in turn leads to the consistency of the test given in Algorithm 2 based on the empirical version (7) of $T_{\mathcal{F},r}$, as shown in the next section.

3.2 General consistency theory

In the following, we will denote by $N(A, \delta, \|\cdot\|_*)$ the δ -covering number (e.g. Wainwright, 2019) of the set A with respect to the norm $\|\cdot\|_*$.

Theorem 6. Fix $\alpha \in (0, 1)$ and $H > \lceil 1/\alpha - 1 \rceil$. Let $\mathcal{H}$ be a dense subset of $C_b^0(\mathcal{X})$ with respect to $\|\cdot\|_\infty$, and suppose there exists fixed $\gamma > 0$ such that $\|\cdot\|_\infty \leq \gamma \|\cdot\|_{\mathcal{H}}$. Further, suppose $N(\{\|\varphi\|_{\mathcal{H}} \leq 1\}, \delta, \|\cdot\|_\infty)$ is finite for all $\delta > 0$ and define

$$T(z_1, \dots, z_{n+m}) := \sup_{\|\varphi\|_{\mathcal{H}} \leq 1} \frac{n+m}{nm} \left| \sum_{i=1}^n \frac{\hat{\lambda}mr(z_i)}{n + \hat{\lambda}mr(z_i)} \varphi(z_i) - \sum_{j=n+1}^{n+m} \frac{n}{n + \hat{\lambda}mr(z_j)} \varphi(z_j) \right|, \quad (7)$$

where $\hat{\lambda}$ satisfies $\sum_{i=1}^{n+m} \{n + \hat{\lambda}mr(z_i)\}^{-1} = 1$. Fix f, g such that H_0 is not true and, letting $W \sim n(n+m)^{-1}f + m(n+m)^{-1}g$, suppose that we have $\mathbb{E}[\{r(W)\}^2] \mathbb{E}[\{r(W)\}^{-2}] \leq V_0$ for some fixed $V_0 \geq 1$. Further suppose that, writing $\tau = \max(m/n, n/m)$, we have $S/\{\tau \log(m+n)\} \rightarrow \infty$. Then the test associated to the p -value returned by Algorithm 2 with parameters S, H and test statistic (7) is consistent, i.e. for all $\alpha \in (0, 1)$ we have $\mathbb{P}\{p \leq \alpha\} \rightarrow 1$ as $n, m \rightarrow \infty$.

Here $\hat{\lambda}$ acts as an estimator of λ_0 ; theoretical results quantifying the estimation error can be found in the proof of this theorem and in Lemma 14 in Appendix A.2. Note that $\sum_{i=1}^{n+m} \{n + \hat{\lambda}mr(Z_i)\}^{-1} = 1$ is equivalent to

$$\sum_{i=1}^n \frac{\hat{\lambda}mr(Z_{\sigma(i)})}{n + \hat{\lambda}mr(Z_{\sigma(i)})} = \sum_{j=n+1}^{n+m} \frac{n}{n + \hat{\lambda}mr(Z_{\sigma(j)})} \quad (8)$$

for all permutations $\sigma \in \mathcal{S}_{n+m}$; this fact is useful throughout our proofs and allows us to see $\hat{\lambda}$ as a normalization factor for r in any division of the empirical distribution $\hat{H}_{n,m}$ into two samples. As for the second moment condition on r , this is satisfied with $V_0 = C^2/c^2$ if $c \leq r(x) \leq C$ for all $x \in \mathcal{X}$. However, we require something significantly weaker. To illustrate this point, let $\phi(\cdot)$ be the density of the standard Gaussian distribution and take $f(x) = \phi(x)$ and $g(x) = \phi(x - \mu)$ with $\mu > 0$. Then, for $r(x) = g(x)/f(x) = \exp\{\mu x - \mu^2/2\}$ we have $\lambda_0 = 1$ and $\mathbb{E}\{r^2(Z)\} \mathbb{E}\{1/r^2(Z)\} = (n+m)^{-2}(ne^{\mu^2} + me^{3\mu^2})(ne^{3\mu^2} + me^{\mu^2}) \leq e^{6\mu^2}$, so that the assumption holds by taking $V_0 = e^{6\mu^2}$. Also, the assumption $\|\cdot\|_\infty \leq \gamma \|\cdot\|_{\mathcal{H}}$ is not overly restrictive and it is satisfied, e.g. when $\mathcal{H}$ is a reproducing kernel Hilbert space based on a uniformly bounded kernel.

The proof of Theorem 6 is based on the following lemma, which gives an approximation to the empirical distribution of our permuted data.

Lemma 7. Fix $S \in \mathbb{N}$ and let σ_S be the result of running Algorithm 1, with acceptance probabilities given in (5), for S steps. Assume that $\max(m, n) \geq 3$ and that we have

$$\left\{ \frac{1}{n+m} \sum_{i=1}^{n+m} r(Z_i) \right\} \left\{ \frac{1}{n+m} \sum_{i=1}^{n+m} \frac{1}{r(Z_i)} \right\} \leq \kappa \quad (9)$$

for some $\kappa \geq 1$. Further, define the empirical measure $\hat{H}_{n,m} = \sum_{i=1}^{n+m} \{n + \hat{\lambda}mr(Z_i)\}^{-1} \delta_{Z_i}$, where $\hat{\lambda} > 0$ is chosen such that $\sum_{i=1}^{n+m} \{n + \hat{\lambda}mr(Z_i)\}^{-1} = 1$. Writing $\tau = \max(m/n, n/m)$, for all $\varphi \in \mathcal{C}_b^0(\mathcal{X})$ we have

$$\mathbb{E} \left[\left(\frac{1}{n} \sum_{i=1}^n \varphi(Z_{\sigma_S(i)}) - \int \varphi d\hat{H}_{n,m} \right)^2 \middle| Z \right] \leq \frac{64\kappa \|\varphi\|_\infty^2 \min(m, n)}{n^2} + \|\varphi\|_\infty^2 \exp\left(-\frac{S}{8\tau\kappa}\right). \quad (10)$$

Remark 3. We see that, provided $S \gtrsim \tau\kappa \log n$, the approximation error due to running the chain only for a finite number of steps is of the same order as the error we would have if the chain had been run for an infinite number of steps. This notion of the convergence of the Markov chain is sufficient for our testing context and allows us to prove that tests that run in finite time have high power.

Besides being crucial in proving consistency, Lemma 7 shows us that the empirical distribution of the first n permuted samples converges in distribution to the limiting version of $hd\mu$ defined in Lemma 4. This extends (6) for general density ratios r satisfying the moment assumptions of Theorem 6. Finally, it is instructive to elaborate on the proof of this result, as the underlying strategy is novel and serves as a key component in the proof of other results, such as Theorem 9 below. To this end, let $S_n^t := \sum_{i=1}^n \varphi(Z_{\sigma_t(i)}) - n \int \varphi d\hat{H}_{n,m}$, where σ_t is the permutation at time t generated by Algorithm 1. The proof strategy is to show that this can be seen as Lyapunov function (e.g. Hairer, 2021) satisfying the geometric drift condition $\mathbb{E}[(n^{-1}S_n^{t+1})^2 \mid Z, \sigma_t] - (n^{-1}S_n^t)^2 \leq A - B(n^{-1}S_n^t)^2$ for appropriate constants A, B . In proving this bound it is very convenient to use acceptance probabilities $\tilde{p}_{i_k, j_k}^t$ as defined in (5). Given this drift condition the result follows using standard machinery, with the first term on the right-hand side of (10) corresponding to A/B and the second term capturing the convergence of the chain to stationarity.

4 Shifted maximum mean discrepancy and minimax optimality

4.1 Shifted maximum mean discrepancy

While Theorem 6 holds under mild assumptions, its practical utility depends on whether the supremum can be explicitly computed and a closed-form expression for $T_{\mathcal{F}, r}$ can be derived. To make this feasible, we focus on the specific case where $\mathcal{H}$ is a reproducing kernel Hilbert space. In this setting, we establish the following representation for the integral probability metric $T_{\mathcal{F}, r}$.

Proposition 8. Let $k(\cdot, \cdot)$ be a measurable kernel satisfying $\int_{\mathcal{X}} \sqrt{k(x, x)} \frac{(n+m)(\lambda_0 r(x)f(x) + g(x))}{n+m\lambda_0 r(x)} d\mu(x) < \infty$ and let $\mathcal{H}$ be the associated reproducing kernel Hilbert space.

$$T_{\mathcal{F}, r}^2(f, g) = \left(\sup_{\|\varphi\|_{\mathcal{H}} \leq 1} \left| \int \frac{n+m}{n+\lambda_0 mr} (\lambda_0 r f - g) \varphi d\mu \right| \right)^2$$

$$\begin{aligned}
&= \mathbb{E}_{X, X' \sim f} \left[\frac{\lambda_0^2 (n+m)^2 r(X) r(X') k(X, X')}{\{n + \lambda_0 m r(X)\} \{n + \lambda_0 m r(X')\}} \right] + \mathbb{E}_{Y, Y' \sim g} \left[\frac{(n+m)^2 k(Y, Y')}{\{n + \lambda_0 m r(Y)\} \{n + \lambda_0 m r(Y')\}} \right] \\
&- 2 \mathbb{E}_{\substack{X \sim f \\ Y \sim g}} \left[\frac{\lambda_0 (n+m)^2 r(X) k(X, Y)}{\{n + \lambda_0 m r(X)\} \{n + \lambda_0 m r(Y)\}} \right].
\end{aligned}$$

First, when $r \equiv 1$ we have $\lambda_0 = 1$, leading to $T_{\mathcal{F}, r=1}^2(f, g) = \text{MMD}_k^2(f, g)$, where $\text{MMD}_k(\cdot, \cdot)$ denotes the maximum mean discrepancy based on the kernel $k(\cdot, \cdot)$; see Equation (1) and Lemma 6 in [Gretton et al. \(2012\)](#). Consequently, our integral probability metric $T_{\mathcal{F}, r}$, which we will now refer to as the shifted maximum mean discrepancy and denote by $\text{MMD}_{r,k}(f, g)$, generalizes the classical one to account for the presence of a shift factor, while reducing to it in the case of a constant r . A similar quantity to $\text{MMD}_{r,k}(f, g)$, but without the denominators and λ_0 , was proposed in [Lee et al. \(2024\)](#) for conditional two-sample testing, where its definition was motivated by importance weighting. In contrast, we naturally derive $\text{MMD}_{r,k}(f, g)$ from Lemma 4, and see that the presence of the denominator plays a crucial role, as it is linked to the invariance under relabelling the samples in a similar spirit to Remark 2.

To ensure that $\text{MMD}_{r,k}(f, g)$ characterizes the null hypothesis, in line with Lemma 5, restrictions on $\mathcal{H}$ must ensure that $\text{MMD}_{r,k}(f, g) = 0$ if and only if H_0 holds. This requirement has been extensively studied in the statistical literature, and reproducing kernel Hilbert spaces satisfying this property are referred to as characteristic, with their defining kernels termed universal. For a comprehensive discussion see [Sriperumbudur et al. \(2010\)](#) and references therein. Notably, reproducing kernel Hilbert spaces induced by Gaussian and Laplacian kernels on $\mathbb{R}^d$ are characteristic.

Turning now to sample versions of the shifted maximum mean discrepancy, applying the same arguments as in Proposition 8 to empirical measures shows that the test statistic in (7) becomes

$$\begin{aligned}
V(x_1, \dots, x_n, y_1, \dots, y_m) &= \frac{(n+m)^2}{n^2 m^2} \sum_{i,j=1}^n \frac{\hat{\lambda}^2 r(x_i) r(x_j) k(x_i, x_j)}{\{n/m + \hat{\lambda} r(x_i)\} \{n/m + \hat{\lambda} r(x_j)\}} \\
&+ \frac{(n+m)^2}{m^4} \sum_{i,j=1}^m \frac{k(y_i, y_j)}{\{n/m + \hat{\lambda} r(y_i)\} \{n/m + \hat{\lambda} r(y_j)\}} - \frac{2(n+m)^2}{nm^3} \sum_{i=1}^n \sum_{j=1}^m \frac{\hat{\lambda} r(x_i) k(x_i, y_j)}{\{n/m + \hat{\lambda} r(x_i)\} \{n/m + \hat{\lambda} r(y_j)\}},
\end{aligned} \tag{11}$$

where $\hat{\lambda}$ satisfies (8). Ignoring the dependence on $\hat{\lambda}$, this estimator is a V-statistic and is asymptotically unbiased for $\text{MMD}_{r,k}^2(f, g)$ since $\hat{\lambda} \xrightarrow{\mathbb{P}} \lambda_0$, as shown in the proof of Theorem 6 in Appendix A.2. Following Theorem 6, the test given in Algorithm 2 using (11) with $k(\cdot, \cdot)$ from a characteristic reproducing kernel Hilbert space is consistent, provided that $\|\cdot\|_\infty \leq \gamma \|\cdot\|_{\mathcal{H}}$ and $N(\|\varphi\|_{\mathcal{H}} \leq 1, \delta, \|\cdot\|_\infty)$ is finite for all $\delta > 0$. The former condition is verified for uniformly bounded kernels, as the Cauchy–Schwarz inequality and the representer theorem imply $|\varphi(x)| = |\langle \varphi, k(\cdot, x) \rangle_{\mathcal{H}}| \leq \sqrt{k(x, x)} \|\varphi\|_{\mathcal{H}}$. This is a common assumption in the reproducing kernel Hilbert space literature (e.g. [Gretton et al., 2012](#), Definition 30), and it is satisfied by many standard choices of $k(\cdot, \cdot)$. The latter condition is also mild, and it is satisfied for common kernels such as the exponential kernel (e.g. [Yang et al., 2020](#), Lemma D.2).

4.2 Minimax rate optimality

While Theorem 6 shows the consistency of our test under mild assumptions, we can further show its minimax rate optimality when $\mathcal{X} = \mathbb{R}^d$ under the extra assumption that $\psi_r := (n+m)m^{-1}(f-h)$ lies in a Sobolev ball. We shall also require that r is uniformly bounded, i.e. there exist $c, C > 0$ such that $c \leq r(x) \leq C$ for all $x \in \mathcal{X}$, and balanced sample sizes; these are stronger assumptions than those needed to prove consistency and might not be necessary, but simplify an already involved analysis. It is also more convenient to consider the test statistic

$$\begin{aligned} U(x_1, \dots, x_n, y_1, \dots, y_m) &= \frac{(n+m)^2}{n^2 m^2} \sum_{i \neq j=1}^n \frac{\hat{\lambda}^2 r(x_i) r(x_j) k_\zeta(x_i, x_j)}{\{n/m + \hat{\lambda} r(x_i)\} \{n/m + \hat{\lambda} r(x_j)\}} \\ &+ \frac{(n+m)^2}{m^4} \sum_{i \neq j=1}^m \frac{k_\zeta(y_i, y_j)}{\{n/m + \hat{\lambda} r(y_i)\} \{n/m + \hat{\lambda} r(y_j)\}} - \frac{2(n+m)^2}{nm^3} \sum_{i=1}^n \sum_{j=1}^m \frac{\hat{\lambda} r(x_i) k_\zeta(x_i, y_j)}{\{n/m + \hat{\lambda} r(x_i)\} \{n/m + \hat{\lambda} r(y_j)\}}, \end{aligned} \quad (12)$$

where $k_\zeta(\cdot, \cdot)$ is a multivariate kernel with bandwidth $\zeta \geq 1$ which will be defined later. Note that apart from a slightly unusual normalization and the dependence on $\hat{\lambda}$, (12) is essentially a U-statistic which serves as an estimator of $\text{MMD}_{r, k_\zeta}^2(f, g)$. This is a slight modification of (11), with diagonal terms removed, that we expect to have similar performance but which is simpler to analyze. Coming back to the definition of k_ζ , let $K : \mathbb{R} \rightarrow \mathbb{R}$ be a bounded, continuous, even, characteristic positive semidefinite kernel with $K \in L^1(\mathbb{R}) \cap L^4(\mathbb{R})$, $\int_{\mathbb{R}} K(u) du = 1$, and $0 < K(0) < \infty$. Given a bandwidth $\zeta \geq 1$, we define a characteristic kernel on $\mathbb{R}^d$ by $k_\zeta(x, y) := \zeta^d \prod_{i=1}^d K(\zeta(x_i - y_i))$ for $x, y \in \mathbb{R}^d$. For example, choosing $K(u) = \frac{1}{\sqrt{\pi}} e^{-u^2}$ recovers the Gaussian kernel in $\mathbb{R}^d$, while choosing $K(u) = \frac{1}{2} e^{-|u|}$ yields the Laplace kernel. Further useful properties of k_ζ can be found in the proof of Theorem 9. Extensions to separate kernels $K_i(\cdot)$ with distinct bandwidths ζ_i along each dimension are straightforward and can be obtained by following similar arguments as in Schrab et al. (2023).

We now characterize the optimality of the testing procedure within the minimax framework, aiming to identify the smallest separation between the null and alternative hypotheses that allows for reliable discrimination. Smoothness is assumed under the alternative but not under the null, as we already showed in Section 2 that Algorithm 2 test guarantees uniform, non-asymptotic control of the type I error. For fixed $r : \mathcal{X} \rightarrow \mathbb{R}_+$ such that $0 < c \leq r(x) \leq C$ for all $x \in \mathcal{X}$, and for $\rho > 0$, we are interested in testing

$$H_0 : g \propto r f \quad \text{vs.} \quad H_1^r(\rho) : \|\psi_r\|_2 > \rho, \quad (13)$$

and aim to find the smallest value of ρ such that there exists a test with uniform error control. Arguing as in Section 3.1, it appears more natural in our setting to define the separation in terms of ψ_r . Nonetheless, when r is uniformly bounded, alternative choices such as $\|g - \bar{r}f\|_2$ with $\bar{r} = r(\int r f d\mu)^{-1}$ are also valid; we prove in Proposition 18 in Appendix A.4 that these norms are equivalent when $\mathcal{X}$ is compact.

Let Ψ denote the collection of randomized tests based on the combined dataset $(X_1, \dots, X_n, Y_1, \dots, Y_m)$. Each test is represented by a function $\varphi : \mathcal{X}^{n+m} \rightarrow [0, 1]$, where, upon observing $(x_1, \dots, x_n, y_1, \dots, y_m)$, the null hypothesis H_0 is rejected with probability $\varphi(x_1, \dots, x_n, y_1, \dots, y_m)$. Additionally, define $\Psi(\alpha)$ as the subset of Ψ consisting of tests with size α , where $\alpha \in (0, 1)$. Define the d -dimensional Sobolev ball with

smoothness parameter $s > 0$ and radius $L > 0$ as

$$\mathcal{S}_d^s(L) := \left\{ p \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d) : \int_{\mathbb{R}^d} \|\xi\|_2^{2s} |\widehat{p}(\xi)|^2 d\xi \leq (2\pi)^d L^2 \right\}, \quad (14)$$

where $\widehat{p}$ denotes the Fourier transform of p , i.e. $\widehat{p}(\xi) := \int_{\mathbb{R}^d} p(x) e^{-i\langle x, \xi \rangle} dx$ for all $\xi \in \mathbb{R}^d$. Throughout this section we will also assume $m \leq n \leq \tau m$ for some $\tau \in [1, \infty)$, which is to say that n and m are of the same order. This assumption is commonly employed in the literature on two-sample testing (e.g. [Schrab et al., 2023](#); [Li and Yuan, 2024](#)). For fixed $r : \mathcal{X} \rightarrow \mathbb{R}_+$ such that $0 < c \leq r(x) \leq C$ for all $x \in \mathcal{X}$, we may then define the minimax separation to be $\rho_r^* \equiv \rho_r^*(n, m, \theta, \alpha, \beta) := \inf \left\{ \rho > 0 : \inf_{\varphi \in \Psi(\alpha)} \sup_{(f, g) \in \mathcal{S}_\theta^r(\rho)} \mathbb{E}_P(1 - \varphi) \leq \beta \right\}$, where $P = P_f^{\otimes n} \otimes P_g^{\otimes m}$, $\theta = (d, \tau, M, s, L) \in \mathbb{N}_+ \times \mathbb{R}_{\geq 1} \times \mathbb{R}_+^3$ and

$$\mathcal{S}_\theta^r(\rho) := \{(f, g) : \max(\|f\|_\infty, \|g\|_\infty) \leq M, \|\psi_r\|_2 > \rho \text{ and } \psi_r \in \mathcal{S}_d^s(L)\}. \quad (15)$$

We can prove an upper bound on the minimax separation by showing that the test in Algorithm 2, based on the test statistic (12) with bandwidth $\zeta = n^{\frac{2}{4s+d}}$, has good power when ρ is sufficiently large.

Theorem 9. *Let $\mathcal{X} = \mathbb{R}^d$ and fix $\alpha, \beta \in (0, 1)$ such that $\alpha + \beta < 1$. There exists $C_1 = C_1(c, C, \theta, \alpha, \beta) > 0$ such that $\rho_r^* \leq C_1 n^{-2s/(4s+d)}$.*

Regarding optimality, a matching lower bound has been established for the classical two-sample testing problem over Sobolev balls, showing that there exists a constant $c_1 = c_1(\theta, \alpha, \beta)$ such that $\rho_{r \equiv 1}^* \geq c_1 n^{-2s/(4s+d)}$; for example, see Theorems 3 and 5 in [Li and Yuan \(2024\)](#). In our setting, the minimax separation rate ρ_r^* will typically depend on the specific choice of r . Nevertheless, we derive the following matching lower bound under the assumptions that r is bounded above and below.

Theorem 10. *Fix $\alpha, \beta \in (0, 1)$ such that $\alpha + \beta < 1$ and suppose that $s \in \mathbb{N}_+$. There exists $c_1 = c_1(c, C, \theta, \alpha, \beta) > 0$ such that for all $r : \mathcal{X} \rightarrow \mathbb{R}_+$ satisfying $0 < c \leq r(\cdot) \leq C$, we have $\rho_r^* \geq c_1 n^{-2s/(4s+d)}$.*

When r is uniformly bounded, the combined results of Theorems 9 and 10 demonstrate that $\rho_r^* \asymp n^{-2s/(4s+d)}$, confirming that Algorithm 2 with test statistic (12) achieves optimality. Future work could study the dependence of ρ_r^* on r , though by analogy with the goodness-of-fit testing problem for densities, this is likely to be technically demanding. Indeed, it is known (e.g. [Balakrishnan and Wasserman, 2019](#)) that minimax rates of convergence for testing the null hypothesis that $f = f_0$ given $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} f$ intricately depend on the specific choice of f_0 , with the hardest version of the problem being when f_0 is a uniform density. Although the simulations in Section 6.1 and the results for binary data in Appendix D seem to suggest that the hardest shifts to test are those closer to $r \equiv 1$, it is still unclear if this is the case in full generality. Here, hardness is meant in the information-theoretic sense; the impact of r on the computational runtime of our procedures is discussed in Remark 3.

Finally, we note that the assumption $s \in \mathbb{N}_+$ is common in the statistical literature ([Li and Yuan, 2024](#)), and plays a critical role in the lower bound construction. In particular, it ensures that the bump functions introduced in Lemma 16 in Appendix A.2 are orthogonal with respect to the Sobolev inner product $\langle \phi_1, \phi_2 \rangle_{\mathcal{S}_d^s} := \int_{\mathbb{R}^d} \|\xi\|_2^{2s} \widehat{\phi_1}(\xi) \widehat{\phi_2}(\xi) d\xi$. This orthogonality holds when $s \in \mathbb{N}_+$ because the bump functions are supported on disjoint sets. However, this argument breaks down for general $s > 0$, as disjoint support

no longer guarantees orthogonality in the Sobolev inner product. Extending the construction to non-integer smoothness may still be possible using techniques from Butucea (2007) and Albert et al. (2022).

5 Extensions to unknown density ratios

We move beyond the case of a simple null, and consider the harder problem of testing

$$H_0^{\mathcal{R}} : \text{there exists } r_{\star} \in \mathcal{R} \text{ such that } g \propto r_{\star} f, \quad (16)$$

where $\mathcal{R}$ is a suitable function class. If $\mathcal{R}$ is compact in a topology with respect to which $r \mapsto \|\psi_r\|_2$ is continuous, this is equivalent to saying that $\inf_{r \in \mathcal{R}} \|\psi_r\|_2 = 0$. We will focus on cases where $\mathcal{R}$ is either a parametric family or a class whose members act only on a proper subset of the variables. In the former case, (16) can serve as an important preliminary tool for model selection. For instance, it is common to assume that the density ratio belongs to a specific parametric family of functions (Qin, 1998; Kanamori et al., 2010), with maximum likelihood estimation used for inference. See Qin (1998) and the references therein for a comprehensive overview of applications that rely on this modelling assumption.

Furthermore, when $\mathcal{R}$ consists of functions acting on a proper subset of variables, i.e. $r(x) = \tilde{r}(x_S)$ for $S \subset [d]$, (16) specializes to tests for structured shifts such as marginal or conditional changes, thus providing useful diagnostics for practitioners. As a concrete example in this setting, we consider the conditional two-sample testing problem (e.g., Lee et al., 2024), where we observe two independent samples $(X_i^{(1)}, Y_i^{(1)})_{i=1}^{n_1}$ drawn independently from a joint density $f_{XY}^{(1)}$, and $(X_i^{(2)}, Y_i^{(2)})_{i=1}^{n_2}$ drawn independently from $f_{XY}^{(2)}$, and we test the equality of conditional densities $f_{Y|X}^{(1)}(y | x) = f_{Y|X}^{(2)}(y | x)$ for $f_X^{(1)}$ -almost every x . Because this is equivalent to $f_{XY}^{(2)}(x, y) = (f_X^{(2)}(x)/f_X^{(1)}(x)) f_{XY}^{(1)}(x, y)$, conditional two-sample testing can be cast as a hypothesis test on density ratios. Since the marginal density ratio is unknown and must be estimated in practice, this yields an instance of the composite problem (16). Even if this ratio were known, as would be the case in a trial with perfect compliance (e.g. Lei and Candès, 2021) or a Model-X setting (Candès et al., 2018), the resulting test would reduce to the simpler null (1), which remains non-trivial and can be addressed via Algorithm 2.

To tackle (16), we first estimate the density ratio on a training sample using density-ratio estimation methods (Kanamori et al., 2012; Sugiyama et al., 2012), and then feed this estimate into Algorithm 2, which is run on an independent sample. At the testing stage, however, plugging an estimated ratio into Algorithm 2 can inflate the type I error. In this regard, we show that if $H_0^{\mathcal{R}}$ is satisfied and $\hat{r}$ is a good approximation of the true $r_{\star}$, then the excess type I error of Algorithm 2 is bounded by the total variation distance between the product measures of the normalized versions of $\hat{r}f$ and $r_{\star}f$. For any two distributions P_1, P_2 defined on the same probability space, the total variation distance is defined as $\text{TV}(P_1, P_2) = \sup_A |P_1(A) - P_2(A)|$, where the supremum is taken over all measurable sets. In the result below, we assume that both the approximation $\hat{r}$ of the true $r_{\star}$ and the test statistic $T : \mathcal{X}^{n+m} \rightarrow \mathbb{R}$ are deterministic, meaning that they are selected independently of Z .

Proposition 11. *Assume $H_0^{\mathcal{R}}$ is true, and let $r_{\star}$ be such that $g \propto r_{\star} f$. Write $\hat{r}$ for an approximation of $r_{\star}$, and define $p_{\hat{r}}$ to the output of Algorithm 2 based on $\hat{r}$ with fixed parameters $S \geq 1, H \geq 1$. Calling $\tilde{r} = (\int \hat{r} f d\mu)^{-1} \hat{r}$, $\bar{r}_{\star} = (\int r_{\star} f d\mu)^{-1} r_{\star}$, for all $\alpha \in [0, 1]$ we have $\mathbb{P}\{p_{\hat{r}} \leq \alpha\} \leq \alpha + \text{TV}(\{\tilde{r} f\}^{\otimes m}, \{\bar{r}_{\star} f\}^{\otimes m})$.*

For Proposition 11 to have practical implications, we need $\text{TV}(\{\tilde{r}f\}^{\otimes(n-N)}, \{\tilde{r}_\star f\}^{\otimes(n-N)}) \xrightarrow{\mathbb{P}} 0$. We discuss conditions under which this holds for conditional two-sample testing in Example 1 in Appendix B, where we construct separate training and test samples via sample splitting. As in Theorem 4 of Berrett et al. (2020), however, Proposition 11 provides a worst-case guarantee taken over an arbitrary test statistic T , which may be chosen adversarially to maximize sensitivity to errors in estimating $r_\star$. In practice, for the concrete choices of T considered in Subsection 4.1, the procedure appears substantially more robust than this worst-case bound suggests; see Figure 1(c) for supporting simulations. Nonetheless, without structural assumptions ensuring that the total variation term is negligible, Proposition 11 does not yield universal finite-sample validity. In the case of conditional two-sample testing this limitation is intrinsic: Theorem 1 in Lee et al. (2024) shows that any valid conditional two-sample test has no power against any alternative, echoing the hardness result of Shah and Peters (2020) for conditional independence testing. Thus, in this setting the lack of universal validity is not a deficiency of our method, but a consequence of the problem's inherent difficulty. Whether analogous hardness phenomena hold for other instances of (16) remains an open question.

For power results, we can use Theorem 6 to show that whenever (16) does not hold, the test given in Algorithm 2 and based on an approximation $\hat{r} \in \mathcal{R}$ has high power, provided that the key moment condition of Theorem 6 is satisfied.

Proposition 12. *Assume that $\hat{r} \in \mathcal{R}$ satisfies the moment condition of Theorem 6 with $\hat{r}$ in place of r , and that the remaining conditions hold. Write $p_{\hat{r}}$ for the output of Algorithm 2 with parameters $S \geq 1, H \geq 1$, run with $\hat{r}$ and based on (7). If (16) does not hold, for all $\alpha \in (0, 1)$ we have $\mathbb{P}\{p_{\hat{r}} \leq \alpha\} \rightarrow 1$ as $n, m \rightarrow \infty$.*

This result is an immediate consequence of Theorem 6. In particular, if (16) does not hold, no element of $\mathcal{R}$ is proportional to g/f . Therefore, applying Theorem 6 to the test based on $\hat{r} \in \mathcal{R}$ shows that our procedure has high power whenever the theorem's assumptions are met. In practice, if $\hat{r}$ is an estimate based on a held out sample that satisfies $\mathbb{P}(\mathbb{E}[\{\hat{r}(W)\}^2 \mid \hat{r}] \mathbb{E}[\{\hat{r}(W)\}^{-2} \mid \hat{r}] \leq V_0) \geq 1 - \eta$ for $\eta \in (0, 1)$, $V_0 \geq 1$ and $W \sim n(n+m)^{-1}f + m(n+m)^{-1}g$, we can use the previous result to show that for all $\alpha \in (0, 1)$ we have $\liminf_{n, m \rightarrow \infty} \mathbb{P}\{p_{\hat{r}} \leq \alpha\} \geq 1 - \eta$. Now, the applicability of Proposition 12 simply requires that $\hat{r}$ lies in $\mathcal{R}$ and satisfies the two-sided second-moment condition. If we also assume that $\hat{r}$ is close to a suitable $r_\star \in \mathcal{R}$, we can bound the excess type II error in finite samples. This is based on the following result.

Proposition 13. *Write $\hat{r}$ for a deterministic approximation of $r_\star$, and define $\eta(Z) := \max_{i \in [n+m]} |\log \hat{r}(Z_i) - \log r_\star(Z_i)|$. Let $p_{\hat{r}}$ denote the p -value in (2) computed from $(Z_{\hat{r}}^{(1)}, \dots, Z_{\hat{r}}^{(H)})$, where $Z_{\hat{r}}^{(h)} := Z_{\sigma^{(h)}}$ and $(\sigma^{(1)}, \dots, \sigma^{(H)})$ are i.i.d. samples from (3) with $\hat{r}$ used as the density ratio. Define $p_{r_\star}$ analogously, replacing $\hat{r}$ by $r_\star$ throughout. For all $\alpha \in (0, 1)$ we have $\mathbb{P}\{p_{\hat{r}} \leq \alpha\} \geq \mathbb{P}\{p_{r_\star} \leq \alpha\} - H \mathbb{E}_Z[\sqrt{1 - \exp\{-2m\eta(Z)\}}]$.*

Proposition 13 immediately leads to the consistency of our procedure provided that $r_\star$ satisfies $\|\psi_{r_\star}\|_2 > 0$ and the assumptions of Theorem 6, and the estimator $\hat{r}$, computed on an held-out training data of size N_{train} , is such that $H \mathbb{E}_Z[\sqrt{1 - e^{-2m\eta(Z)}} \mid \hat{r}] \xrightarrow{\mathbb{P}} 0$. For example, for a univariate logistic regression model with Gaussian data we expect $\mathbb{E}_Z[\sqrt{\eta(Z)} \mid \hat{r}] \lesssim \log^{1/4}(n+m) N_{\text{train}}^{-1/4}$ with high probability, suggesting that the error term vanish as long as $N_{\text{train}} \gg H^4 m^2 \log(n+m)$. The assumption that $\hat{r}$ remains close to $r_\star$ even when the model $\mathcal{R}$ is misspecified parallels existing theory on density-ratio estimation, which ensures that, under suitable regularity conditions, density-ratio estimators converge to the projection of the true density

ratio g/f onto $\mathcal{R}$ with respect to an appropriate divergence (Kanamori et al., 2010; Sugiyama et al., 2012). Analogous consistency guarantees continue to hold when the p -values are computed via Algorithm 2. More importantly, under additional smoothness assumptions on both the null and the alternative, Propositions 11 and 13 provide finite-sample control of the excess type I and type II errors for the composite null (16); combining this with our finite-sample guarantees available for the simple null, such as those in Section 4.2, can lead to stronger non-asymptotic power results. This general approach was adopted in Zheng et al. (2025) for the conditional two-sample testing problem.

6 Simulations and real-data applications

6.1 Synthetic data

Code and dataset access for reproducing the simulations are available at <https://github.com/abordino/DRPT>. An implementation of Algorithm 2 is provided in the R-package DRPT (Bordino and Berrett, 2025). We now define variants of Algorithm 2, each with $S = 50$ and $H = 99$, that differ in their choice of test statistic, as well as classical permutation tests applied to resampled data. (E1) corresponds to Algorithm 2 with the empirical shifted maximum mean discrepancy (12) and Gaussian kernel $k(x, y) = \zeta^d e^{-\zeta^2 \|x-y\|_2^2}$; the bandwidth ζ is chosen via the median heuristic. (E2) uses rejection sampling on $X_{1:n}$ to obtain $W_{1:n'} \sim \bar{r}f$, then applies a uniform permutation test based on the classical maximum mean discrepancy with Gaussian kernel on $(W_{1:n'}, Y_{1:m})$, using the median heuristic for the bandwidth. (E3) corresponds to Algorithm 2 with V-statistic (11) and collision kernel $k(x, y) = \sum_{j=0}^J \mathbb{1}\{x = j\} \mathbb{1}\{y = j\}$ for fixed $J \geq 1$.

In this section, we empirically validate the performance of Algorithm 2 on synthetic data. Fig. 1(a) shows the performance of Algorithm 2 in the case of bivariate data. Here, we let P_q be an absolutely continuous distribution on $[0, 1]$ with density $q(x) = 2x$, and choose $P_f = \text{Unif}([0, 1]^2)$ and $P_g = (1 + \eta)^{-1} P_q^{\otimes 2} + \eta(1 + \eta)^{-1} \text{Beta}(\frac{1}{2}, \frac{1}{2})^{\otimes 2}$ for $\eta \in \{0, \dots, 0.80\}$. When $\eta = 0$, g satisfies the null hypothesis with $r(x, y) = 4xy$, while larger values of η correspond to greater departures from the null. The purple curve corresponds to (E1), and we compare it against (E2), shown in green. We also consider a similar setting, shown in orange and blue, where we fix $P_f = \text{Unif}([0, 1]^2)$ and $P_g = (1 + \eta)^{-1} P_q \otimes \text{Unif}([0, 1]) + \eta(1 + \eta)^{-1} \text{Beta}(\frac{1}{2}, \frac{1}{2})^{\otimes 2}$; this implies that the null is satisfied for $r'(x, y) = 2x$ when $\eta = 0$. We fix $n = m = 150$ and repeat each test 500 times in each setting. Algorithm 2 demonstrates superior performance compared to the rejection sampling-based procedure, which is to be expected due to the reduction in effective sample size inherent in such resampling methods. This justifies our choice to forego rejection-sampling schemes and utilize the full sample directly. Furthermore, less extreme shifts, such as r' , seem harder to test, as Algorithm 2 shows greater power with more peaked density ratios; this trend is also reflected in the empirical results for binary data in Appendix D.

In the case of binary data, we fix $n = 100$ and $m \in \{100, 200, 500, 2000\}$, $r = (1, 3)$ and choose $f = \text{Ber}(1/2)$ and $g = \text{Ber}(3(1 - \eta)/4 + \eta/4)$ with $\eta \in \{0, \dots, 0.90\}$. The null hypothesis holds for $\eta = 0$, with increasing values of η representing greater deviations from the null. We run (E3) 5000 times in each setting and report the average decision. The results, illustrated in Fig. 1(b), indicate that our methodology performs well even with imbalanced sample sizes, achieving greatest power in the most imbalanced case, which corresponds to the largest sample sizes in this setting.

We also empirically verify that Proposition 11 characterizes a worst-case scenario, while in practice the

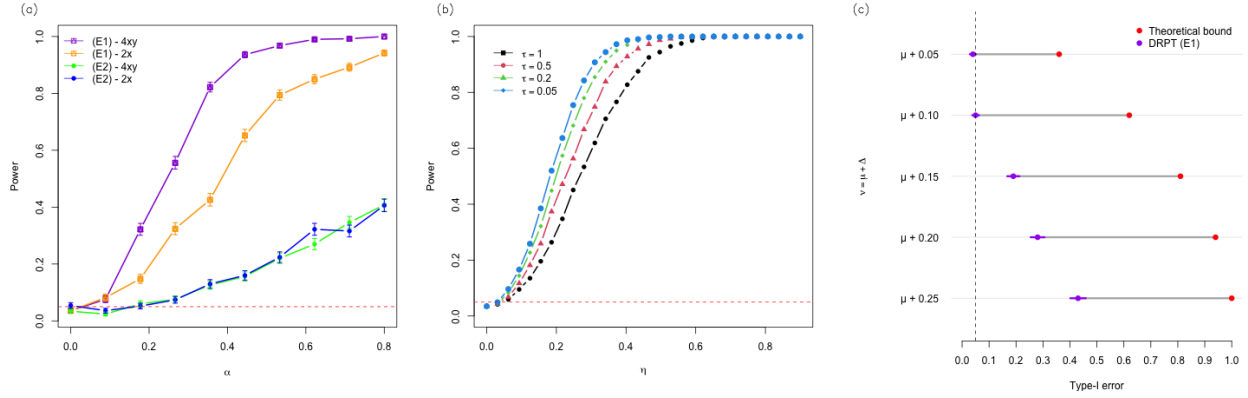


Figure 1: (a) Empirical power of (E1) for bivariate data on the unit square: purple, $r(x, y) = 4xy$; orange, $r'(x, y) = 2x$. Green and blue: alternative approaches based on (E2). (b) Estimated power of (E3) for binary data with varying sample sizes and $r = (1, 3)$; $\tau := n/m$. (c) Empirical validation of Proposition 11 for Gaussian data with misspecified mean: red, theoretical bound $\alpha + \text{TV}(\mathcal{N}(\mu, 1)^{\otimes m}, \mathcal{N}(\nu, 1)^{\otimes m}) = \alpha + 2\Phi(\sqrt{m}|\mu - \nu|/2) - 1$; purple, estimate of type I error over 300 runs of (E1) based on $\hat{r}$. In all panels, error bars denote ± 1 standard error.

type I error increases only mildly when $\hat{r}$ is sufficiently accurate and T is chosen appropriately. Let ϕ denote the density of a standard Gaussian and Φ its cumulative distribution function, and set $f(x) = \phi(x)$ and $g(x) = \phi(x - \mu)$ with $\mu > 0$ so that $r_\star = \exp\{\mu x - \mu^2/2\}$. If we run Algorithm 2 using $\hat{r}(x) = \exp\{\nu x - \nu^2/2\}$ with $\nu > 0$, meaning that the mean is misspecified, theory predicts that the excess type I error is upper bounded by $\text{TV}(\mathcal{N}(\mu, 1)^{\otimes m}, \mathcal{N}(\nu, 1)^{\otimes m}) = 2\Phi(\sqrt{m}|\mu - \nu|/2) - 1$. An approximation of this quantity when $n = m = 250$, $\mu = 1$, $\nu \in \mu + \{0.05, 0.10, 0.15, 0.20, 0.25\}$ is shown in red in Fig. 1(c). In contrast, running (E1) under the same setup and using the average decision over 300 repetitions in each setting to estimate the size yields a much smaller inflation of the type I error in practice, as shown by the purple dots.

Finally, we study a scenario motivated by propensity weighting in causal inference. Let covariates $Z \sim \mathcal{N}(0, \Sigma)$ with $\Sigma \in \mathbb{R}^{10 \times 10}$, and assign treatment $D \in \{0, 1\}$ according to $\mathbb{P}(D = 1 | Z) = \{1 + \exp(\eta_\gamma(Z))\}^{-1}$ with $\eta_\gamma(z) = \beta_0 + \beta^\top z + \gamma\{\sin(10z_1) + z_2 z_3\}$ and $\gamma \in \{0, 0.25, 0.5, 1, 2\}$. Writing $f(z)$ and $g(z)$ for the conditional densities of Z given $D = 0$ and $D = 1$, respectively, and $\pi := \mathbb{P}(D = 1) \in (0, 1)$, the density ratio is $r(z) := g(z)/f(z) = \frac{1-\pi}{\pi} \exp\{\eta_\gamma(z)\}$. In medical applications, such ratios are often used to reweight untreated patients when estimating causal effects. For example, let $n_0 := \#\{i : D_i = 0\}$ and $n_1 := \#\{i : D_i = 1\}$, and assume well-defined potential outcomes $(Y(0), Y(1))$, $Y = DY(1) + (1 - D)Y(0)$, and $(Y(0), Y(1)) \perp\!\!\!\perp D | Z$. Then $n_1^{-1} \sum_{i: D_i=1} Y_i - n_0^{-1} \sum_{i: D_i=0} r(Z_i) Y_i$ is an unbiased estimator of the average treatment effect on the treated, $\mathbb{E}[Y(1) - Y(0) | D = 1]$. In the simulation study, we estimate r for each $\gamma \in \{0, 0.25, 0.5, 1, 2\}$ using linear logistic or kernel logistic regression (e.g. Sugiyama et al., 2010) on a training sample of size 1000, apply (E1) to an independent test sample of size 200, repeat this 200 times, and report the average rejection rate as an estimate of power. The results in Fig. 2(a) show that kernel logistic regression performs well for all values of γ , as expected from its flexibility, whereas linear logistic regression is misspecified for $\gamma > 0$, and our test has high power in detecting the inadequacy of this estimator.

6.2 Real-world application

6.2.1 New York-frisk and Stroop-effect datasets

When working with simple classes of distributional shifts, our hypothesis tests can be inverted in the usual way to furnish confidence sets, offering useful insights to practitioners. Formally, writing C_α for the set of all functions r that are not rejected by our test at level α , we have $\mathbb{P}(g/f \in C_\alpha) \geq 1 - \alpha$. To illustrate this, we consider two practical scenarios, starting with the New York-frisk dataset for the years 2011 and 2012, which contains detailed records of police stop-and-frisk encounters. Restricting attention to stops labelled “criminal possession of a weapon” and using a binary indicator for whether any weapon was found, we partition the sample into X_i ’s, Black individuals with $n_B = 82\,626$, and Y_i ’s, White individuals with $n_W = 6\,383$. A conjecture in the literature, illustrated in Fig. 1(b) of [Goel et al. \(2016\)](#) and recalled in Section 8.1 in [Koh et al. \(2020\)](#), suggests that Black individuals are approximately five times less likely to possess a weapon compared to White individuals. We examine this claim using (E3) and test $w_1/w_0 = rb_1/b_0$ for $r \in \{0.1, \dots, 7\}$, where w_1 and b_1 denote the probabilities of weapon possession for White and Black individuals respectively. The blue curve in Fig. 2(b) shows that (E3) does not reject for $r \in [5, 6]$, consistent with Black individuals being about five times less likely than White individuals to be found carrying a weapon conditional on being frisked. Repeating the analysis for the years 2015–2016, where $n'_W=138$ and $n'_B=2\,298$, (E3) does not reject for $r \in [2, 4]$, potentially suggesting a change in practices over time. To illustrate the informativeness of confidence intervals based on Algorithm 2, we also report Wald-type confidence intervals computed as follows. Define $z_{1-\alpha/2}$ to be the $(1 - \alpha/2)$ quantile of the standard normal distribution and $\text{logit}(p) := \log\{p/(1 - p)\}$ for $p \in (0, 1)$. Let $\hat{w}_1 := n_W^{-1} \sum_i Y_i$ with $\sqrt{n_W}(\hat{w}_1 - w_1) \xrightarrow{d} \mathcal{N}(0, (1 - w_1)w_1)$, so by the delta method $\text{Var}\{\text{logit}(\hat{w}_1)\} \approx \{n_W w_1(1 - w_1)\}^{-1} \approx \{n_W \hat{w}_1\}^{-1} + \{n_W(1 - \hat{w}_1)\}^{-1}$. Similarly, for $\hat{b}_1 := n_B^{-1} \sum_i X_i$, we have $\text{Var}\{\text{logit}(\hat{b}_1)\} \approx \{n_B \hat{b}_1\}^{-1} + \{n_B(1 - \hat{b}_1)\}^{-1}$. Since $\log r = \text{logit}(w_1) - \text{logit}(b_1)$, we estimate $\widehat{\log r} := \text{logit}(\hat{w}_1) - \text{logit}(\hat{b}_1)$ and form the Wald confidence interval $\widehat{\log r} \pm z_{1-\alpha/2} \sqrt{\{n_W \hat{w}_1\}^{-1} + \{n_W(1 - \hat{w}_1)\}^{-1} + \{n_B \hat{b}_1\}^{-1} + \{n_B(1 - \hat{b}_1)\}^{-1}}$. Exponentiating yields the intervals for r shown in the figure; its asymmetry reflects the log-scale construction. We see that the Wald intervals essentially coincide with ours in this simple setting, confirming the practical utility of our test.

In a similar spirit, we also consider an application dealing with the well-known Stroop effect ([Stroop, 1935](#)). This refers to the cognitive interference observed when an individual attempts to name the colour of the ink of a word that spells a different colour, compared to naming the colour of the ink when the word and ink colour match. Typically, reaction times and error rates differ significantly between these congruent and incongruent conditions, indicating distinct underlying distributions for concordant and non-concordant stimuli. The data set consists of observations from 131 individuals, each with two recorded values: X , representing the time taken to name a set of concordant pairs and Y , the time taken to name an equal number of discordant pairs. We standardize the data, and test the hypothesis $g(y) \propto e^{y/\eta} f(y)$ for varying $\eta \in \{0.01, \dots, 0.30\}$ using (E1). The results in Fig. 2(c) show that there are values for which such modelling assumptions cannot be ruled out, especially in a neighbourhood of 0.20. Given the small sample size compared to the New York frisk dataset, we conduct a sensitivity analysis by running (E1) on 10 random subsamples of size $n = m = 100$. The resulting thin black lines align closely with the purple line corresponding to the full sample.

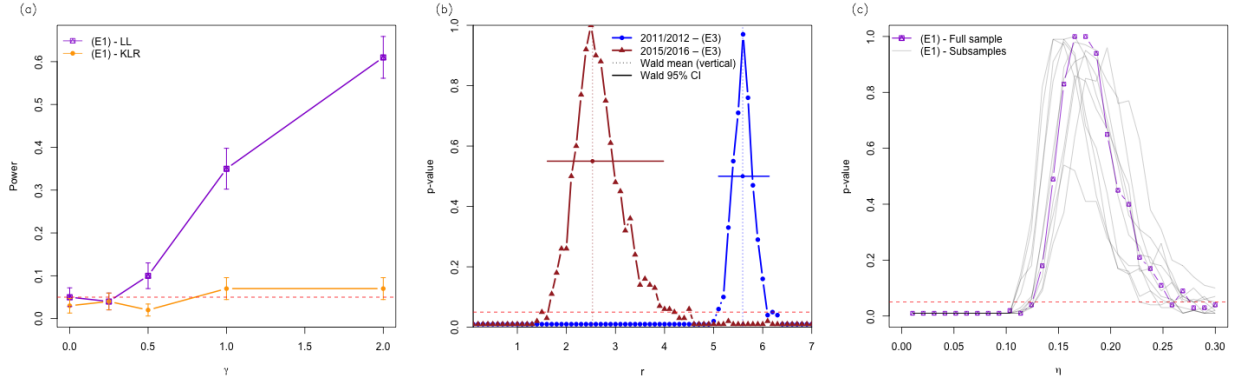


Figure 2: (a) Empirical power of (E1) in the causal inference setting, using estimates of the true density ratio $r(z) = \frac{1-\pi}{\pi} \exp\{\beta_0 + \beta^\top z + \gamma(\sin(10z_1) + z_2 z_3)\}$ obtained via linear logistic (LL) and kernel logistic (KLR) regression; error bars show ± 1 standard error. (b) p -values of (E3) for testing $w_1/w_0 = r b_1/b_0$ for $r \in \{0.1, \dots, 7\}$ on the New York-frisk datasets for 2011/2012 (blue) and 2015/2016 (brown); horizontal lines give Wald-type confidence intervals. (c) p -values of (E1) for testing $g(y) \propto e^{y/\eta} f(y)$ for $\eta \in \{0.01, \dots, 0.3\}$ on the Stroop data; sensitivity analysis over 10 random subsamples of size $n = m = 100$.

6.2.2 Simulation Based Inference on the Two Moons dataset

We illustrate how our methodology provides a practical diagnostic for simulation based inference, yielding formal decision rules for the goodness-of-fit of posterior-to-prior/likelihood-to-evidence estimates. As noted in the Introduction, when the likelihood is intractable yet a simulator allows sampling from $p(x | \theta)$, many simulation based inference pipelines estimate the ratio $r(\theta, x) := p(\theta | x)/p(x) = p(x | \theta)/p(x) = p(\theta, x)/p(\theta)p(x)$, for example by training a classifier to distinguish pairs (θ, x) drawn from the joint $p(\theta, x)$ versus the product $p(\theta)p(x)$. We now show how to use Algorithm 2 to formally test the quality of estimates of $r(\theta, x)$. Concretely, simulate $\{(\theta_i, X_i)\}_{i=1}^N \stackrel{\text{i.i.d.}}{\sim} p(\theta, x)$ by drawing $\theta_i \sim p(\theta)$ and $X_i | \theta_i$ from the simulator, and simulate $\{(\tilde{\theta}_i, \tilde{X}_i)\}_{i=1}^N \stackrel{\text{i.i.d.}}{\sim} p(\theta)p(x)$ by repeating the same two-step procedure independently and retaining one component from each run. Given an approximation $\hat{r}(\theta, x)$ of $r(\theta, x)$, trained on an independent split of data, we test the null $p(\theta, x) \propto \hat{r}(\theta, x)p(x)p(\theta)$. Applying Algorithm 2 only requires $\hat{r}(\theta, x)$ being pointwise evaluable for fixed (θ, x) , in addition to having access to the simulator, which is standard in this setting.

We validate the procedure on the Two Moons Dataset, a standard simulation based inference benchmark (Greenberg et al., 2019; Lueckmann et al., 2021). Samples are of the form $(\theta_1, \theta_2, X_1, X_2)$, where $(\theta_1, \theta_2)^\top \sim \text{Unif}([-1, 1]^2)$ and $(X_1, X_2) | (\theta_1, \theta_2) = (r \cos \alpha + 0.25, r \sin \alpha)^\top + (-2^{-1/2}|\theta_1 + \theta_2|, 2^{-1/2}(-\theta_1 + \theta_2))^\top$, with $\alpha \sim \text{Unif}(-\pi/2, \pi/2)$ and $r \sim \mathcal{N}(0.1, 0.01^2)$. This simulator induces a highly non-Gaussian posterior over (θ_1, θ_2) , which is bimodal with two separated crescent-shaped modes for typical observations (Chatterjee et al., 2024, Figure 7), making it a challenging benchmark for posterior approximation methods. We consider two ratio estimators, $\hat{r}_{\text{NRE}}(\theta, x)$ and $\hat{r}_{\text{BNRE}}(\theta, x)$ (Delaunoy et al., 2022), trained on $N_{\text{train}} \in \{2^4, 2^7, 2^9, 2^{12}, 2^{15}\}$ independent samples. Details on the the training phase are provided in Appendix C.2. For each of these, we run (E1) based on such estimates of the density ratio on $n = m = 200$ independent test samples, repeat this 300 times per setting, and report the proportion of rejections as an estimate of power. Fig. 3(a) shows our test rejects the null for $\hat{r}_{\text{BNRE}}$ more often, in line with its conservative, prior-leaning design that shrinks estimates toward 1, trading accuracy for coverage (Delaunoy et al., 2022).

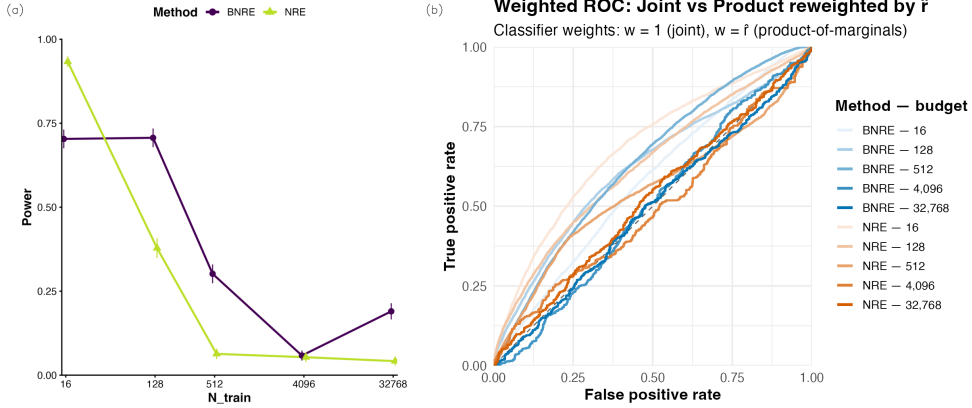


Figure 3: (a) Empirical power of (E1) for testing $p(\theta, x) \propto \hat{r}(\theta, x)p(x)p(\theta)$ with $\hat{r} \in \{\hat{r}_{\text{NRE}}, \hat{r}_{\text{BNRE}}\}$, the estimators in Delaunoy et al. (2022) trained on $N_{\text{train}} \in \{2^4, 2^7, 2^9, 2^{12}, 2^{15}\}$ independent samples; error bars show ± 1 standard error. (b) Receiver operating characteristic curves for classifiers distinguishing joint samples from product-of-marginals samples weighted by $\hat{r}_{\text{NRE}}$ and $\hat{r}_{\text{BNRE}}$; curves closer to the diagonal indicate better estimators.

For completeness, Fig. 3(b) reports receiver operating characteristic curves for classifiers distinguishing $5 \cdot 10^4$ joint samples from an equal number of product-of-marginals samples weighted by $\hat{r}_{\text{NRE}}$ and $\hat{r}_{\text{BNRE}}$. Better estimators produce curves closer to the diagonal; no option clearly dominates, while Algorithm 2 offers clearer guidance in this setting.

The same idea yields a test for an approximate posterior $\hat{p}(\theta | x)$, rather than $\hat{r}(\theta, x)$, as it is sufficient to replace $\hat{r}(\theta, x)$ by $\hat{p}(\theta | x)/p(\theta)$. Again, Algorithm 2 only requires pointwise evaluation of $\hat{p}(\theta | x)$. There are many formal tests for assessing the quality of posterior approximation. Compared to these, Algorithm 2 avoids training a classifier on held out data (Lopez-Paz and Oquab, 2017; Linhart et al., 2023) and does not require sampling from the approximate posterior, as in Chatterjee et al. (2024). While sampling from a learned posterior is often feasible, when this is impractical Algorithm 2 provides a viable alternative.

6.2.3 Conditional two-sample testing on the diamonds datasets

Finally, we assess the performance of Algorithm 2 in the conditional two-sample testing problem, as outlined in Section 5. Our analysis utilizes the diamonds dataset, which is available in the R-package `ggplot2`, and contains 53,490 observations with 10 features. Following Lee et al. (2024), we designate the price variable as Y and use the six numerical variables (carat, depth, table, x, y, z) as predictors X . Prior to the analysis, we standardize both X and Y . To introduce covariate shift, we implement biased sampling procedures: $X^{(1)}$ is sampled uniformly from the original feature space, while $X^{(2)}$ is sampled with probabilities proportional to $\exp(-x_1^2)$, with x_1 representing the first feature of X . Under the null hypothesis, the response variable Y is uniformly sampled for both $Y^{(1)}$ and $Y^{(2)}$. Under the alternative hypothesis, $Y^{(1)}$ remains uniformly sampled, while $Y^{(2)}$ is sampled with probabilities proportional to $\exp(-y)$, where y corresponds to the dataset’s Y values. We compare our method with two tests from Lee et al. (2024): the single-split classifier test and the linear-time maximum mean discrepancy. More extensive simulations with additional alternative methods are provided in Appendix C.1. In our simulation, we used an 80/20 split, allocating 80% of the total sampling budget $N \in \{200, 400, 800, 1200, 1600, 2000\}$ to marginal density ratio estimation and

20% to testing; the other methods use an even split, as in their simulation study. Consistently with the other methods, the marginal density ratio is estimated either with the linear logistic or the kernel logistic regression. The testing phase was performed using (E1) and the average decision over 200 repetitions is reported. All three methods control type I error, with the possible exception of the classifier-based test, likely due to high estimation error in the classifier. The maximum mean discrepancy method matches ours in size but shows slightly lower power, as expected from its linear-time statistic, which trades power for computational speed-up.

Table 1: Results for the conditional two-sample testing problem on the diamonds dataset using (E1), compared with alternative methods. LL: linear logistic regression; KLR: kernel logistic regression; CLF: single-split classifier test; MMD- l : linear-time maximum mean discrepancy test. N denotes the total sampling budget.

Estimator	Hypothesis	Test	$N = 200$	$N = 400$	$N = 800$	$N = 1200$	$N = 1600$	$N = 2000$
LL	Null	CLF	0.0900	0.0650	0.0750	0.0450	0.0675	0.0425
LL	Null	MMD- l	0.0750	0.0650	0.0750	0.0500	0.0575	0.0575
LL	Null	DRPT (E1)	0.0600	0.0550	0.0850	0.0600	0.0500	0.0350
LL	Alternative	CLF	0.1575	0.2050	0.2650	0.3600	0.3925	0.4800
LL	Alternative	MMD- l	0.0650	0.0675	0.0850	0.0825	0.0975	0.0850
LL	Alternative	DRPT (E1)	0.1500	0.1800	0.4150	0.5450	0.6350	0.7150
KLR	Null	CLF	0.0750	0.0575	0.0675	0.0350	0.0475	0.0450
KLR	Null	MMD- l	0.0600	0.0625	0.0725	0.0575	0.0550	0.0600
KLR	Null	DRPT (E1)	0.0400	0.0350	0.0600	0.0500	0.0600	0.0300
KLR	Alternative	CLF	0.0975	0.1525	0.2600	0.3550	0.3675	0.4450
KLR	Alternative	MMD- l	0.0725	0.0675	0.0800	0.0900	0.0950	0.1050
KLR	Alternative	DRPT (E1)	0.1250	0.1750	0.3750	0.5000	0.6300	0.6650

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References

- Mélanie Albert, Béatrice Laurent, Amandine Marrel, and Anouar Meynaoui. Adaptive test of independence based on HSIC measures. *The Annals of Statistics*, 50(2):858–879, 2022.
- ATLAS Collaboration. An implementation of neural simulation-based inference for parameter estimation in ATLAS. *Reports on Progress in Physics*, 88:067801, 2025.
- Peter C. Austin and Elizabeth A. Stuart. Moving Towards Best Practice When Using Inverse Probability of Treatment Weighting (IPTW) Using the Propensity Score to Estimate Causal Treatment Effects in Observational Studies. *Statistics in Medicine*, 34:3661–3679, 2015.
- Sivaraman Balakrishnan and Larry Wasserman. Hypothesis testing for densities and high-dimensional multinomials: Sharp local minimax rates. *Ann. Statist.*, 47:1893–1927, 2019.

- Thomas B. Berrett. Efficient estimation with incomplete data via generalised ANOVA decomposition, 2024. URL <https://arxiv.org/abs/2409.05729>.
- Thomas B Berrett, Yi Wang, Rina Foygel Barber, and Richard J Samworth. The conditional permutation test for independence while controlling for confounders. *J. Roy. Statist. Soc. Ser. B*, 82:175–197, 2020.
- Thomas B Berrett, Ioannis Kontoyiannis, and Richard J Samworth. Optimal rates for independence testing via U -statistic permutation tests. *Ann. Statist.*, 49:2457–2490, 2021.
- Julian Besag and Peter Clifford. Generalized Monte Carlo Significance Tests. *Biometrika*, 76(4):633–642, 1989.
- Alberto Bordino and Thomas B. Berrett. *DRPT: Density Ratio Permutation Test*, 2025. URL <https://CRAN.R-project.org/package=DRPT>. R package version 1.1.
- J. Bretagnolle and C. Huber. Estimation des densités: risque minimax. In *Séminaire de Probabilités XII*, volume 649 of *Lecture Notes in Mathematics*, pages 342–363. Springer, Berlin, 1978. Univ. Strasbourg, Strasbourg, 1976/1977.
- Cristina Butucea. Goodness-of-fit testing and quadratic functional estimation from indirect observations. *Annals of Statistics*, 35(5):1907–1930, 2007.
- Emmanuel Candès, Yingying Fan, Lucas Janson, and Jinchi Lv. Panning for gold: Model-X knockoffs for high-dimensional controlled variable selection. *J. Roy. Statist. Soc. Ser. B*, 80(3):551–577, 2018.
- Anirban Chatterjee, Ziang Niu, and Bhaswar B. Bhattacharya. A Kernel-Based Conditional Two-Sample Test Using Nearest Neighbors (with Applications to Calibration, Regression Curves, and Simulation-Based Inference), 2024. URL <https://arxiv.org/abs/2407.16550>.
- Siu Lun Chau, Antonin Schrab, Arthur Gretton, Dino Sejdinovic, and Krikamol Muandet. Credal Two-Sample Tests of Epistemic Ignorance. In Yingzhen Li, Stephan Mandt, Shipra Agrawal, and Emtiyaz Khan, editors, *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics (AISTATS 2025)*, volume 258 of *Proceedings of Machine Learning Research*, pages 127–135, 2025.
- Jiahua Chen and Yukun Liu. Quantile and quantile-function estimations under density ratio model. *The Annals of Statistics*, 41(3):1669 – 1692, 2013.
- Yuchen Chen and Jing Lei. De-Biased Two-Sample U-Statistics With Application To Conditional Distribution Testing. *Machine Learning*, 114(2):33, 2025.
- David R Cox. Some sampling problems in technology. *New Developments in Survey Sampling*, 1:506–527, 1969.
- Michael Deistler, Jan Boelts, Peter Steinbach, Guy Moss, Thomas Moreau, Manuel Gloeckler, Pedro L. C. Rodrigues, Julia Linhart, Janne K. Lappalainen, Benjamin Kurt Miller, Pedro J. Gonçalves, Jan-Matthis Lueckmann, Cornelius Schröder, and Jakob H. Macke. Simulation-Based Inference: A Practical Guide, 2025.

- Arnaud Delaunoy, Joeri Hermans, François Rozet, Antoine Wehenkel, and Gilles Louppe. Towards Reliable Simulation-Based Inference with Balanced Neural Ratio Estimation. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho, editors, *Advances in Neural Information Processing Systems*, volume 35, pages 20025–20037, 2022.
- R. M. Dudley. *Real Analysis and Probability*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2 edition, 2002.
- Sam Efromovich. Density estimation for biased data. *The Annals of Statistics*, 32(3):1137–1161, 2004.
- Sharad Goel, Justin M. Rao, and Ravi Shroff. Precinct or prejudice? Understanding racial disparities in New York City’s stop-and-frisk policy. *The Annals of Applied Statistics*, 10(1):365–394, 2016.
- David Greenberg, Marcel Nonnenmacher, and Jakob Macke. Automatic Posterior Transformation for Likelihood-Free Inference. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2404–2414. PMLR, 2019.
- Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A Kernel Two-Sample Test. *Journal of Machine Learning Research*, 13(25):723–773, 2012.
- Hao Guan and Mingxia Liu. Domain Adaptation for Medical Image Analysis: A Survey. *IEEE Transactions on Biomedical Engineering*, 69(3):1173–1185, 2022.
- Martin Hairer. Convergence of Markov processes, 2021.
- Joeri Hermans, Volodimir Begy, and Gilles Louppe. Likelihood-free MCMC with Amortized Approximate Ratio Estimators. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, 2020.
- Joeri Hermans, Arnaud Delaunoy, François Rozet, Antoine Wehenkel, Volodimir Begy, and Gilles Louppe. A Crisis In Simulation-Based Inference? Beware, Your Posterior Approximations Can Be Unfaithful. *Transactions on Machine Learning Research*, 2022. ISSN 2835-8856.
- Wassily Hoeffding. The Large-Sample Power of Tests Based on Permutations of Observations. *The Annals of Mathematical Statistics*, 23(2):169 – 192, 1952.
- D. G. Horvitz and D. J. Thompson. A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47(260):663–685, 1952.
- Xiaoyu Hu and Jing Lei. A Two-Sample Conditional Distribution Test Using Conformal Prediction and Weighted Rank Sum. *Journal of the American Statistical Association*, 119:1136 – 1154, 2020.
- Ming-Yueh Huang, Jing Qin, and Chiung-Yu Huang. Efficient data integration under prior probability shift. *Biometrics*, 80(2):ujae035, 2024.
- Yu. I. Ingster. Minimax Testing of Nonparametric Hypotheses on a Distribution Density in the l_p Metrics. *Theory of Probability and Its Applications*, 31(2):333–337, 1987.

- Yuri Ingster and Irina Suslina. *Nonparametric Goodness-of-Fit Testing Under Gaussian Models*, volume 169. Springer Science & Business Media, 2003.
- Wenlong Ji, Weizhe Yuan, Emily Getzen, Kyunghyun Cho, Michael I. Jordan, Song Mei, Jason E Weston, Weijie J. Su, Jing Xu, and Linjun Zhang. An Overview of Large Language Models for Statisticians, 2025. URL <https://arxiv.org/abs/2502.17814>.
- Ying Jin and Emmanuel J. Candès. Model-free selective inference under covariate shift via weighted conformal p-values. *Biometrika*, 2025. Advance article.
- Herman Kahn and Theodore E. Harris. Estimation of particle transmission by random sampling. Technical Report 12, National Bureau of Standards, 1951.
- Takafumi Kanamori, Taiji Suzuki, and Masashi Sugiyama. Theoretical Analysis of Density Ratio Estimation. *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, 93-A:787–798, 2010.
- Takafumi Kanamori, Taiji Suzuki, and Masashi Sugiyama. Statistical analysis of kernel-based least-squares density-ratio estimation. *Machine Learning*, 86:335–367, 2012.
- Qing Kang and Paul I. Nelson. Permutation tests from biased samples for the equality of two distributions. *Journal of Nonparametric Statistics*, 21(3):305–319, 2009.
- Ilmun Kim, Sivaraman Balakrishnan, and Larry Wasserman. Minimax optimality of permutation tests. *Ann. Statist.*, 50(1):225–251, 2022.
- Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanus Phillips, Irena Gao, Tony Lee, Etienne David, Ian Stavness, Wei Guo, Berton A. Earnshaw, Imran S. Haque, Sara Meghan Beery, Jure Leskovec, Anshul Kundaje, Emma Pierson, Sergey Levine, Chelsea Finn, and Percy Liang. WILDS: A Benchmark of in-the-Wild Distribution Shifts. In *International Conference on Machine Learning*, 2020.
- A. J. Lee. *U-Statistics: Theory and Practice*. Routledge, New York, 1990.
- Seongchan Lee, Suman Cha, and Ilmun Kim. General Frameworks for Conditional Two-Sample Testing, 2024. URL <https://arxiv.org/abs/2410.16636>.
- Erich L Lehmann and Joseph P Romano. *Testing Statistical Hypotheses*. Springer Science & Business Media, 2005.
- Lihua Lei and Emmanuel J. Candès. Conformal inference of counterfactuals and individual treatment effects. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 83(5):911–938, 2021.
- Dan Leonte, Raphaël Huser, and Almut E. D. Veraart. Simulation-based inference via telescoping ratio estimation for trawl processes, 2025. URL <https://arxiv.org/abs/2510.04042>.
- Tong Li and Ming Yuan. On the Optimality of Gaussian Kernel Based Nonparametric Tests against Smooth Alternatives. *Journal of Machine Learning Research*, 25(334):1–62, 2024.

- Julia Linhart, Alexandre Gramfort, and Pedro L. C. Rodrigues. ℓ -C2ST: Local Diagnostics for Posterior Approximations in Simulation-Based Inference. In *Advances in Neural Information Processing Systems* 36, 2023.
- David Lopez-Paz and Maxime Oquab. Revisiting Classifier Two-Sample Tests. In *International Conference on Learning Representations*, 2017.
- Jan-Matthis Lueckmann, Jan Boelts, David Greenberg, Pedro J. Gonçalves, and Jakob H. Macke. Benchmarking Simulation-Based Inference. In Arindam Banerjee and Kenji Fukumizu, editors, *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics*, volume 130 of *Proceedings of Machine Learning Research*, pages 343–351. PMLR, 2021.
- Cong Ma, Reese Pathak, and Martin J. Wainwright. Optimally tackling covariate shift in RKHS-based nonparametric regression. *The Annals of Statistics*, 51(2):738 – 761, 2023.
- P. McCullagh and J.A. Nelder. *Generalized Linear Models, Second Edition*. Chapman and Hall/CRC Monographs on Statistics and Applied Probability Series. Chapman & Hall, 1989.
- Benjamin Kurt Miller, Christoph Weniger, and Patrick Forré. Contrastive Neural Ratio Estimation. In *Advances in Neural Information Processing Systems*, 2022.
- XuanLong Nguyen, Martin J. Wainwright, and Michael I. Jordan. Estimating Divergence Functionals and the Likelihood Ratio by Convex Risk Minimization. *IEEE Transactions on Information Theory*, 56:5847–5861, 2008.
- Art B. Owen and Yi Zhou. Safe and Effective Importance Sampling. *Journal of the American Statistical Association*, 95(449):135–143, 2000.
- Drew Prinster, Samuel Don Stanton, Anqi Liu, and Suchi Saria. Conformal Validity Guarantees Exist for Any Data Distribution (and How to Find Them). In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 41086–41118, 2024.
- Jing Qin. Empirical likelihood in biased sample problems. *The Annals of Statistics*, 21(3):1182–1196, 1993.
- Yongsong Qin. Inferences for case-control and semiparametric two-sample density ratio models. *Biometrika*, 85:619–630, 1998.
- Aaditya Ramdas, Rina Foygel Barber, Emmanuel J. Candès, and Ryan J. Tibshirani. Permutation Tests Using Arbitrary Permutation Distributions. *Sankhya A*, 85:1156 – 1177, 2022.
- James M. Robins, Miguel A. Hernán, and Babette A. Brumback. Marginal Structural Models and Causal Inference in Epidemiology. *Epidemiology*, 11:550–560, 2000.
- Alex Rogozhnikov. Reweighting with boosted decision trees. *Journal of Physics: Conference Series*, 762(1):012036, 2016.
- Antonin Schrab, Ilmun Kim, Mélisande Albert, Béatrice Laurent, Benjamin Guedj, and Arthur Gretton. MMD Aggregated Two-Sample Test. *Journal of Machine Learning Research*, 24(194):1–81, 2023.

- Rajen D Shah and Jonas Peters. The hardness of conditional independence testing and the generalised covariance measure. *Ann. Statist.*, 48:1514–1538, 2020.
- Bharath K. Sriperumbudur, Arthur Gretton, Kenji Fukumizu, Bernhard Schölkopf, and Gert R.G. Lanckriet. Hilbert Space Embeddings and Metrics on Probability Measures. *Journal of Machine Learning Research*, 11:1517–1561, 2010.
- Amos Storkey. When Training and Test Sets Are Different: Characterizing Learning Transfer. In *Dataset Shift in Machine Learning*, Neural Information Processing Series, pages 3–28. MIT Press, 2009.
- J. Ridley Stroop. Studies of Interference in Serial Verbal Reactions. *Journal of Experimental Psychology*, 18(6):643–662, 1935.
- Masashi Sugiyama, Taiji Suzuki, Shinichi Nakajima, Hisashi Kashima, Paul von Büna, and Motoaki Kawanabe. Direct importance estimation for covariate shift adaptation. *Annals of the Institute of Statistical Mathematics*, 60:699–746, 2008.
- Masashi Sugiyama, Taiji Suzuki, and Takafumi Kanamori. Density Ratio Estimation: A Comprehensive Review. *RIMS Kokyuroku*, pages 10–31, 2010.
- Masashi Sugiyama, Teruyuki Suzuki, and Takafumi Kanamori. Density-ratio matching under the Bregman divergence: a unified framework of density-ratio estimation. *Annals of the Institute of Statistical Mathematics*, 64:1009–1044, 2012.
- Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. A Bradford Book, 2018.
- Yaniv Tenzer, Micha Mandel, and Or Zuk. Testing independence under biased sampling. *Journal of the American Statistical Association*, 117(540):2194–2206, 2022.
- Nikolaj Thams, Sorawit Saengkyongam, Niklas Pfister, and Jonas Peters. Statistical testing under distributional shifts. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 85(3):597–663, 2023.
- Owen Thomas, Ritabrata Dutta, Jukka Corander, Samuel Kaski, and Michael U. Gutmann. Likelihood-Free Inference by Ratio Estimation. *Bayesian Analysis*, 17(1):1–31, 2022.
- Ryan J Tibshirani, Rina Foygel Barber, Emmanuel Candes, and Aaditya Ramdas. Conformal Prediction Under Covariate Shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Surya T. Tokdar and Robert E. Kass. Importance Sampling: A Review. *Wiley Interdisciplinary Reviews: Computational Statistics*, 2(1):54–60, 2010.
- Martin J Wainwright. *High-dimensional Statistics: A Non-asymptotic Viewpoint*. Cambridge University Press, 2019.
- Karl Weiss, Taghi M. Khoshgoftaar, and Dingding Wang. A Survey of Transfer Learning. *Journal of Big Data*, 3(1):9, 2016.

- Daniel Westreich, Jessie K. Edwards, Catherine R. Lesko, Elizabeth Stuart, and Stephen R. Cole. Transportability of Trial Results Using Inverse Odds of Sampling Weights. *American Journal of Epidemiology*, 186(8):1010–1014, 2017.
- Jian Yan, Zhuoxi Li, and Xianyang Zhang. Distance and Kernel-Based Measures for Global and Local Two-Sample Conditional Distribution Testing, 2024. URL <https://arxiv.org/abs/2210.08149>.
- Zhuoran Yang, Chi Jin, Zhaoran Wang, Mengdi Wang, and Michael I. Jordan. On Function Approximation in Reinforcement Learning: Optimism in the Face of Large State Spaces. In *Advances in Neural Information Processing Systems*, volume 33, pages 13903–13916, 2020.
- Siming Zheng, Meifang Lan, Tong Wang, and Yuanyuan Lin. A Generative Conditional Distribution Equality Testing Framework and Its Minimax Analysis, 2025. URL <https://arxiv.org/abs/2509.17729>.
- Tianhui Zhou, Guangyu Tong, Fan Li, Laine E. Thomas, and Fan Li. PSweight: An R Package for Propensity Score Weighting Analysis. *The R Journal*, 14:282–300, 2022. ISSN 2073-4859. <https://doi.org/10.32614/RJ-2022-011>.

Appendices

Appendix A contains the proofs of all results stated in the main text, while Appendix B presents and proves some results that were only briefly mentioned there. Appendix C expands on the simulation studies, and Appendix D analyzes Algorithm 2 in the discrete, finite-support setting.

Appendix A Proofs

A.1 Proofs for Section 2

Proof of Theorem 1. To establish exchangeability under the null, we adapt ideas from Berrett et al. (2020) and consider an equivalent formulation of the permutation scheme. Let $Z_{() } = (Z_{(1)}, \dots, Z_{(n+m)})$ be the order statistics of Z . When $\mathcal{X} \subseteq \mathbb{R}$, we naturally use the standard ordering on $\mathbb{R}$. In the general case, we can select any total ordering on $\mathcal{X}$; the specific choice does not matter, as its sole purpose is to let us examine the set of Z -values without needing to know which value is associated with which data point. Define also $Z_{(p)} = (Z_{(p(1))}, \dots, Z_{(p(n+m))})$ for each $p \in \mathcal{S}_{n+m}$, and let $P \in \mathcal{S}_{n+m}$ be the permutation given by the ranks of the true observed vector Z , so that $Z = Z_{(P)}$. Under $g \propto r f$, we can use Bayes theorem to show that the distribution of the true ranks P , conditional on the order statistics $Z_{() }$, is given by

$$\mathbb{P} \{P = p \mid Z_{() }\} = \frac{\prod_{i \in \{n+1, \dots, n+m\}} r(Z_{(p(i))})}{\sum_{\tilde{p} \in \mathcal{S}_{n+m}} \prod_{i \in \{n+1, \dots, n+m\}} r(Z_{(\tilde{p}(i))})}. \quad (17)$$

As a result, if $P^{(1)}, \dots, P^{(H)}$ are drawn independently from (17), under H_0 the vectors $Z, Z_{(P^{(1)})}, \dots, Z_{(P^{(H)})}$ are conditionally independent and identically distributed given $Z_{() }$. Consequently, after marginalising over $Z_{() }$, the sequence $(Z, Z_{(P^{(1)})}, \dots, Z_{(P^{(H)})})$ is exchangeable.

We now conclude the proof by showing that (3) and (17) are equivalent. For fixed Z , and thus P , the map $\sigma \mapsto p := \sigma \circ P$ is a bijection over $\mathcal{S}_{n+m}$. This, together with $Z_{\sigma(i)} = Z_{(\sigma \circ P)(i)} = Z_{(p(i))}$, allows showing that the weight $\prod_{i=n+1}^{n+m} r(Z_{\sigma(i)})$ in (3) can be rewritten as $\prod_{i=n+1}^{n+m} r(Z_{(p(i))})$, which corresponds to the weight in (17). Therefore, under the change of variables $p = \sigma \circ P$, the distribution of $\sigma^{(h)}$ given Z in (3) induces the distribution of $P^{(h)} = \sigma^{(h)} \circ P$ given $Z_{(P)}$ in (17), and conversely. In particular, $(\sigma^{(1)}, \dots, \sigma^{(H)})$ and $(P^{(1)}, \dots, P^{(H)})$ have corresponding laws via this bijection, and since $Z_{\sigma^{(h)}} = Z_{(\sigma^{(h)} \circ P)} = Z_{(P^{(h)})}$ for each $h \in [H]$, the joint law of $(Z, Z_{\sigma^{(1)}}, \dots, Z_{\sigma^{(H)}})$ is the same as that of $(Z, Z_{(P^{(1)})}, \dots, Z_{(P^{(H)})})$. As we have already shown that the latter is exchangeable, this implies the finite-sample validity of (2) for every test statistic T . $\square$

Proof of Proposition 2. The proof consists of simply checking the detailed balance equations for the Markov chain defined by the algorithm. Denote with $\mathcal{P}$ the collection of all K couples of indices $\{(i_1, j_1), \dots, (i_K, j_K)\}$ such that $(i_1, \dots, i_K)$ contains distinct elements from $[n]$, and $(j_1, \dots, j_K)$ contains distinct elements from $\{n+1, \dots, n+m\}$. For any $\tau \in \mathcal{P}$ and any permutations p, p' , we write $p \sim_\tau p'$ if p can be transformed to p' by swapping any subset of the pairs in τ . We now compute the transition probability matrix of the Markov chain defined by Algorithm 1. Every probability sign has to be intended conditionally on Z . For all $t \in \mathbb{N}_+$ and any permutations p, p' , we have

$$\mathbb{P}\{\sigma_t = p' \mid \sigma_{t-1} = p\} = \frac{1}{|\mathcal{P}|} \sum_{\tau \in \mathcal{P}} \mathbb{P}\{\sigma_t = p' \mid \sigma_{t-1} = p, \tau_t = \tau\},$$

since at each time t , Step 3 of the algorithm corresponds to drawing $\tau_t \in \mathcal{P}$ uniformly at random. Next, given $\tau_t = \tau := \{(i_1, j_1), \dots, (i_K, j_K)\}$ and $\sigma_{t-1} = p$, it must be the case that σ_t satisfies $\sigma_t \sim_\tau p$ by definition of Steps 4-5 of the algorithm. In light of the definition of the odds ratio for each B_{i_k, j_k}^t in (4), we see that for any $p', p'' \sim_\tau p$, we have

$$\begin{aligned} \frac{\mathbb{P}\{\sigma_t = p' \mid \sigma_{t-1} = p, \tau_t = \tau\}}{\mathbb{P}\{\sigma_t = p'' \mid \sigma_{t-1} = p, \tau_t = \tau\}} &= \prod_{j \in \{j_1, \dots, j_K\}} \frac{r(Z_{p'(j)})}{r(Z_{p''(j)})} = \prod_{j \in \{j_1, \dots, j_K\}} \frac{r(Z_{p'(j)})}{r(Z_{p''(j)})} \prod_{j \notin \{j_1, \dots, j_K\}} \frac{r(Z_{p'(j)})}{r(Z_{p''(j)})} \\ &= \prod_{j \in \{n+1, \dots, n+m\}} \frac{r(Z_{p'(j)})}{r(Z_{p''(j)})} = \frac{\mathbb{P}\{\sigma = p'\}}{\mathbb{P}\{\sigma = p''\}}, \end{aligned} \quad (18)$$

where in the second equality we used the fact that $r(Z_{p'(j)})/r(Z_{p''(j)}) = 1$ for all $j \notin \{j_1, \dots, j_K\}$, while in the last step we used the definition of the distribution (3). Therefore,

$$\mathbb{P}\{\sigma_t = p' \mid \sigma_{t-1} = p\} = \frac{1}{|\mathcal{P}|} \sum_{\tau \in \mathcal{P}} \frac{\mathbb{1}\{p' \sim_\tau p\} \cdot \mathbb{P}\{\sigma = p'\}}{\sum_{p''} \mathbb{1}\{p'' \sim_\tau p\} \cdot \mathbb{P}\{\sigma = p''\}}.$$

This, together with the fact that $\sim_\tau$ defines an equivalence relation on $\mathcal{P}$, shows that

$$\begin{aligned} \mathbb{P}\{\sigma = p\} \cdot \mathbb{P}\{\sigma_t = p' \mid \sigma_{t-1} = p\} &= \frac{1}{|\mathcal{P}|} \sum_{\tau \in \mathcal{P}} \mathbb{P}\{\sigma = p\} \cdot \frac{\mathbb{1}\{p' \sim_\tau p\} \cdot \mathbb{P}\{\sigma = p'\}}{\sum_{p''} \mathbb{1}\{p'' \sim_\tau p\} \cdot \mathbb{P}\{\sigma = p''\}} \\ &= \frac{1}{|\mathcal{P}|} \sum_{\tau \in \mathcal{P}} \mathbb{P}\{\sigma = p'\} \cdot \frac{\mathbb{1}\{p \sim_\tau p'\} \cdot \mathbb{P}\{\sigma = p\}}{\sum_{p''} \mathbb{1}\{p'' \sim_\tau p'\} \cdot \mathbb{P}\{\sigma = p''\}} = \mathbb{P}\{\sigma = p'\} \cdot \mathbb{P}\{\sigma_t = p \mid \sigma_{t-1} = p'\}. \end{aligned} \quad (19)$$

This verifies the detailed balance equations, and so the Markov chain is reversible and has (3) as stationary distribution. Moreover, since $r(x)$ is assumed to be positive for all $x \in \mathcal{X}$, it follows that the chain is aperiodic and irreducible, ensuring the uniqueness of the stationary distribution. $\square$

Proof of Theorem 3. The proof relies fundamentally on the reversibility of the Markov chain defined in Algorithm 1, a property established in the proof of Proposition 2. This reversibility permits an alternative but distributionally equivalent sampling procedure whenever the input σ is drawn from the stationary distribution: we may first sample σ_* from distribution (3) conditional on Z , and subsequently generate $(\sigma, \sigma^{(1)}, \dots, \sigma^{(H)})$ through $H + 1$ independent applications of Algorithm 1, each running for S steps and initialized at σ_* . The independence of these runs, combined with their shared initialization point σ_* , ensures that $(\sigma, \sigma^{(1)}, \dots, \sigma^{(H)})$ are independently and identically distributed when conditioned on σ_* and Z . This conditional independence directly implies the exchangeability of $(Z, Z^{(1)}, \dots, Z^{(H)})$ since, under H_0 , the distribution of $Z_{\sigma_0} \mid Z_{(0)}$, with $\sigma_0 = \text{id}_{n+m}$, coincides with the distribution of $Z_\sigma \mid Z_{(0)}$, where σ is sampled from (3). This concludes the proof. $\square$

A.2 Proofs for Section 3

Proof of Lemma 4. We begin by proving that there exists a unique λ_0 such that $h = \frac{nf+mg}{n+\lambda_0 mr}$ is a density function. Define the function

$$F : \begin{cases} \mathbb{R}_+ \rightarrow \mathbb{R}_+ \\ \lambda \mapsto \int \frac{nf+mg}{n+\lambda mr} d\mu \end{cases} \quad (20)$$

which can easily be seen to be continuous. It also straightforward to see that F is strictly decreasing, since if $\lambda_2 > \lambda_1 > 0$ we have

$$\frac{nf(x) + mg(x)}{n + \lambda_2 mr(x)} \leq \frac{nf(x) + mg(x)}{n + \lambda_1 mr(x)} \quad \text{for all } x \in \mathcal{X},$$

and this inequality is strict in the support of f and g as we are assuming that $r(x) > 0$ for all $x \in \mathcal{X}$. It is clear that $\lim_{\lambda \rightarrow 0^+} F(\lambda) = 1 + m/n > 1$, while $\lim_{\lambda \rightarrow \infty} F(\lambda) = 0 < 1$. Thus, by the intermediate value theorem and the strict monotonicity of F we have established our first claim that λ_0 exists and is unique.

It can now be seen that h satisfies the requirements of the result. Indeed, we have

$$\begin{aligned} 1 &= \int h d\mu = \int \frac{nf + mg}{n + \lambda_0 mr} d\mu = \frac{1}{n} \int \frac{n}{n + \lambda_0 mr} (nf + mg) d\mu \\ &= \frac{1}{n} \int \left(1 - \frac{\lambda_0 mr}{n + \lambda_0 mr} \right) (nf + mg) d\mu = \frac{1}{n} \left(\int (nf + mg) d\mu - \int (\lambda_0 mr) \frac{nf + mg}{n + \lambda_0 mr} d\mu \right) \\ &= \frac{1}{n} \left(n + m - \lambda_0 m \int r h d\mu \right) = 1 + \frac{m}{n} - \lambda_0 \frac{m}{n} \int r h d\mu, \end{aligned}$$

which implies that $\lambda_0 = (\int r h d\mu)^{-1}$. This shows us that $nh + m(\int r h d\mu)^{-1}rh = nh + \lambda_0 mrh = nf + mg$, so that h gives rise to null distributions for which the distribution of the combined sample matches that of our data, concluding the existence part of the result.

As for its uniqueness, observe that any density h preserving the combined distribution under the null

hypothesis must be of the form h for some λ_0 positive, since

$$nf + mg = nh + m \frac{rh}{\int r f d\mu} = \left\{ n + mr \left(\int r h d\mu \right)^{-1} \right\} h = (n + \lambda_0 mr) h.$$

But the λ_0 such that h integrates to 1 is unique. This completes the proof. $\square$

Proof of Lemma 5. Suppose that the null hypothesis holds, so that $g = rf / \int r f d\mu$. It follows from Lemma 4 and its proof that $f = h$ and that $\lambda_0 = (\int r f d\mu)^{-1}$. We therefore see that $\lambda_0 r f - g = 0$, so indeed $T_{\mathcal{F},r}(f, g) = 0$.

We now turn to the reverse implication. Assuming that f and g are such that $T_{\mathcal{F},r}(f, g) = 0$, we will show that H_0 must be true. By assumption, $\mathcal{H}$ is dense in $C_b^0(\mathcal{X})$ with respect to $\|\cdot\|_\infty$, so that for all $\phi \in C_b^0(\mathcal{X})$ and all $\epsilon > 0$ there exists $\varphi \in \mathcal{H}$ such that $\|\varphi - \phi\|_\infty \leq \epsilon$. Thus

$$\begin{aligned} & \frac{n+m}{m} \left| \int \frac{\lambda_0 r f - g}{n/m + \lambda_0 r} \phi d\mu \right| \\ & \leq \frac{n+m}{m} \left| \int \frac{\lambda_0 r f}{n/m + \lambda_0 r} (\phi - \varphi) d\mu \right| + \frac{n+m}{m} \left| \int \frac{\lambda_0 r f - g}{n/m + \lambda_0 r} \varphi d\mu \right| + \frac{n+m}{m} \left| \int \frac{g}{n/m + \lambda_0 r} (\phi - \varphi) d\mu \right| \\ & \leq \int \frac{n+m}{n + \lambda_0 mr} \lambda_0 r f |\phi - \varphi| d\mu + \|\varphi\|_{\mathcal{H}} T_{\mathcal{F},r}(f, g) + \int \frac{n+m}{n + \lambda_0 mr} g |\phi - \varphi| d\mu \\ & = \int \frac{n+m}{n + \lambda_0 mr} \lambda_0 r f |\phi - \varphi| d\mu + \int \frac{n+m}{n + \lambda_0 mr} g |\phi - \varphi| d\mu \\ & \leq \epsilon \int \frac{\lambda_0 r}{\min(1, \lambda_0 r)} f d\mu + \epsilon \int \frac{1}{\min(1, \lambda_0 r)} g d\mu = \epsilon (\mathbb{E}[\max\{1, \lambda_0 r(X)\}] + \mathbb{E}[\max\{1, 1/\lambda_0 r(Y)\}]) \\ & \leq \epsilon (2 + \mathbb{E}[\lambda_0 r(X)] + \mathbb{E}[1/\lambda_0 r(Y)]) = \epsilon \left(2 + \lambda_0 \int r f d\mu + \lambda_0^{-1} \int g/r d\mu \right), \end{aligned}$$

where in the first equality we used our assumption that $T_{\mathcal{F},r}(f, g) = 0$. Because $\epsilon > 0$ was arbitrary, and since $\int r f d\mu < \infty$, $\int g/r d\mu < \infty$ by assumption, and $\lambda_0 < \infty$ by Proposition 17 in Appendix B, we now see that $\frac{n+m}{m} \int \frac{\lambda_0 r f - g}{n/m + \lambda_0 r} \phi d\mu = 0$ for all $\phi \in C_b^0(\mathcal{X})$, which further implies

$$\frac{mg}{n + \lambda_0 mr} = \frac{\lambda_0 mr f}{n + \lambda_0 mr} \text{ a.s.}$$

by Lemma 9.3.2 in Dudley (2002). Simplifying this inequality we see that $g = \lambda_0 r f \propto rf$, as required. $\square$

Proof of Theorem 6. We assume throughout the proof that H_0 does not hold and show that $\mathbb{P}\{p > \alpha\} \rightarrow 0$ as $n, m \rightarrow \infty$. Recall our notation $\bar{R}$ and $\check{R}$ for the arithmetic and harmonic means of the r values, respectively, and write A for the event that $\bar{R} \leq 2\mathbb{E}(\bar{R})$ and $1/\check{R} \leq 2\mathbb{E}(1/\check{R})$. On this event, we have

$$\bar{R}/\check{R} \leq 4\mathbb{E}(\bar{R})\mathbb{E}(\check{R}^{-1}) \leq 4\sqrt{\mathbb{E}\{r(Z)^2\}\mathbb{E}\{r(Z)^{-2}\}} \leq 4\sqrt{V_0},$$

which will allow us to apply Lemma 7. Since we assume that $H > \lceil 1/\alpha - 1 \rceil$ we have $\alpha(1 + H) - 1 > 0$, and

we can therefore use Markov's inequality to see that

$$\begin{aligned}
\mathbb{P}\{p > \alpha\} &= \mathbb{P}\left\{1 + \sum_{h=1}^H \mathbb{1}\{T(Z_{\sigma^{(h)}}) \geq T(Z)\} > \alpha(1+H)\right\} \\
&\leq \frac{\mathbb{E}\left[\sum_{h=1}^H \mathbb{1}\{T(Z_{\sigma^{(h)}}) \geq T(Z)\}\right]}{\alpha(1+H) - 1} = \frac{H}{\alpha(1+H) - 1} \mathbb{P}\{T(Z_{\sigma^{(1)}}) \geq T(Z)\} \\
&\leq \frac{H}{\alpha(1+H) - 1} [\mathbb{P}\{T(Z_{\sigma^{(1)}}) \geq T(Z), A\} + \mathbb{P}(A^c)],
\end{aligned}$$

where the penultimate step follows from exchangeability. We begin by controlling $\mathbb{P}(A^c)$ by combining Chebyshev's inequality with the fact that $\max(\mathbb{E}[\{\lambda_0 r(Z)\}^2], \mathbb{E}[\{\lambda_0 r(Z)\}^{-2}]) \leq V_0$. In particular, this follows from $\mathbb{E}[\{r(Z)\}^2] \mathbb{E}[\{r(Z)\}^{-2}] \leq V_0$ and

$$\frac{1}{\mathbb{E}(1/\check{R})} = \frac{n+m}{n \int f/r d\mu + m \int g/r d\mu} \leq \frac{1}{\lambda_0} \leq \frac{1}{n+m} \left(n \int r f d\mu + m \int r g d\mu \right) = \mathbb{E}(\bar{R}),$$

where the latter holds by Jensen's inequality. Together, these allow showing

$$\mathbb{P}(A^c) \leq \mathbb{P}\left(\frac{\bar{R}}{\mathbb{E}(\bar{R})} > 2\right) + \mathbb{P}\left(\frac{1}{\check{R} \mathbb{E}(1/\check{R})} > 2\right) \leq \frac{\text{Var}(\bar{R})}{\mathbb{E}^2(\bar{R})} + \frac{\text{Var}(1/\check{R})}{\mathbb{E}^2(1/\check{R})} \leq \text{Var}(\lambda_0 \bar{R}) + \text{Var}\left(\frac{1}{\lambda_0 \check{R}}\right) \leq \frac{2V_0}{m+n}. \quad (21)$$

Using the shorthand $\sigma = \sigma^{(1)}$ it now suffices to consider $T(Z_\sigma)$ and $T(Z)$ on the event A .

We will first prove that $\mathbb{E}[T(Z_\sigma) \mathbb{1}_A] = \mathbb{E}[T(Z) \mathbb{1}_A] \rightarrow 0$. Write

$$W_\varphi^\sigma := \frac{1}{n} \left(\sum_{i=1}^n \frac{\hat{\lambda} m r(Z_{\sigma(i)})}{n + \hat{\lambda} m r(Z_{\sigma(i)})} \varphi(Z_{\sigma(i)}) - \sum_{j=n+1}^{n+m} \frac{n}{n + \hat{\lambda} m r(Z_{\sigma(j)})} \varphi(Z_{\sigma(j)}) \right),$$

so that we have $T(Z_\sigma) = \frac{n+m}{m} \sup_{\|\varphi\|_{\mathcal{H}} \leq 1} |W_\varphi^\sigma|$. In particular, W_φ^σ coincides with the term inside the conditional expectation in the statement of Lemma 7. Write $N \equiv N(\delta) \equiv N(\{\|\varphi\|_{\mathcal{H}} \leq 1\}, \delta, \|\cdot\|_\infty)$ for the covering number of $\{\|\varphi\|_{\mathcal{H}} \leq 1\}$ with respect to $\|\cdot\|_\infty$ and let $\{\psi_1, \dots, \psi_N\}$ be an associated δ -cover. Now, for a generic function $\varphi_0 \in \mathcal{H}$ such that $\|\varphi_0\|_{\mathcal{H}} \leq 1$ we have

$$\begin{aligned}
\mathbb{E}\left[\sup_{\|\varphi\|_{\mathcal{H}} \leq 1} |W_\varphi^\sigma| \mathbb{1}_A\right] &= \mathbb{E}\left[\sup_{\|\varphi\|_{\mathcal{H}} \leq 1} (|W_\varphi^\sigma| - |W_{\varphi_0}^\sigma|) \mathbb{1}_A\right] + \mathbb{E}[|W_{\varphi_0}^\sigma| \mathbb{1}_A] \\
&\leq \mathbb{E}\left[\sup_{\substack{\|\varphi_1\|_{\mathcal{H}} \leq 1 \\ \|\varphi_2\|_{\mathcal{H}} \leq 1}} (|W_{\varphi_1}^\sigma| - |W_{\varphi_2}^\sigma|) \mathbb{1}_A\right] + \mathbb{E}[|W_{\varphi_0}^\sigma| \mathbb{1}_A] \leq \mathbb{E}\left[\sup_{\substack{\|\varphi_1\|_{\mathcal{H}} \leq 1 \\ \|\varphi_2\|_{\mathcal{H}} \leq 1}} |W_{\varphi_1}^\sigma - W_{\varphi_2}^\sigma| \mathbb{1}_A\right] + \mathbb{E}[|W_{\varphi_0}^\sigma| \mathbb{1}_A] \\
&\leq 2\mathbb{E}\left[\sup_{\|\varphi_1 - \varphi_2\|_\infty \leq \delta} |W_{\varphi_1}^\sigma - W_{\varphi_2}^\sigma| \mathbb{1}_A\right] + 2\mathbb{E}\left[\max_{i \in [N]} |W_{\psi_i}^\sigma - W_{\psi_i}^\sigma| \mathbb{1}_A\right] + \mathbb{E}[|W_{\varphi_0}^\sigma| \mathbb{1}_A] \\
&\leq 2\mathbb{E}\left[\sup_{\|\varphi\|_\infty \leq \delta} |W_\varphi^\sigma| \mathbb{1}_A\right] + (4N+1) \sup_{\|\varphi\|_{\mathcal{H}} \leq 1} \mathbb{E}[|W_\varphi^\sigma| \mathbb{1}_A], \quad (22)
\end{aligned}$$

where the penultimate inequality follows from a one-step discretization argument as in [Wainwright \(2019, Equation 5.34\)](#) and the final inequality follows on bounding the maximum by a sum, using the fact that $W_{\varphi_1}^\sigma - W_{\varphi_2}^\sigma = W_{\varphi_1 - \varphi_2}^\sigma$ for any φ_1, φ_2 , and using the triangle inequality to say that $\max_{i \in [N]} \|\psi_1 - \psi_i\|_{\mathcal{H}} \leq 2$. We will now bound each term on the right-hand side of (22) separately. As for the first, observe that by (8) we have

$$\begin{aligned} |W_\varphi^\sigma| &= \frac{1}{n} \left| \sum_{i=1}^n \frac{\widehat{\lambda} m r(Z_{\sigma(i)})}{n + \widehat{\lambda} m r(Z_{\sigma(i)})} \varphi(Z_{\sigma(i)}) - \sum_{j=n+1}^{n+m} \frac{n}{n + \widehat{\lambda} m r(Z_{\sigma(j)})} \varphi(Z_{\sigma(j)}) \right| \\ &\leq \|\varphi\|_\infty \left\{ \frac{1}{n} \sum_{i=1}^n \frac{\widehat{\lambda} m r(Z_{\sigma(i)})}{n + \widehat{\lambda} m r(Z_{\sigma(i)})} + \sum_{j=n+1}^{n+m} \frac{1}{n + \widehat{\lambda} m r(Z_{\sigma(j)})} \right\} \\ &= \frac{2}{n} \|\varphi\|_\infty \min \left\{ \sum_{i=1}^n \frac{\widehat{\lambda} m r(Z_{\sigma(i)})}{n + \widehat{\lambda} m r(Z_{\sigma(i)})}, \sum_{j=n+1}^{n+m} \frac{n}{n + \widehat{\lambda} m r(Z_{\sigma(j)})} \right\} \leq \frac{2 \min(m, n)}{n} \|\varphi\|_\infty \end{aligned}$$

for any $\varphi \in \mathcal{H}$, which implies that

$$\mathbb{E} \left[\sup_{\|\varphi\|_\infty \leq \delta} |W_\varphi^\sigma| \right] \leq \frac{2 \min(m, n)}{n} \delta.$$

We now turn to the second term in (22). With our assumption that $\|\cdot\|_\infty \leq \gamma \|\cdot\|_{\mathcal{H}}$ for a constant γ , it follows from the Cauchy-Schwarz inequality, (25) and Lemma 7 that for any φ such that $\|\varphi\|_{\mathcal{H}} \leq 1$ we have

$$\mathbb{E}[|W_{\varphi_0}^\sigma| \mathbb{1}_A] \leq \sqrt{\mathbb{E}[(W_\varphi^\sigma)^2 \mathbb{1}_A]} \leq \gamma \left\{ \frac{16 V_0^{1/4} \min(m^{1/2}, n^{1/2})}{n} + \exp\left(-\frac{S}{64\tau V_0^{1/2}}\right) \right\}.$$

Putting these bounds together and using $T(Z_\sigma) = \frac{n+m}{m} \sup_{\|\varphi\|_{\mathcal{H}} \leq 1} |W_\varphi^\sigma|$, we see that for any $\delta > 0$ we have

$$\begin{aligned} \mathbb{E}[T(Z_\sigma) \mathbb{1}_A] &\leq \frac{4 \min(m, n)(n+m)}{nm} \delta + (4N(\delta) + 1) \gamma \left\{ \frac{16 V_0^{1/4} \sqrt{\min(n, m)}(n+m)}{nm} + \frac{n+m}{m} \exp\left(-\frac{S}{64\tau V_0^{1/2}}\right) \right\} \\ &\leq 8\delta + (4N(\delta) + 1) \gamma \left\{ \frac{32 V_0^{1/4}}{\sqrt{\min(n, m)}} + \frac{n+m}{m} \exp\left(-\frac{S}{64\tau V_0^{1/2}}\right) \right\}. \end{aligned} \tag{23}$$

Since we assume that $N(\{\|\varphi\|_{\mathcal{H}} \leq 1\}, \delta, \|\cdot\|_\infty)$ is finite for all $\delta > 0$, and that $S \gg \tau \log(m+n)$, there exists a sequence $(\delta_n)_{n \in \mathbb{N}_+}$ such that this right-hand side converges to zero.

We now turn to the unpermuted statistic $T(Z)$. Recall the population quantity

$$T_{\mathcal{F}, r} \equiv T_{\mathcal{F}, r}(f, g) = \sup_{\|\varphi\|_{\mathcal{H}} \leq 1} |I_\varphi|, \quad \text{where} \quad I_\varphi := \frac{n+m}{m} \int \frac{\lambda_0 r f - g}{n/m + \lambda_0 r} \varphi d\mu$$

with λ_0 such that $\int \frac{n f + m g}{n + \lambda_0 m r} d\mu = 1$. As we are working under the alternative hypothesis, we know by Lemma 5 and Proposition 17 in Appendix B that $T_{\mathcal{F}, r} > 0$ and we may fix φ such that $\|\varphi\|_{\mathcal{H}} \leq 1$ and

$I_\varphi \geq T_{\mathcal{F},r}/2$. Then we have

$$\begin{aligned}
T(Z) &\geq \frac{n+m}{nm} \sum_{i=1}^n \frac{\widehat{\lambda}mr(X_i)}{n + \widehat{\lambda}mr(X_i)} \varphi(X_i) - \frac{n+m}{nm} \sum_{j=n+1}^{n+m} \frac{n}{n + \widehat{\lambda}mr(Y_j)} \varphi(Y_j) \\
&\geq \frac{1}{2}T_{\mathcal{F},r} + \frac{n+m}{m} \left(\frac{1}{n} \sum_{i=1}^n \frac{\widehat{\lambda}mr(X_i)\varphi(X_i)}{n + \widehat{\lambda}mr(X_i)} - \int \frac{\lambda_0mr}{n + \lambda_0mr} \varphi f - \frac{1}{m} \sum_{j=n+1}^{n+m} \frac{m\varphi(Y_j)}{n + \widehat{\lambda}mr(Y_j)} + \int \frac{m}{n + \lambda_0mr} \varphi g \right) \\
&\geq \frac{1}{2}T_{\mathcal{F},r} - \gamma \left| \frac{\widehat{\lambda}}{\lambda_0} - 1 \right| \frac{n+m}{m} \sum_{i=1}^{n+m} \frac{\lambda_0mr(Z_i)}{(n + \lambda_0mr(Z_i))(n + \widehat{\lambda}mr(Z_i))} \\
&\quad - \frac{n+m}{m} \left| \frac{1}{n} \sum_{i=1}^n \frac{\lambda_0mr(X_i)\varphi(X_i)}{n + \lambda_0mr(X_i)} - \int \frac{\lambda_0mr}{n + \lambda_0mr} \varphi f - \frac{1}{m} \sum_{j=n+1}^{n+m} \frac{m\varphi(Y_j)}{n + \lambda_0mr(Y_j)} + \int \frac{m}{n + \lambda_0mr} \varphi g \right| \\
&\geq \frac{1}{2}T_{\mathcal{F},r} - \gamma \left| \frac{\widehat{\lambda}}{\lambda_0} - 1 \right| \frac{1}{n+m} \sum_{i=1}^{n+m} \frac{\max\{1, \lambda_0r(Z_i)\}}{\min\{1, \widehat{\lambda}r(Z_i)\}} \\
&\quad - \frac{n+m}{m} \left| \frac{1}{n} \sum_{i=1}^n \frac{\lambda_0mr(X_i)\varphi(X_i)}{n + \lambda_0mr(X_i)} - \int \frac{\lambda_0mr}{n + \lambda_0mr} \varphi f - \frac{1}{m} \sum_{j=n+1}^{n+m} \frac{m\varphi(Y_j)}{n + \lambda_0mr(Y_j)} + \int \frac{m}{n + \lambda_0mr} \varphi g \right| \\
&=: \frac{1}{2}T_{\mathcal{F},r} - \gamma \left| \frac{\widehat{\lambda}}{\lambda_0} - 1 \right| \frac{1}{n+m} \sum_{i=1}^{n+m} \frac{\max\{1, \lambda_0r(Z_i)\}}{\min\{1, \widehat{\lambda}r(Z_i)\}} - R_1.
\end{aligned}$$

The final term is a sum of independent mean-zero random variables, so that we may use Chebyshev's inequality to control the probability of this event. Indeed, using again the boundedness assumption on the second moments of r we can see that

$$\begin{aligned}
&\mathbb{P}(R_1 \geq \epsilon) \\
&= \mathbb{P} \left(\frac{n+m}{m} \left| \frac{1}{n} \sum_{i=1}^n \frac{\lambda_0mr(X_i)\varphi(X_i)}{n + \lambda_0mr(X_i)} - \int \frac{\lambda_0mr}{n + \lambda_0mr} \varphi f - \frac{1}{m} \sum_{j=n+1}^{n+m} \frac{m\varphi(Y_j)}{n + \lambda_0mr(Y_j)} + \int \frac{m}{n + \lambda_0mr} \varphi g \right| \geq \epsilon \right) \\
&\leq \frac{(n+m)^2}{\epsilon^2 m^2} \left\{ n^{-1} \text{Var} \left(\frac{\lambda_0mr(X)\varphi(X)}{n + \lambda_0mr(X)} \right) + m^{-1} \text{Var} \left(\frac{m\varphi(Y)}{n + \lambda_0mr(Y)} \right) \right\} \\
&\leq \frac{\gamma^2(n+m)^2}{\epsilon^2 m^2} \left\{ n^{-1} \min \left(\frac{m^2}{n^2} \mathbb{E}[\lambda_0^2 r^2(X)], 1 \right) + \frac{m}{n^2} \min \left(1, \frac{n^2}{m^2} \mathbb{E}[\lambda_0^{-2} r^{-2}(Y)] \right) \right\} \\
&\leq \frac{\gamma^2(n+m)^2}{\epsilon^2 m^2} \left\{ \frac{1}{n} \min \left(\frac{m^2(m+n)}{n^3} V_0, 1 \right) + \frac{m}{n^2} \min \left(1, \frac{n^2(m+n)}{m^3} V_0 \right) \right\} \\
&\leq \frac{\gamma^2(n+m)^2}{\epsilon^2 m^2} V_0 \left\{ \mathbb{1}_{\{n \leq m\}} \left(\frac{1}{n} + \frac{m+n}{m^2} \right) + \mathbb{1}_{\{n > m\}} \left(\frac{m^2(m+n)}{n^4} + \frac{m}{n^2} \right) \right\} \leq \frac{12\gamma^2 V_0}{\epsilon^2 \min(n, m)}. \tag{24}
\end{aligned}$$

It remains for us to control the term depending on $\widehat{\lambda}/\lambda_0$ under the event A . We first note that our assumption on r gives us bounds on $\widehat{\lambda}$ and λ_0 . Exactly as in the proof of Lemma 7, on the event A we have that $1/\{2\mathbb{E}(\bar{R})\} \leq \widehat{\lambda} \leq 2\mathbb{E}(\bar{R}^{-1})$. As also claimed above, by a very similar argument we also have that $1/\mathbb{E}(\bar{R}) \leq \lambda_0 \leq \mathbb{E}(\bar{R}^{-1})$. We may view $\widehat{\lambda}$ as an M -estimator of λ_0 . Indeed, writing $\phi_\lambda(z) = (n+m)/\{n+m\lambda r(z)\} - 1$,

we defined $\hat{\lambda}$ to be the solution to

$$\hat{S}_{m,n}(\lambda) := \frac{1}{n+m} \sum_{i=1}^{n+m} \phi_{\lambda}(Z_i) = 0.$$

Similarly, λ_0 can be thought of as the solution to $S_{m,n}(\lambda) := \mathbb{E}\{\hat{S}_{m,n}(\lambda)\} = 0$. By Jensen's inequality we have that

$$-\hat{S}'_{m,n}(\lambda) = \sum_{i=1}^{n+m} \frac{mr(Z_i)}{\{n + \lambda mr(Z_i)\}^2} \geq \frac{(n+m)^2}{\sum_{i=1}^{n+m} \frac{\{n + \lambda mr(Z_i)\}^2}{mr(Z_i)}} = \frac{m(n+m)}{n^2 \check{R}^{-1} + 2\lambda mn + \lambda^2 m^2 \bar{R}},$$

which is decreasing in λ . Thus, for any $\epsilon \in (0, 1)$ we have

$$\begin{aligned} \mathbb{P}(|\hat{\lambda}/\lambda_0 - 1| \geq \epsilon, A) &= \mathbb{P}(|\hat{\lambda} - \lambda_0| \geq \epsilon\lambda_0, A) \leq \mathbb{P}\left(|\hat{S}_{m,n}(\hat{\lambda}) - \hat{S}_{m,n}(\lambda_0)| \geq \epsilon\lambda_0 |\hat{S}'_{m,n}((1+\epsilon)\lambda_0)|, A\right) \\ &\leq \mathbb{P}\left(|\hat{S}_{m,n}(\lambda_0)| \geq \epsilon\lambda_0 |\hat{S}'_{m,n}(2\lambda_0)|, A\right) \leq \mathbb{P}\left(|\hat{S}_{m,n}(\lambda_0)| \geq \frac{(m+n)m\lambda_0\epsilon}{2\{n^2\mathbb{E}(\check{R}^{-1}) + 2\lambda_0 mn + \lambda_0^2 m^2 \mathbb{E}(\bar{R})\}}\right) \\ &\leq \mathbb{P}\left(|\hat{S}_{m,n}(\lambda_0)| \geq \frac{(m+n)m\epsilon}{2} \min\left[\frac{1}{n^2\mathbb{E}(\check{R}^{-1})\mathbb{E}(\bar{R}) + 2mn + m^2}, \frac{1}{n^2 + 2mn + m^2\mathbb{E}(\bar{R})\mathbb{E}(\check{R}^{-1})}\right]\right) \\ &\leq \mathbb{P}\left(|\hat{S}_{m,n}(\lambda_0)| \geq \frac{m\epsilon}{2V_0^{1/2}(m+n)}\right), \end{aligned}$$

where the last step follows from the boundedness assumption. Now $\hat{S}_{m,n}(\lambda_0)$ is a sum of independent mean-zero random variables, so that we may again use Chebyshev's inequality to control the probability of this event. Indeed, it is straightforward to see that for $\delta = m\epsilon/\{2V_0^{1/2}(m+n)\}$ we have

$$\begin{aligned} \mathbb{P}(|\hat{S}_{m,n}(\lambda_0)| \geq \delta) &\leq \delta^{-2} \text{Var}(\hat{S}_{m,n}(\lambda_0)) = \frac{1}{\delta^2(m+n)^2} \{n \text{Var} \phi_{\lambda_0}(X) + m \text{Var} \phi_{\lambda_0}(Y)\} \\ &= \frac{4V_0}{\epsilon^2} \left\{ n \text{Var} \left(\frac{1 - \lambda_0 r(X)}{n + m\lambda_0 r(X)} \right) + m \text{Var} \left(\frac{1 - \lambda_0 r(Y)}{n + m\lambda_0 r(Y)} \right) \right\} \\ &\leq \frac{4V_0}{\epsilon^2(m+n)^2} \left[n \mathbb{E} \left\{ \frac{\{1 - \lambda_0 r(X)\}^2}{\min^2(1, \lambda_0 r(X))} \right\} + m \mathbb{E} \left\{ \frac{\{1 - \lambda_0 r(Y)\}^2}{\min^2(1, \lambda_0 r(Y))} \right\} \right] \\ &\leq \frac{4V_0}{\epsilon^2(m+n)^2} \left[n \mathbb{E} \left\{ (\lambda_0 r(X))^2 + \left(\frac{1}{\lambda_0 r(X)} \right)^2 \right\} + m \mathbb{E} \left\{ (\lambda_0 r(Y))^2 + \left(\frac{1}{\lambda_0 r(Y)} \right)^2 \right\} \right] \\ &\leq \frac{8V_0^2}{\epsilon^2(m+n)}. \end{aligned}$$

Finally, since on the event A we have $\bar{R} \leq 2\mathbb{E}(\bar{R})$, $\check{R}^{-1} \leq 2\mathbb{E}(\check{R}^{-1})$, $1/\{2\mathbb{E}(\bar{R})\} \leq \hat{\lambda} \leq 2\mathbb{E}(\check{R}^{-1})$, $1/\mathbb{E}(\bar{R}) \leq \lambda_0 \leq \mathbb{E}(\check{R}^{-1})$ and $\mathbb{E}(\bar{R})\mathbb{E}(\check{R}^{-1}) \leq V_0$, it follows that

$$\begin{aligned} \frac{1}{n+m} \sum_{i=1}^{n+m} \frac{\max\{1, \lambda_0 r(Z_i)\}}{\min\{1, \hat{\lambda} r(Z_i)\}} &\leq \frac{1}{n+m} \sum_{i=1}^{n+m} (1 + \lambda_0 r(Z_i)) \left(1 + \frac{1}{\hat{\lambda} r(Z_i)} \right) \\ &\leq \frac{1}{n+m} \sum_{i=1}^{n+m} \left(1 + \lambda_0 r(Z_i) + \frac{1}{\hat{\lambda} r(Z_i)} + \frac{\lambda_0}{\hat{\lambda}} \right) \leq \frac{1}{n+m} \sum_{i=1}^{n+m} \left(1 + \mathbb{E}(\check{R}^{-1})r(Z_i) + \frac{2\mathbb{E}(\bar{R})}{r(Z_i)} + 2\mathbb{E}(\check{R}^{-1})\mathbb{E}(\bar{R}) \right) \end{aligned}$$

$$\begin{aligned}
&= 1 + \mathbb{E}(\check{R}^{-1}) \bar{R} + 2\mathbb{E}(\bar{R}) \check{R}^{-1} + 2\mathbb{E}(\check{R}^{-1})\mathbb{E}(\bar{R}) \leq 1 + 2\mathbb{E}(\check{R}^{-1}) \mathbb{E}(\bar{R}) + 4\mathbb{E}(\bar{R}) \mathbb{E}(\check{R}^{-1}) + 2\mathbb{E}(\check{R}^{-1})\mathbb{E}(\bar{R}) \\
&\leq 1 + 8\mathbb{E}(\check{R}^{-1})\mathbb{E}(\bar{R}) \leq 1 + 8V_0^{1/2}.
\end{aligned}$$

This, in conjunction with (24) and the previous display, gives

$$\begin{aligned}
&\mathbb{P}(T(Z) < \tfrac{1}{4}T_{\mathcal{F},r}, A) \\
&\leq \mathbb{P}\left(\tfrac{1}{2}T_{\mathcal{F},r} - \gamma(1 + 8V_0^{1/2}) \left| \frac{\hat{\lambda}}{\lambda_0} - 1 \right| - R_1 < \tfrac{1}{4}T_{\mathcal{F},r}, A\right) \leq \mathbb{P}\left(\gamma(1 + 8V_0^{1/2}) \left| \frac{\hat{\lambda}}{\lambda_0} - 1 \right| + R_1 > \tfrac{1}{4}T_{\mathcal{F},r}, A\right) \\
&\leq \mathbb{P}\left(\gamma(1 + 8V_0^{1/2}) \left| \frac{\hat{\lambda}}{\lambda_0} - 1 \right| > \tfrac{1}{8}T_{\mathcal{F},r}, A\right) + \mathbb{P}(R_1 > \tfrac{1}{8}T_{\mathcal{F},r}, A) \leq \frac{2^9 V_0^2 \gamma^2 (1 + 8V_0^{1/2})^2}{T_{\mathcal{F},r}^2 (n+m)} + \frac{2^{10} \gamma^2 V_0}{T_{\mathcal{F},r}^2 \min(n, m)}.
\end{aligned}$$

We can thus bound

$$\begin{aligned}
\mathbb{P}(T(Z_\sigma) \geq T(Z), A) &\leq \mathbb{P}(T(Z_\sigma) \geq \tfrac{1}{4}T_{\mathcal{F},r}, A) + \mathbb{P}(T(Z) < \tfrac{1}{4}T_{\mathcal{F},r}, A) \\
&\leq \frac{4\mathbb{E}[T(Z_\sigma)\mathbb{1}_A]}{T_{\mathcal{F},r}} + \frac{2^9 V_0^2 \gamma^2 (1 + 8V_0^{1/2})^2}{T_{\mathcal{F},r}^2 (n+m)} + \frac{2^{10} \gamma^2 V_0}{T_{\mathcal{F},r}^2 \min(n, m)},
\end{aligned}$$

and use (23) to show that the last display goes to zero. This, combined with (21), concludes the proof. $\square$

Proof of Lemma 7. Let $(\sigma_0, \sigma_1, \dots, \sigma_S)$ be the permutations generated through Algorithm 1. For $t \in \{0, 1, \dots, S\}$ define

$$S_n^t := \sum_{i=1}^n \varphi(Z_{\sigma_t(i)}) - n \int \varphi d\hat{H}_{n,m},$$

Introduce the shorthand $r_i^t := r(Z_{\sigma_t(i)})$ and $\varphi_i^t := \varphi(Z_{\sigma_t(i)})$ for $i \in [n+m]$ and define $q_i^t := \hat{\lambda}mr_i^t/(n + \hat{\lambda}mr_i^t)$. Then we see that

$$\begin{aligned}
\frac{S_n^t}{n} &= \frac{1}{n} \sum_{i=1}^n \varphi_i^t - \int \varphi d\hat{H}_{n,m} = \frac{1}{n} \sum_{i=1}^n \varphi_i^t - \sum_{i=1}^{n+m} \frac{\varphi(Z_i)}{n + \hat{\lambda}mr(Z_i)} = \frac{1}{n} \sum_{i=1}^n \varphi_i^t - \sum_{i=1}^{n+m} \frac{\varphi_i^t}{n + \hat{\lambda}mr_i^t} \\
&= \frac{1}{n} \left(\sum_{i=1}^n \frac{\hat{\lambda}mr_i^t}{n + \hat{\lambda}mr_i^t} \varphi_i^t - \sum_{j=n+1}^{n+m} \frac{n}{n + \hat{\lambda}mr_j^t} \varphi_j^t \right) = \frac{1}{n} \left\{ \sum_{i=1}^n q_i^t \varphi_i^t - \sum_{j=n+1}^{n+m} (1 - q_j^t) \varphi_j^t \right\}. \tag{25}
\end{aligned}$$

Recall that we write $K = \min(m, n)$ and, for each $t \geq 0$, we generate σ_{t+1} by sampling $I_1, \dots, I_K \in [n]$ and $J_1, \dots, J_K \in \{n+1, \dots, n+m\}$ uniformly at random without replacement, and switching $Z_{\sigma_t(I_\ell)}$ with $Z_{\sigma_t(J_\ell)}$ with probability

$$\tilde{p}_{I_\ell, J_\ell}^t = \frac{\hat{\lambda}nmr_{I_\ell}^t}{(n + \hat{\lambda}mr_{I_\ell}^t)(n + \hat{\lambda}mr_{J_\ell}^t)} = q_{I_\ell}^t(1 - q_{J_\ell}^t).$$

Given $(I_\ell), (J_\ell)$, for $\ell \in [K]$ let $B_{I_\ell, J_\ell} \sim \text{Ber}(\tilde{p}_{I_\ell, J_\ell}^t)$ determine whether the ℓ th switch occurs.

Having introduced the necessary notation, we prove the result by analysing the change in S_n^t over a single step of the algorithm, showing that S_n^t can be regarded as a Lyapunov function satisfying a geometric drift

condition. We have that

$$\begin{aligned}\mathbb{E}\{(S_n^{t+1})^2 | \sigma_t, Z\} - (S_n^t)^2 &= \mathbb{E}\left[\left\{S_n^t + \sum_{\ell=1}^K B_{I_\ell, J_\ell}(\varphi_{J_\ell}^t - \varphi_{I_\ell}^t)\right\}^2 \middle| \sigma_t, Z\right] - (S_n^t)^2 \\ &= 2S_n^t \mathbb{E}\left[\sum_{\ell=1}^K B_{I_\ell, J_\ell}(\varphi_{J_\ell}^t - \varphi_{I_\ell}^t) \middle| \sigma_t, Z\right] + \mathbb{E}\left[\left\{\sum_{\ell=1}^K B_{I_\ell, J_\ell}(\varphi_{J_\ell}^t - \varphi_{I_\ell}^t)\right\}^2 \middle| \sigma_t, Z\right].\end{aligned}\quad (26)$$

We treat each of the two terms in this sum separately. For the first, we may calculate that

$$\begin{aligned}\mathbb{E}\left[\sum_{\ell=1}^K B_{I_\ell, J_\ell}(\varphi_{J_\ell}^t - \varphi_{I_\ell}^t) \middle| \sigma_t, Z\right] &= \sum_{\ell=1}^K \mathbb{E}\{\tilde{p}_{I_\ell, J_\ell}^t(\varphi_{J_\ell}^t - \varphi_{I_\ell}^t) \mid \sigma_t, Z\} = K \mathbb{E}\{\tilde{p}_{I_1, J_1}^t(\varphi_{J_1}^t - \varphi_{I_1}^t) \mid \sigma_t, Z\} \\ &= \frac{K}{mn} \sum_{i=1}^n \sum_{j=n+1}^{n+m} q_i^t(1 - q_j^t)(\varphi_j^t - \varphi_i^t) = \frac{K}{mn} \left(\sum_{i=1}^n q_i^t\right) \left\{\sum_{j=n+1}^{n+m} (1 - q_j^t)\varphi_j^t - \sum_{i=1}^n q_i^t \varphi_i^t\right\} = -\frac{K}{mn} \left(\sum_{i=1}^n q_i^t\right) S_n^t,\end{aligned}\quad (27)$$

where the penultimate equality follows from (8). The second term in (26) is controlled by writing

$$\begin{aligned}\mathbb{E}\left[\left\{\sum_{\ell=1}^K B_{I_\ell, J_\ell}(\varphi_{J_\ell}^t - \varphi_{I_\ell}^t)\right\}^2 \middle| \sigma_t, Z\right] &= \sum_{\ell=1}^K \mathbb{E}\{B_{I_\ell, J_\ell}(\varphi_{J_\ell}^t - \varphi_{I_\ell}^t)^2 \mid \sigma_t, Z\} + \sum_{\ell_1 \neq \ell_2} \mathbb{E}\{B_{I_{\ell_1}, J_{\ell_1}} B_{I_{\ell_2}, J_{\ell_2}}(\varphi_{I_{\ell_1}}^t - \varphi_{J_{\ell_1}}^t)(\varphi_{I_{\ell_2}}^t - \varphi_{J_{\ell_2}}^t) \mid \sigma_t, Z\} \\ &= \frac{K}{mn} \sum_{i=1}^n \sum_{j=n+1}^{n+m} \tilde{p}_{ij}^t(\varphi_i^t - \varphi_j^t)^2 + \frac{K(K-1)}{n(n-1)m(m-1)} \sum_{i_1 \neq i_2} \sum_{j_1 \neq j_2} \tilde{p}_{i_1 j_1}^t \tilde{p}_{i_2 j_2}^t (\varphi_{i_1}^t - \varphi_{j_1}^t)(\varphi_{i_2}^t - \varphi_{j_2}^t) \\ &= \frac{K}{mn} \sum_{i=1}^n \sum_{j=n+1}^{n+m} \tilde{p}_{ij}^t(\varphi_i^t - \varphi_j^t)^2 + \frac{K(K-1)}{n(n-1)m(m-1)} \left[\sum_{i_1, i_2=1}^n \sum_{j_1, j_2=n+1}^{n+m} \tilde{p}_{i_1 j_1}^t \tilde{p}_{i_2 j_2}^t (\varphi_{i_1}^t - \varphi_{j_1}^t)(\varphi_{i_2}^t - \varphi_{j_2}^t) \right. \\ &\quad - \sum_{i_1, i_2=1}^n \sum_{j=n+1}^{n+m} \tilde{p}_{i_1 j}^t \tilde{p}_{i_2 j}^t (\varphi_{i_1}^t - \varphi_j^t)(\varphi_{i_2}^t - \varphi_j^t) - \sum_{i=1}^n \sum_{j_1, j_2=n+1}^{n+m} \tilde{p}_{i j_1}^t \tilde{p}_{i j_2}^t (\varphi_i^t - \varphi_{j_1}^t)(\varphi_i^t - \varphi_{j_2}^t) \\ &\quad \left. + \sum_{i=1}^n \sum_{j=n+1}^{n+m} (\tilde{p}_{ij}^t)^2 (\varphi_i^t - \varphi_j^t)^2 \right] \\ &= \frac{K}{mn} \sum_{i=1}^n \sum_{j=n+1}^{n+m} \tilde{p}_{ij}^t \left\{1 + \frac{K-1}{(m-1)(n-1)} \tilde{p}_{ij}^t\right\} (\varphi_i^t - \varphi_j^t)^2 + \frac{K(K-1)}{n(n-1)m(m-1)} \left[\left(\sum_{i=1}^n q_i^t\right)^2 (S_n^t)^2 \right. \\ &\quad - \sum_{j=n+1}^{n+m} (1 - q_j^t)^2 \left\{\sum_{i=1}^n q_i^t (\varphi_i^t - \varphi_j^t)\right\}^2 - \sum_{i=1}^n (q_i^t)^2 \left\{\sum_{j=n+1}^{n+m} (1 - q_j^t)(\varphi_i^t - \varphi_j^t)\right\}^2 \left. \right] \\ &\leq \left(\sum_{i=1}^n q_i^t\right)^2 \left\{\frac{8K\|\varphi\|_\infty^2}{mn} + \frac{K(K-1)}{n(n-1)m(m-1)} (S_n^t)^2\right\},\end{aligned}\quad (28)$$

where for the final inequality we again use (8). Combining (26), (27) and (28), using our assumption that

$\max(m, n) \geq 3$ and using the fact that by (8) we have $\sum_{i=1}^n q_i^t \leq K$, we see that

$$\begin{aligned} \mathbb{E}\{(S_n^{t+1})^2 | \sigma_t, Z\} - (S_n^t)^2 &\leq -\frac{2K}{mn} \left(\sum_{i=1}^n q_i^t \right) (S_n^t)^2 + \left(\sum_{i=1}^n q_i^t \right)^2 \left\{ \frac{8K \|\varphi\|_\infty^2}{mn} + \frac{K(K-1)}{n(n-1)m(m-1)} (S_n^t)^2 \right\} \\ &= -\frac{2K}{mn} \left(\sum_{i=1}^n q_i^t \right) \left\{ 1 - \frac{K-1}{2(m-1)(n-1)} \sum_{i=1}^n q_i^t \right\} (S_n^t)^2 + \frac{8K \|\varphi\|_\infty^2}{mn} \left(\sum_{i=1}^n q_i^t \right)^2 \\ &\leq -\frac{K}{2mn} \left(\sum_{i=1}^n q_i^t \right) (S_n^t)^2 + \frac{8K \|\varphi\|_\infty^2}{mn} \left(\sum_{i=1}^n q_i^t \right)^2. \end{aligned}$$

We now control $\sum_i q_i^t$ using our boundedness assumption on r . Before doing so we establish some basic consequences of this assumption. First, we see that it implies bounds on $\hat{\lambda}$. Indeed, writing $\bar{R} = (n+m)^{-1} \sum_{i=1}^{n+m} r(Z_i)$ for the arithmetic mean of the observed r values, by Jensen's inequality we have that

$$1 = \sum_{i=1}^{n+m} \frac{1}{n + \hat{\lambda} m r(Z_i)} \geq \left[\frac{1}{(n+m)^2} \sum_{i=1}^{n+m} \{n + \hat{\lambda} m r(Z_i)\} \right]^{-1} = \left(\frac{n}{n+m} + \frac{m}{n+m} \hat{\lambda} \bar{R} \right)^{-1}.$$

Similarly, writing $\check{R} = \{(n+m)^{-1} \sum_{i=1}^{n+m} r(Z_i)^{-1}\}^{-1}$ for the harmonic mean of the observed r values, we have that

$$1 = \sum_{i=1}^{n+m} \frac{1}{m + \hat{\lambda}^{-1} n r(Z_i)^{-1}} \geq \left(\frac{m}{n+m} + \frac{n}{m+n} \hat{\lambda}^{-1} \check{R}^{-1} \right)^{-1},$$

where the first identity follows by the symmetry of our problem, as discussed in Remark 2. These two bounds together imply that $\bar{R}^{-1} \leq \hat{\lambda} \leq \check{R}^{-1}$. Together with our assumption that $\bar{R}/\check{R} \leq \kappa$, this implies that $\max(\hat{\lambda}^{-1} \check{R}^{-1}, \hat{\lambda} \bar{R}) \leq (\hat{\lambda}^{-1} \check{R}^{-1}) \cdot (\hat{\lambda} \bar{R}) = \bar{R}/\check{R} \leq \kappa$. Having bounds on $\hat{\lambda}$, we now turn to the control of $\sum_i q_i^t$ itself. Let $(\cdot)$ be a reordering of our data such that $r(Z_{(1)}) \leq r(Z_{(2)}) \leq \dots \leq r(Z_{(m+n)})$. Then, again using Jensen's inequality, we have that

$$\begin{aligned} \sum_{i=1}^n q_i^t &\geq \sum_{i=1}^n \frac{m \hat{\lambda} r(Z_{(i)})}{n + m \hat{\lambda} r(Z_{(i)})} \geq n \left\{ \frac{1}{n} \sum_{i=1}^n \frac{n + m \hat{\lambda} r(Z_{(i)})}{m \hat{\lambda} r(Z_{(i)})} \right\}^{-1} = \frac{mn}{m + \hat{\lambda}^{-1} \sum_{i=1}^n r(Z_{(i)})^{-1}} \\ &\geq \frac{mn}{m + (n+m) \hat{\lambda}^{-1} \check{R}^{-1}} \geq \frac{mn}{m + (n+m) \kappa} \geq \frac{1}{2} \frac{mn}{(m+n) \kappa}. \end{aligned}$$

Having bounded $\sum_i q_i^t$ we may now conclude the proof. Writing $\tau = \max(m, n)/\min(m, n)$ and arguing inductively for the third inequality, we now see that

$$\begin{aligned} n^{-2} \mathbb{E}\{(S_n^{t+1})^2 | Z\} &\leq n^{-2} \mathbb{E}\{(S_n^t)^2 | Z\} + n^{-2} \mathbb{E}\left\{ -\frac{K}{2mn} \left(\sum_{i=1}^n q_i^t \right) (S_n^t)^2 + \frac{8K \|\varphi\|_\infty^2}{mn} \left(\sum_{i=1}^n q_i^t \right)^2 \mid Z \right\} \\ &\leq n^{-2} \left\{ 1 - \frac{K}{4(m+n)\kappa} \right\} \mathbb{E}\{(S_n^t)^2 | Z\} + \frac{8 \|\varphi\|_\infty^2 K^3}{mn^3} \\ &\leq \frac{8 \|\varphi\|_\infty^2 K^3}{mn^3} \sum_{s=0}^t \left(1 - \frac{1}{8\tau\kappa} \right)^s + \left(1 - \frac{1}{8\tau\kappa} \right)^{t+1} n^{-2} \mathbb{E}\{(S_n^0)^2 | Z\} \\ &\leq \frac{64 \|\varphi\|_\infty^2 K^3}{mn^3} \tau\kappa + \|\varphi\|_\infty^2 \exp\left(-\frac{t+1}{8\tau\kappa}\right) \leq \frac{64 \|\varphi\|_\infty^2 \kappa K}{n^2} + \|\varphi\|_\infty^2 \exp\left(-\frac{t+1}{8\tau\kappa}\right), \end{aligned}$$

and the result follows upon taking $t = S - 1$. □

A.3 Proofs for Section 4

Proof of Proposition 8. This proof borrows ideas from the proof of Lemma 4 in [Gretton et al. \(2012\)](#). Define the linear operator

$$T_{rf} : \begin{cases} \mathcal{H} \rightarrow \mathbb{R} \\ \varphi \mapsto \int \frac{n+m}{n+\lambda_0 mr} \lambda_0 r f \varphi d\mu. \end{cases}$$

Write $d\mu_x = d\mu(x)$. Using the reproducing property of the reproducing kernel Hilbert space, i.e. $\varphi(x) = \langle \varphi, k(\cdot, x) \rangle_{\mathcal{H}}$, we can show that this operator is bounded, since for all $\varphi \in \mathcal{H}$ we have

$$\begin{aligned} |T_{rf}\varphi| &\leq \int_{\mathcal{X}} \frac{(n+m)\lambda_0 r(x)f(x)}{n+\lambda_0 mr(x)} |\varphi(x)| d\mu_x = \int_{\mathcal{X}} \frac{(n+m)\lambda_0 r(x)f(x)}{n+\lambda_0 mr(x)} |\langle \varphi, k(\cdot, x) \rangle_{\mathcal{H}}| d\mu_x \\ &\leq \|\varphi\|_{\mathcal{H}} \int_{\mathcal{X}} \sqrt{k(x, x)} \frac{(n+m)\lambda_0 r(x)f(x)}{n+\lambda_0 mr(x)} d\mu_x, \end{aligned}$$

which shows that $|T_{rf}\varphi|/\|\varphi\|_{\mathcal{H}}$ is bounded uniformly in φ . The same is true for the linear operator

$$T_g : \begin{cases} \mathcal{H} \rightarrow \mathbb{R} \\ \varphi \mapsto \int \frac{(n+m)g}{n+\lambda_0 mr} \varphi d\mu, \end{cases}$$

hence the Riesz representation theorem implies that there exist $m_{rf}, m_g \in \mathcal{H}$ such that $T_{rf}\varphi = \langle m_{rf}, \varphi \rangle_{\mathcal{H}}$ and $T_g\varphi = \langle m_g, \varphi \rangle_{\mathcal{H}}$. Furthermore, using again the reproducing property of $\mathcal{H}$, we have that

$$m_{rf}(t) = \langle m_{rf}, k(t, \cdot) \rangle_{\mathcal{H}} = T_{rf}k(t, \cdot) = \int_{\mathcal{X}} \frac{(n+m)\lambda_0 r(x)f(x)}{n+\lambda_0 mr(x)} k(x, t) d\mu_x$$

and, similarly, $m_g(t) = \int_{\mathcal{X}} \frac{(n+m)g(x)}{n+\lambda_0 mr(x)} k(x, t) d\mu_x$. This implies that

$$\begin{aligned} T_{\mathcal{F},r}^2(f, g) &= \left(\sup_{\|\varphi\|_{\mathcal{H}} \leq 1} \left| \int \frac{(n+m)(\lambda_0 r f - g)}{n+\lambda_0 mr} \varphi d\mu \right| \right)^2 = \left(\sup_{\|\varphi\|_{\mathcal{H}} \leq 1} |T_{rf}\varphi - T_g\varphi| \right)^2 \\ &= \left(\sup_{\|\varphi\|_{\mathcal{H}} \leq 1} |\langle m_{rf} - m_g, \varphi \rangle_{\mathcal{H}}| \right)^2 = \|m_{rf} - m_g\|_{\mathcal{H}}^2 = \langle m_{rf}, m_{rf} \rangle_{\mathcal{H}} + \langle m_g, m_g \rangle_{\mathcal{H}} - 2\langle m_{rf}, m_g \rangle_{\mathcal{H}} \\ &= \int_{\mathcal{X}} \frac{(n+m)\lambda_0 r(t)f(t)}{n+\lambda_0 mr(t)} \left(\int_{\mathcal{X}} \frac{(n+m)\lambda_0 r(x)f(x)}{n+\lambda_0 mr(x)} k(x, t) d\mu_x \right) d\mu_t \\ &\quad + \int_{\mathcal{X}} \frac{(n+m)g(t)}{n+\lambda_0 mr(t)} \left(\int_{\mathcal{X}} \frac{(n+m)g(x)}{n+\lambda_0 mr(x)} k(x, t) d\mu_x \right) d\mu_t \\ &\quad - 2 \int_{\mathcal{X}} \frac{(n+m)\lambda_0 r(t)f(t)}{n+\lambda_0 mr(t)} \left(\int_{\mathcal{X}} \frac{(n+m)g(x)}{n+\lambda_0 mr(x)} k(x, t) d\mu_x \right) d\mu_t \\ &= \mathbb{E}_{X, X'} \left[\frac{\lambda_0^2 (n+m)^2 r(X) r(X')}{\{n+\lambda_0 mr(X)\} \{n+\lambda_0 mr(X')\}} k(X, X') \right] \end{aligned}$$

$$\begin{aligned}
& + \mathbb{E}_{Y,Y'} \left[\frac{(n+m)^2}{\{n + \lambda_0 m r(Y)\} \{n + \lambda_0 m r(Y')\}} k(Y, Y') \right] \\
& - 2 \mathbb{E}_{X,Y} \left[\frac{\lambda_0 (n+m)^2 r(X)}{\{n + \lambda_0 m r(X)\} \{n + \lambda_0 m r(Y)\}} k(X, Y) \right],
\end{aligned}$$

where $X, X' \stackrel{\text{i.i.d.}}{\sim} f$ independently of $Y, Y' \stackrel{\text{i.i.d.}}{\sim} g$, and concludes the proof. $\square$

Proof of Theorem 9. Consider a kernel function $K : \mathbb{R} \rightarrow \mathbb{R}$ that satisfies the assumptions in Subsection 4.2. In particular, $K \in L^1(\mathbb{R}) \cap L^4(\mathbb{R})$ implies that $K \in L^2(\mathbb{R})$ using the basic inequality $t^2 \leq t + t^4$ for $t \geq 0$. Based on this, we define the constants

$$\kappa_j(d) := \left(\int_{\mathbb{R}} |K(u)|^j du \right)^d < \infty \quad \text{for } j \in \{1, 2, 4\},$$

which are finite by assumption. Since the kernel is fixed in advance, we omit its dependence from the notation. We frequently use the analytical properties of the product kernel k_ζ with bandwidth $\zeta \geq 1$, which satisfies

$$\int_{\mathbb{R}^d} |k_\zeta(x, y)|^j dy = \zeta^{(j-1)d} \left(\int_{\mathbb{R}} |K(u)|^j du \right)^d = \kappa_j(d) \zeta^{(j-1)d} \quad \text{for all } x \in \mathbb{R}^d \text{ and } j \in \{1, 2, 4\}.$$

As a result, using the fact that $\max(\|f\|_\infty, \|g\|_\infty) \leq M$, it follows that for any $x \in \mathbb{R}^d$ we have

$$\mathbb{E}[|k_\zeta(x, X)|^j] = \int_{\mathbb{R}^d} |k_\zeta(x, y)|^j f(y) dy \leq M \int_{\mathbb{R}^d} |k_\zeta(x, y)|^j dy = M \kappa_j(d) \zeta^{(j-1)d} \quad \text{for } j \in \{1, 2, 4\},$$

and similarly for the expectation of $|k_\zeta(x, Y)|^j$. Furthermore, these bounds imply the same bounds on the quantities $\mathbb{E}[|k_\zeta(X_1, X_2)|^j]$, $\mathbb{E}[|k_\zeta(Y_1, Y_2)|^j]$ and $\mathbb{E}[|k_\zeta(X, Y)|^j]$ for all $j \in \{1, 2, 4\}$. Now, we already know that Algorithm 2 controls the type I error at a nominal level α , so we can bound the minimax separation ρ_r^* by controlling its type II error. In order to do this, fix $\beta \in (0, 1 - \alpha)$, choose $H \geq 2 \lceil \frac{1}{\alpha\beta} - 1 \rceil$, and suppose $(f, g) \in \mathcal{S}_\theta^r(\rho)$ satisfies

$$\text{MMD}_{r, k_\zeta}^2(f, g) \geq \max \left\{ 2 |\mathbb{E}[U(Z) - U(Z_\sigma)] - \text{MMD}_{r, k_\zeta}^2(f, g)|, \left(\frac{8}{\alpha\beta} \text{Var}[U(Z_\sigma) - U(Z)] \right)^{1/2} \right\}. \quad (29)$$

Then, a double application of Markov's inequality shows that

$$\begin{aligned}
\mathbb{P}\{p > \alpha\} &= \mathbb{P} \left(1 + \sum_{h=1}^H \mathbb{1}\{U(Z_{\sigma(h)}) \geq U(Z)\} > (1+H)\alpha \right) \leq \frac{1 + H \mathbb{P}\{U(Z_\sigma) \geq U(Z)\}}{(1+H)\alpha} \\
&\leq \frac{1}{(1+H)\alpha} \left(1 + \frac{H \text{Var}[U(Z_\sigma) - U(Z)]}{\{\mathbb{E}[U(Z_\sigma) - U(Z)]\}^2} \right) \leq \frac{1}{(1+H)\alpha} \left(1 + \frac{H\alpha\beta}{2} \right) \leq \beta.
\end{aligned}$$

Bounding the terms on the right-hand side of (29) is therefore sufficient to establish an upper bound on the minimax separation with respect to the squared shifted maximum mean discrepancy metric. However, since our goal is to characterize the separation in terms of the L^2 distance defined in (13), we must also relate $\text{MMD}_{r, k_\zeta}^2$ to this L^2 norm. The proof proceeds in two main steps: first, we analyze the expectation and

variance terms appearing on the right-hand side of (29) and derive appropriate upper bounds. In the second step, we express the squared shifted maximum mean discrepancy as the sum of the square of the separation metric in (13) and the bias term $\|\psi_r - \varphi_\zeta * \psi_r\|_2^2$, where we define $\varphi_\zeta(u) := \zeta^d \prod_{i=1}^d K(\zeta u_i)$ for $u \in \mathbb{R}^d$, so that $k_\zeta(x, y) = \varphi_\zeta(x - y)$ for all $x, y \in \mathbb{R}^d$. We then control the bias term using the smoothness assumptions associated with the Sobolev class. Combining these two steps yields an upper bound on ρ_r^* . Throughout the following we set $\zeta = n^{\frac{2}{4s+d}}$ so that in particular $n^{-2}\zeta^d = n^{-\frac{8s}{4s+d}} \leq 1$. Also, for parameters $a_1, \dots, a_k$, we denote by $Q(a_1, \dots, a_k)$ a constant that depends only on these parameters. Its value may change from line to line, but it may only depend on $a_1, \dots, a_k$.

• **Mean and variance of $U(Z_\sigma)$**

We begin by analyzing the second moment of the permuted statistic $U(Z_\sigma)$. Throughout this bullet point, we rescale the kernel so that $K(0) = 1$; this normalization is inessential and serves only to simplify notation, and the bound in (36) remains valid after absorbing the resulting scaling factor into Q_0 . Furthermore, sums over i 's are to be intended for i varying in $[n]$, sums over j 's are to be intended for j varying in $\{n+1, \dots, n+m\}$, and sums over k 's are to be intended for k varying in $[n+m]$. The strategy is to relate the distribution of $U_\sigma := U(Z_\sigma)$, where σ is sampled from (3), to the distribution of $U_{\sigma_t} \equiv U_t$, where σ_t is the permutation at time $t \in \mathbb{N}$ of an equivalent version of Algorithm 1 that is initialized at stationarity. More precisely, we consider a procedure that at each time step t samples $i \in [n]$ and $j \in \{n+1, \dots, n+m\}$ uniformly at random, and switches $Z_{\sigma_t(i)}$ with $Z_{\sigma_t(j)}$ with probability

$$\tilde{p}_{i,j}^t := \mathbb{P}\{\text{switch } i \text{ and } j \text{ at time } t \mid i, j \text{ are selected}\} = \frac{\hat{\lambda} n m r_i^t}{(n + \hat{\lambda} m r_i^t)(n + \hat{\lambda} m r_j^t)},$$

where $r_i^t := r(Z_{\sigma_t(i)})$ for all $i \in [n+m]$. As already outlined in Remark 1, this algorithm still targets the distribution (3) since $\tilde{p}_{i,j}^t / \tilde{p}_{j,i}^t = r_i^t / r_j^t$. Write

$$K_{ij}^t := k_\zeta(Z_{\sigma_t(i)}, Z_{\sigma_t(j)}) \text{ and } K_i^t := k_\zeta(Z_{\sigma_t(i)}, \cdot),$$

$$q_i^t := \frac{\hat{\lambda} m r_i^t}{n + \hat{\lambda} m r_i^t} \text{ and } S_n^t := \sum_i q_i^t = \sum_j (1 - q_j^t),$$

where the last equality holds by (8). In particular we have $\tilde{p}_{i,j}^t = q_i^t(1 - q_j^t)$. Observe that under the assumption that $0 < c \leq r(x) \leq C < \infty$ for all $x \in \mathcal{X}$, we have that $C^{-1} \leq \hat{\lambda} \leq c^{-1}$, which implies $mc(mc + nC)^{-1} \leq q_i^t \leq mC(mC + nc)^{-1}$ and $S_n^t \geq nmc(nC + mc)^{-1}$. We define the V-statistic

$$V_t = \frac{(n+m)^2}{n^2 m^2} \left\| \sum_i q_i^t K_i^t - \sum_j (1 - q_j^t) K_j^t \right\|_{\mathcal{H}}^2 =: \|G_t\|_{\mathcal{H}}^2 = \langle G_t, G_t \rangle_{\mathcal{H}},$$

which coincides with (11) evaluated on our data at time t with kernel k_ζ , and will be convenient for our analysis. We further define the U-statistic

$$U_t = V_t - \frac{(n+m)^2 \zeta^d}{n^2 m^2} \sum_i (q_i^t)^2 - \frac{(n+m)^2 \zeta^d}{n^2 m^2} \sum_j (1 - q_j^t)^2$$

$$\begin{aligned}
&= V_t - \frac{(n+m)^2 \zeta^d}{n^2 m^2} \sum_i (q_i^t)^2 - \frac{(n+m)^2 \zeta^d}{n^2 m^2} \sum_j \{1 + (q_j^t)^2 - 2q_j^t\} \\
&= V_t - \frac{(n+m)^2 \zeta^d}{n^2 m^2} \sum_k (q_k^t)^2 + \frac{(n+m)^2 \zeta^d}{n^2 m} - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} \sum_j (1 - q_j^t) \\
&= V_t - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} S_n^t - \underbrace{\frac{(n+m)^2 \zeta^d}{n^2 m^2} \sum_k (q_k^t)^2 + \frac{(n+m)^2 \zeta^d}{n^2 m}}_{\text{permutation independent}},
\end{aligned}$$

which has the same distribution as U_σ for each $t \in \mathbb{N}$. If we swap i and j at time t , then

$$\begin{aligned}
U_{t+1} - U_t &= \left\| G_t + \frac{n+m}{nm} (K_j^t - K_i^t) \right\|_{\mathcal{H}}^2 - \|G_t\|_{\mathcal{H}}^2 - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} (q_j^t - q_i^t) \\
&= \frac{2(n+m)}{nm} \langle G_t, K_j^t - K_i^t \rangle + \frac{(n+m)^2}{n^2 m^2} \|K_j^t - K_i^t\|_{\mathcal{H}}^2 - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} (q_j^t - q_i^t).
\end{aligned}$$

By stationarity, we therefore have

$$\begin{aligned}
0 &= nm \mathbb{E}[U_{t+1}^2 - U_t^2] \\
&= \sum_{i,j} \mathbb{E} \left[q_i^t (1 - q_j^t) \left(\left\{ U_t + \frac{2(n+m)}{nm} \langle G_t, K_j^t - K_i^t \rangle + \frac{(n+m)^2}{n^2 m^2} \|K_j^t - K_i^t\|_{\mathcal{H}}^2 \right. \right. \right. \\
&\quad \left. \left. \left. - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} (q_j^t - q_i^t) \right\}^2 - U_t^2 \right) \right] \\
&= \sum_{i,j} \mathbb{E} \left[q_i^t (1 - q_j^t) \cdot 2U_t \left\{ \frac{2(n+m)}{nm} \langle G_t, K_j^t - K_i^t \rangle + \frac{(n+m)^2}{n^2 m^2} \|K_j^t - K_i^t\|_{\mathcal{H}}^2 - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} (q_j^t - q_i^t) \right\} \right] \\
&\quad + \sum_{i,j} \mathbb{E} \left[q_i^t (1 - q_j^t) \left\{ \frac{2(n+m)}{nm} \langle G_t, K_j^t - K_i^t \rangle + \frac{(n+m)^2}{n^2 m^2} \|K_j^t - K_i^t\|_{\mathcal{H}}^2 - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} (q_j^t - q_i^t) \right\}^2 \right], \tag{30}
\end{aligned}$$

which is the sum of a linear and a quadratic term. In order to simplify this right-hand side we now provide some useful identities. We notice that

$$\begin{aligned}
\sum_{i,j} q_i^t (1 - q_j^t) (K_j^t - K_i^t) &= S_n^t \left\{ \sum_j (1 - q_j^t) K_j^t - \sum_i q_i^t K_i^t \right\} = -\frac{nm}{n+m} S_n^t G_t, \\
\sum_{i,j} q_i^t (1 - q_j^t) \|K_j^t - K_i^t\|_{\mathcal{H}}^2 &= \sum_{i,j} q_i^t (1 - q_j^t) (2\zeta^d - 2K_{ij}^t) = 2\zeta^d (S_n^t)^2 - 2 \sum_{i,j} q_i^t (1 - q_j^t) K_{ij}^t,
\end{aligned}$$

and

$$\sum_{i,j} q_i^t (1 - q_j^t) (q_j^t - q_i^t) = S_n^t \left\{ \sum_j q_j^t (1 - q_j^t) - \sum_i (q_i^t)^2 \right\} = S_n^t \left\{ m - S_n^t - \sum_k (q_k^t)^2 \right\}.$$

Using these identities, the linear term in (30) gives

$$\begin{aligned}
& \sum_{i,j} \mathbb{E} \left[q_i^t (1 - q_j^t) \cdot 2U_t \left\{ \frac{2(n+m)}{nm} \langle G_t, K_j^t - K_i^t \rangle + \frac{(n+m)^2}{n^2 m^2} \|K_j^t - K_i^t\|_{\mathcal{H}}^2 - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} (q_j^t - q_i^t) \right\} \right] \\
&= \mathbb{E} \left[2U_t \left\{ \frac{2(n+m)}{nm} \langle G_t, \sum_{i,j} (K_j^t - K_i^t) q_i^t (1 - q_j^t) \rangle + \frac{(n+m)^2}{n^2 m^2} \sum_{i,j} \|K_j^t - K_i^t\|_{\mathcal{H}}^2 q_i^t (1 - q_j^t) \right. \right. \\
&\quad \left. \left. - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} \sum_{i,j} (q_j^t - q_i^t) q_i^t (1 - q_j^t) \right\} \right] \\
&= \mathbb{E} \left[2U_t \left\{ -2S_n^t \left(U_t + \frac{2(n+m)^2 \zeta^d}{n^2 m^2} S_n^t + \frac{(n+m)^2 \zeta^d}{n^2 m^2} \sum_k (q_k^t)^2 - \frac{(n+m)^2 \zeta^d}{n^2 m} \right) \right. \right. \\
&\quad \left. \left. + \frac{2(n+m)^2 \zeta^d}{n^2 m^2} (S_n^t)^2 - \frac{2(n+m)^2}{n^2 m^2} \sum_{i,j} q_i^t (1 - q_j^t) K_{ij}^t - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} S_n^t \left(m - S_n^t - \sum_k (q_k^t)^2 \right) \right\} \right] \\
&= -4\mathbb{E} [S_n^t U_t^2] - 4\mathbb{E} \left[U_t \frac{(n+m)^2}{n^2 m^2} \sum_{i,j} q_i^t (1 - q_j^t) K_{ij}^t \right]. \tag{31}
\end{aligned}$$

As for the quadratic term, it is useful to define $G_t^{-(i,j)} := G_t - (n+m)n^{-1}m^{-1}\{q_i^t K_i^t - (1 - q_j^t) K_j^t\}$ and write

$$\begin{aligned}
& \frac{2(n+m)}{nm} \langle G_t, K_j^t - K_i^t \rangle + \frac{(n+m)^2}{n^2 m^2} \|K_j^t - K_i^t\|_{\mathcal{H}}^2 - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} (q_j^t - q_i^t) \\
&= \frac{2(n+m)}{nm} \langle G_t - \frac{n+m}{nm} \{q_i^t K_i^t - (1 - q_j^t) K_j^t\}, K_j^t - K_i^t \rangle + \frac{2(n+m)^2}{n^2 m^2} \langle q_i^t K_i^t - (1 - q_j^t) K_j^t, K_j^t - K_i^t \rangle \\
&\quad + \frac{(n+m)^2}{n^2 m^2} \|K_j^t - K_i^t\|_{\mathcal{H}}^2 - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} (q_j^t - q_i^t) \\
&= \frac{2(n+m)}{nm} \langle G_t^{-(i,j)}, K_j^t - K_i^t \rangle + \frac{2(n+m)^2}{n^2 m^2} \langle q_i^t K_i^t - (1 - q_j^t) K_j^t, K_j^t - K_i^t \rangle \\
&\quad + \frac{2(n+m)^2}{n^2 m^2} (\zeta^d - K_{ij}^t) - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} (q_j^t - q_i^t) \\
&= \frac{2(n+m)}{nm} \langle G_t^{-(i,j)}, K_j^t - K_i^t \rangle + \frac{2(n+m)^2}{n^2 m^2} (q_i^t K_{ij}^t - \zeta^d q_i^t - \zeta^d (1 - q_j^t) + (1 - q_j^t) K_{ij}^t) \\
&\quad + \frac{2(n+m)^2}{n^2 m^2} (\zeta^d - K_{ij}^t) - \frac{2(n+m)^2 \zeta^d}{n^2 m^2} (q_j^t - q_i^t) \\
&= \frac{2(n+m)}{nm} \langle G_t^{-(i,j)}, K_j^t - K_i^t \rangle + \frac{2(n+m)^2}{n^2 m^2} (q_i^t - q_j^t) K_{ij}^t. \tag{32}
\end{aligned}$$

Combining (30), (31), (32) gives

$$\begin{aligned}
4\mathbb{E} [S_n^t U_t^2] &= -4\mathbb{E} \left[U_t \frac{(n+m)^2}{n^2 m^2} \sum_{i,j} q_i^t (1 - q_j^t) K_{ij}^t \right] \\
&\quad + \sum_{i,j} \mathbb{E} \left[q_i^t (1 - q_j^t) \left\{ \frac{2(n+m)}{nm} \langle G_t^{-(i,j)}, K_j^t - K_i^t \rangle + \frac{2(n+m)^2}{n^2 m^2} (q_i^t - q_j^t) K_{ij}^t \right\}^2 \right]
\end{aligned}$$

$$\begin{aligned}
&\leq 4\mathbb{E} \left[|U_t| \frac{(n+m)^2}{n^2 m^2} \sum_{i,j} q_i^t (1-q_j^t) |K_{ij}^t| \right] \\
&\quad + \sum_{i,j} \mathbb{E} \left[q_i^t (1-q_j^t) \left\{ \frac{2(n+m)}{nm} \langle G_t^{-(i,j)}, K_j^t - K_i^t \rangle + \frac{2(n+m)^2}{n^2 m^2} (q_i^t - q_j^t) K_{ij}^t \right\}^2 \right] \\
&\leq 4 \sqrt{\mathbb{E}[U_t^2] \mathbb{E} \left[\frac{(n+m)^4}{n^4 m^4} \left\{ \sum_{i,j} q_i^t (1-q_j^t) |K_{ij}^t| \right\}^2 \right]} + \frac{8(n+m)^2}{n^2 m^2} \sum_{i,j} \mathbb{E} \left[q_i^t (1-q_j^t) \langle G_t^{-(i,j)}, K_j^t - K_i^t \rangle^2 \right] \\
&\quad + \frac{8(n+m)^4}{n^4 m^4} \sum_{i,j} \mathbb{E} \left[q_i^t (1-q_j^t) (q_i^t - q_j^t)^2 (K_{ij}^t)^2 \right] \\
&\leq 4 \sqrt{\mathbb{E}[U_t^2] \mathbb{E} \left[\frac{(n+m)^4}{n^3 m^3} \sum_{i,j} \{q_i^t (1-q_j^t) K_{ij}^t\}^2 \right]} + \frac{8(n+m)^2}{n^2 m^2} \sum_{i,j} \mathbb{E} \left[q_i^t (1-q_j^t) \langle G_t^{-(i,j)}, K_j^t - K_i^t \rangle^2 \right] \\
&\quad + \frac{8(n+m)^4}{n^4 m^4} \sum_{i,j} \mathbb{E} \left[q_i^t (1-q_j^t) (q_i^t - q_j^t)^2 (K_{ij}^t)^2 \right] \\
&\leq 12 \max \left\{ \sqrt{\mathbb{E}[U_t^2] \mathbb{E} \left[\frac{(n+m)^4}{n^3 m^3} \sum_{i,j} \{q_i^t (1-q_j^t) K_{ij}^t\}^2 \right]}, \frac{2(n+m)^2}{n^2 m^2} \sum_{i,j} \mathbb{E} \left[q_i^t (1-q_j^t) \langle G_t^{-(i,j)}, K_j^t - K_i^t \rangle^2 \right], \right. \\
&\quad \left. \frac{2(n+m)^4}{n^4 m^4} \sum_{i,j} \mathbb{E} \left[q_i^t (1-q_j^t) (q_i^t - q_j^t)^2 (K_{ij}^t)^2 \right] \right\}. \tag{33}
\end{aligned}$$

We will now bound each of the three quantities inside the maximum separately. If the first quantity reaches the maximum, we can use $m \leq n \leq \tau m$, $q_k^t \in [0, 1]$ and $S_n^t \geq \frac{nm}{nC+mc}$ to show that

$$\begin{aligned}
\frac{nm}{nC+mc} \mathbb{E}[U_t^2] &\leq \mathbb{E}[S_n^t U_t^2] \leq 3 \sqrt{\mathbb{E}[U_t^2] \mathbb{E} \left[\frac{(n+m)^4}{n^3 m^3} \sum_{i,j} \{q_i^t (1-q_j^t) K_{ij}^t\}^2 \right]} \\
&\leq 3 \sqrt{\mathbb{E}[U_t^2] \mathbb{E} \left[\frac{(n+m)^4}{n^3 m^3} \sum_{i,j} (K_{ij}^t)^2 \right]} \leq 3 \sqrt{\mathbb{E}[U_t^2] \mathbb{E} \left[\frac{(n+m)^4}{n^3 m^3} \sum_{k_1 \neq k_2} (K_{k_1, k_2}^t)^2 \right]} \\
&= 3 \sqrt{\mathbb{E}[U_t^2] \mathbb{E} \left[\frac{(n+m)^4}{n^3 m^3} \left\{ \sum_{i_1 \neq i_2} k_\zeta^2(X_{i_1}, X_{i_2}) + \sum_{j_1 \neq j_2} k_\zeta^2(Y_{j_1}, Y_{j_2}) + 2 \sum_{i,j} k_\zeta^2(X_i, Y_j) \right\} \right]} \\
&= 3 \sqrt{\mathbb{E}[U_t^2] \mathbb{E} \left[\frac{(n+m)^4}{n^3 m^3} \left\{ n(n-1)k_\zeta^2(X_1, X_2) + m(m-1)k_\zeta^2(Y_1, Y_2) + 2nmk_\zeta^2(X_1, Y_1) \right\} \right]} \\
&\leq 3 \sqrt{\mathbb{E}[U_t^2] \frac{(n+m)^6}{n^3 m^3} M \kappa_2(d) \zeta^d} \leq 3 \sqrt{(1+\tau)^6 \tau^{-3} \mathbb{E}[U_t^2] M \kappa_2(d) \zeta^d}. \tag{34}
\end{aligned}$$

Note that in the fourth inequality we bound $\sum_{i,j} (K_{ij}^t)^2$ by $\sum_{k_1 \neq k_2} (K_{k_1, k_2}^t)^2$, the latter being more advan-

tageous to analyze due to its permutation invariance. Making (34) explicit for the expectation of U_t^2 shows that there exists a constant $Q_0 \equiv Q_0(c, C, d, \tau, M)$ such that $\mathbb{E}[U_t^2] \leq Q_0 \frac{\zeta^d}{n^2}$, in the case where the first quantity in the maximum in (33) dominates. As for the case when the third quantity dominates, we can argue exactly as in (34) to show that

$$\frac{(n+m)^4}{n^4 m^4} \sum_{i,j} \mathbb{E} [q_i^t (1 - q_j^t) (q_i^t - q_j^t)^2 (K_{ij}^t)^2] \leq \frac{4(n+m)^6}{n^4 m^4} M \kappa_2(d) \zeta^d, \quad (35)$$

which implies that $\mathbb{E}[U_t^2] \leq Q_0 \frac{\zeta^d}{n^3}$. Finally, as for the term involving $G_t^{-(i,j)}$, we have

$$\sum_{i,j} \mathbb{E} \left[q_i^t (1 - q_j^t) \langle G_t^{-(i,j)}, K_j^t - K_i^t \rangle^2 \right] \leq 2 \sum_{i,j} \mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle^2 \right] + 2 \sum_{i,j} \mathbb{E} \left[\langle G_t^{-(i,j)}, K_j^t \rangle^2 \right],$$

and we can analyze the evolution of the right hand side through a version of Algorithm 1 where indices i and j remain fixed. In other words, the Algorithm works the same, but we are just allowed to swap indices $\tilde{i} \neq i$ with $\tilde{j} \neq j$ with probability $q_i^t (1 - q_j^t)$. We show in Lemma 15 that this procedure still preserves the stationary distribution. The two terms above can be bounded by almost identical arguments so we will restrict attention to the first. In analysing the evolution of $\langle G_t^{-(i,j)}, K_i^t \rangle^2$, it is useful to write

$$\begin{aligned} \sum_{\tilde{i} \neq i} \sum_{\tilde{j} \neq j} q_i^t (1 - q_j^t) (K_j^t - K_i^t) &= \left(\sum_{\tilde{i} \neq i} q_i^t \right) \left(\sum_{\tilde{j} \neq j} (1 - q_j^t) K_j^t \right) - \left(\sum_{\tilde{j} \neq j} (1 - q_j^t) \right) \left(\sum_{\tilde{i} \neq i} q_i^t K_i^t \right) \\ &= -\frac{nm}{n+m} S_n^t G_t^{-(i,j)} + (1 - q_j^t) \sum_{\tilde{i} \neq i} q_i^t K_i^t - q_i^t \sum_{\tilde{j} \neq j} (1 - q_j^t) K_j^t. \end{aligned}$$

Under stationarity, we thus have

$$\begin{aligned} 0 &= (n-1)(m-1) \mathbb{E} [\langle G_{t+1}^{-(i,j)}, K_i^t \rangle^2 - \langle G_t^{-(i,j)}, K_i^t \rangle^2] \\ &= \sum_{\tilde{i} \neq i} \sum_{\tilde{j} \neq j} \mathbb{E} \left[q_i^t (1 - q_j^t) \left\{ \langle G_t^{-(i,j)} + \frac{n+m}{nm} (K_j^t - K_i^t), K_i^t \rangle^2 - \langle G_t^{-(i,j)}, K_i^t \rangle^2 \right\} \right] \\ &= \sum_{\tilde{i} \neq i} \sum_{\tilde{j} \neq j} \mathbb{E} \left[\frac{(n+m)^2}{n^2 m^2} q_i^t (1 - q_j^t) \langle K_j^t - K_i^t, K_i^t \rangle^2 \right] + \frac{2(n+m)}{nm} \mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle \left\langle \sum_{\tilde{i} \neq i} \sum_{\tilde{j} \neq j} q_i^t (1 - q_j^t) (K_j^t - K_i^t), K_i^t \right\rangle \right] \\ &= \sum_{\tilde{i} \neq i} \sum_{\tilde{j} \neq j} \mathbb{E} \left[\frac{(n+m)^2}{n^2 m^2} q_i^t (1 - q_j^t) \langle K_j^t - K_i^t, K_i^t \rangle^2 \right] - 2 \mathbb{E} [S_n^t \langle G_t^{-(i,j)}, K_i^t \rangle^2] \\ &\quad + \frac{2(n+m)}{nm} \mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle \left\langle (1 - q_j^t) \sum_{\tilde{i} \neq i} q_i^t K_i^t - q_i^t \sum_{\tilde{j} \neq j} (1 - q_j^t) K_j^t, K_i^t \right\rangle \right] \\ &\leq \frac{2(n+m)^2}{nm^2} \sum_{\tilde{j} \neq j} \mathbb{E} \left[\langle K_j^t, K_i^t \rangle^2 \right] + \frac{2(n+m)^2}{n^2 m} \sum_{\tilde{i} \neq i} \mathbb{E} \left[\langle K_i^t, K_i^t \rangle^2 \right] - 2 \mathbb{E} [S_n^t \langle G_t^{-(i,j)}, K_i^t \rangle^2] \\ &\quad + \frac{2(n+m)}{nm} \mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle \left\langle (1 - q_j^t) \sum_{\tilde{i} \neq i} q_i^t K_i^t - q_i^t \sum_{\tilde{j} \neq j} (1 - q_j^t) K_j^t, K_i^t \right\rangle \right] \end{aligned}$$

$$\begin{aligned}
&\leq \frac{(n+m)^4}{n^2 m^2} \max_{k_1 \neq k_2} \mathbb{E}[k_\zeta^2(Z_{k_1}, Z_{k_2})] - 2\mathbb{E}[S_n^t \langle G_t^{-(i,j)}, K_i^t \rangle^2] \\
&\quad + \frac{2(n+m)}{nm} \mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle \langle (1-q_j^t) \sum_{\tilde{i} \neq i} q_i^t K_i^t - q_i^t \sum_{\tilde{j} \neq j} (1-q_j^t) K_j^t, K_i^t \rangle \right] \\
&\leq \frac{(n+m)^4}{n^2 m^2} M\kappa_2(d)\zeta^d - 2\mathbb{E} \left[S_n^t \langle G_t^{-(i,j)}, K_i^t \rangle^2 \right] \\
&\quad + \frac{2(n+m)}{nm} \sqrt{\mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle^2 \right] \mathbb{E} \left[\langle (1-q_j^t) \sum_{\tilde{i} \neq i} q_i^t K_i^t - q_i^t \sum_{\tilde{j} \neq j} (1-q_j^t) K_j^t, K_i^t \rangle^2 \right]} \\
&\leq \frac{(n+m)^4}{n^2 m^2} M\kappa_2(d)\zeta^d - 2\mathbb{E} \left[S_n^t \langle G_t^{-(i,j)}, K_i^t \rangle^2 \right] \\
&\quad + 2\sqrt{2} \frac{n+m}{nm} \sqrt{\mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle^2 \right] \mathbb{E} \left[\langle \sum_{\tilde{i} \neq i} q_i^t K_i^t, K_i^t \rangle^2 + \langle \sum_{\tilde{j} \neq j} (1-q_j^t) K_j^t, K_i^t \rangle^2 \right]} \\
&\leq \frac{(n+m)^4}{n^2 m^2} M\kappa_2(d)\zeta^d - 2\mathbb{E} \left[S_n^t \langle G_t^{-(i,j)}, K_i^t \rangle^2 \right] + 2\sqrt{2} \frac{(n+m)^2}{nm} \sqrt{\mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle^2 \right] \max_{k_1 \neq k_2} \mathbb{E}[k_\zeta^2(Z_{k_1}, Z_{k_2})]} \\
&\leq \frac{(n+m)^4}{n^2 m^2} M\kappa_2(d)\zeta^d - 2\mathbb{E} \left[S_n^t \langle G_t^{-(i,j)}, K_i^t \rangle^2 \right] + 2\sqrt{2} \frac{(n+m)^2}{nm} \sqrt{M\kappa_2(d)\zeta^d \mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle^2 \right]}.
\end{aligned}$$

Using again the fact that $S_n^t \geq \frac{nmc}{nC+mc}$, we can employ the previous calculations to show that

$$\begin{aligned}
\frac{2nmc}{nC+mc} \mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle^2 \right] &\leq 2\mathbb{E} \left[S_n^t \langle G_t^{-(i,j)}, K_i^t \rangle^2 \right] \\
&\leq \frac{(n+m)^4}{n^2 m^2} M\kappa_2(d)\zeta^d + 2\sqrt{2} \frac{(n+m)^2}{nm} \sqrt{M\kappa_2(d)\zeta^d \mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle^2 \right]} \\
&\leq 2(\tau+1)^4 \tau^{-2} \max \left\{ M\kappa_2(d)\zeta^d, \sqrt{M\kappa_2(d)\zeta^d \mathbb{E} \left[\langle G_t^{-(i,j)}, K_i^t \rangle^2 \right]} \right\}.
\end{aligned}$$

In particular, this yields $\mathbb{E}[\langle G_t^{-(i,j)}, K_i^t \rangle^2] \leq Q_0 \frac{\zeta^d}{n}$, which further implies that $\mathbb{E}[U_t^2] \leq Q_0 \frac{\zeta^d}{n^2}$ also when the second quantity in (33) attains the maximum. Combining this with (34) and (35) gives

$$\max \{ \mathbb{E}^2[|U_\sigma|], \text{Var}[U_\sigma] \} \leq \mathbb{E}[U_\sigma^2] \leq Q_0 \frac{\zeta^d}{n^2}, \quad (36)$$

and concludes the analysis for U_σ .

• Mean and variance of U_{id}

In what follows, we define $U_{\text{id}} := U(Z)$, which targets $\text{MMD}_{r, k_\zeta}^2(f, g)$, and we study its mean and variance. This would be straightforward if we used the normalization factors $\{n(n-1)\}^{-1}$ and $\{m(m-1)\}^{-1}$ instead of n^{-2} and m^{-2} , and if $\hat{\lambda}$ were not random. The former problem is easy to address. Define

$$\tilde{U}_{\text{id}} = \frac{(n+m)^2}{n(n-1)m^2} \sum_{i_1 \neq i_2=1}^n \frac{\hat{\lambda}^2 r(X_{i_1}) r(X_{i_2}) k_\zeta(X_{i_1}, X_{i_2})}{\{\frac{n}{m} + \hat{\lambda} r(X_{i_1})\} \{\frac{n}{m} + \hat{\lambda} r(X_{i_2})\}}$$

$$+ \frac{(n+m)^2}{m^3(m-1)} \sum_{j_1 \neq j_2=1}^m \frac{k_\zeta(Y_{j_1}, Y_{j_2})}{\{\frac{n}{m} + \widehat{\lambda}r(Y_{j_1})\}\{\frac{n}{m} + \widehat{\lambda}r(Y_{j_2})\}} - \frac{2(n+m)^2}{nm^3} \sum_{i=1}^n \sum_{j=1}^m \frac{\widehat{\lambda}r(X_i)k_\zeta(X_i, Y_j)}{\{\frac{n}{m} + \widehat{\lambda}r(X_i)\}\{\frac{n}{m} + \widehat{\lambda}r(Y_j)\}}, \quad (37)$$

and observe that $\tilde{U}_{\text{id}} - U_{\text{id}}$ equals

$$\frac{(n+m)^2}{n^2(n-1)m^2} \sum_{i_1 \neq i_2=1}^n \frac{\widehat{\lambda}^2 r(X_{i_1})r(X_{i_2})k_\zeta(X_{i_1}, X_{i_2})}{\{\frac{n}{m} + \widehat{\lambda}r(X_{i_1})\}\{\frac{n}{m} + \widehat{\lambda}r(X_{i_2})\}} + \frac{(n+m)^2}{m^4(m-1)} \sum_{j_1 \neq j_2=1}^m \frac{k_\zeta(Y_{j_1}, Y_{j_2})}{\{\frac{n}{m} + \widehat{\lambda}r(Y_{j_1})\}\{\frac{n}{m} + \widehat{\lambda}r(Y_{j_2})\}}.$$

Due to the boundedness assumption on r , this implies that there exists a constant $Q_1 \equiv Q_1(c, C, \tau) > 0$ such that

$$\begin{aligned} |\mathbb{E}[U_{\text{id}}] - \text{MMD}_{r, k_\zeta}^2(f, g)| &\leq |\mathbb{E}[\tilde{U}_{\text{id}}] - \text{MMD}_{r, k_\zeta}^2(f, g)| + \mathbb{E}[|U_{\text{id}} - \tilde{U}_{\text{id}}|] \\ &\leq |\mathbb{E}[\tilde{U}_{\text{id}}] - \text{MMD}_{r, k_\zeta}^2(f, g)| + \frac{Q_1}{n^3} \mathbb{E} \left[\sum_{i_1 \neq i_2=1}^n |k_\zeta(X_{i_1}, X_{i_2})| + \sum_{j_1 \neq j_2=1}^m |k_\zeta(Y_{j_1}, Y_{j_2})| \right] \\ &\leq |\mathbb{E}[\tilde{U}_{\text{id}}] - \text{MMD}_{r, k_\zeta}^2(f, g)| + \frac{Q_1 M \kappa_1(d)}{n}, \end{aligned} \quad (38)$$

and

$$\begin{aligned} \text{Var}[U_{\text{id}}] &\leq 2 \text{Var}[\tilde{U}_{\text{id}}] + 2 \text{Var}[U_{\text{id}} - \tilde{U}_{\text{id}}] \\ &\leq 2 \text{Var}[\tilde{U}_{\text{id}}] + \frac{Q_1}{n^2} \left(\max_{i_1 \neq i_2} \mathbb{E}[k_\zeta^2(X_{i_1}, X_{i_2})] + \max_{j_1 \neq j_2} \mathbb{E}[k_\zeta^2(Y_{j_1}, Y_{j_2})] \right) \leq 2 \text{Var}[\tilde{U}_{\text{id}}] + \frac{Q_1 M \kappa_2(d) \zeta^d}{n^2}. \end{aligned} \quad (39)$$

Since $\zeta \geq 1$, we have $1/n \leq \zeta^{d/2}/n = \sqrt{\zeta^d/n^2}$, and therefore the additional mean term in (38) is of no larger order than the second-moment scale given in (36), namely $\sqrt{\mathbb{E}[U_{\text{id}}^2]} \lesssim \sqrt{\zeta^d/n^2}$. Similarly, the additional variance term in (39) matches the order in (36). Thus, it suffices to control the moments of $\tilde{U}_{\text{id}}$.

• Mean of $\tilde{U}_{\text{id}}$

We now address the harder problem of having $\widehat{\lambda}$ in $\tilde{U}_{\text{id}}$, instead of the non-random quantity λ_0 appearing in the definition of $\text{MMD}_{r, k_\zeta}^2(f, g)$. To overcome this issue, for $\lambda > 0$ define

$$\begin{aligned} G(\lambda) &:= \frac{(n+m)^2}{n(n-1)m^2} \sum_{i_1 \neq i_2=1}^n \frac{\lambda^2 r(X_{i_1})r(X_{i_2})k_\zeta(X_{i_1}, X_{i_2})}{\{n/m + \lambda r(X_{i_1})\}\{n/m + \lambda r(X_{i_2})\}} \\ &\quad + \frac{(n+m)^2}{m^3(m-1)} \sum_{j_1 \neq j_2=1}^m \frac{k_\zeta(Y_{j_1}, Y_{j_2})}{\{n/m + \lambda r(Y_{j_1})\}\{n/m + \lambda r(Y_{j_2})\}} \\ &\quad - \frac{2(n+m)^2}{nm^3} \sum_{i=1}^n \sum_{j=1}^m \frac{\lambda r(X_i)k_\zeta(X_i, Y_j)}{\{n/m + \lambda r(X_i)\}\{n/m + \lambda r(Y_j)\}}, \end{aligned} \quad (40)$$

so that $\tilde{U}_{\text{id}} = G(\widehat{\lambda})$. By expanding G around λ_0 using a Taylor sum up to the second order we get $G(\widehat{\lambda}) = G(\lambda_0) + (\widehat{\lambda} - \lambda_0) G'(\lambda_0) + \frac{1}{2}(\widehat{\lambda} - \lambda_0)^2 G''(\tilde{\lambda})$, where the random $\tilde{\lambda}$ is in between $\widehat{\lambda}$ and λ_0 . We can use this

identity to bound the mean of $\tilde{U}_{\text{id}}$ as follows:

$$\begin{aligned}
& |\mathbb{E}[G(\hat{\lambda})] - \text{MMD}_{r, k_\zeta}^2(f, g)| \\
&= |\mathbb{E}[G(\lambda_0)] - \text{MMD}_{r, k_\zeta}^2(f, g) + \mathbb{E}[(\hat{\lambda} - \lambda_0)G'(\lambda_0)] + \frac{1}{2}\mathbb{E}[(\hat{\lambda} - \lambda_0)^2G''(\check{\lambda})]| \\
&\leq |\mathbb{E}[(\hat{\lambda} - \lambda_0)G'(\lambda_0)]| + \left| \frac{1}{2}\mathbb{E}[(\hat{\lambda} - \lambda_0)^2G''(\check{\lambda})] \right| \leq \sqrt{\mathbb{E}[(\hat{\lambda} - \lambda_0)^2]\mathbb{E}[\{G'(\lambda_0)\}^2]} + \frac{1}{2}\sqrt{\mathbb{E}[(\hat{\lambda} - \lambda_0)^4]\mathbb{E}[\{G''(\check{\lambda})\}^2]} \\
&\leq Q_1\sqrt{\frac{1}{n}\mathbb{E}[\{G'(\lambda_0)\}^2]} + Q_1\sqrt{\frac{1}{n^2}\mathbb{E}[\{G''(\check{\lambda})\}^2]} \\
&= Q_1\sqrt{\frac{1}{n}\{\mathbb{E}[G'(\lambda_0)]\}^2 + \frac{1}{n}\text{Var}[G'(\lambda_0)]} + Q_1\sqrt{\frac{1}{n^2}\mathbb{E}[\{G''(\check{\lambda})\}^2]}. \tag{41}
\end{aligned}$$

Note that in the first bound we used the triangle inequality together with the fact that the U-statistic $G(\lambda_0)$ is unbiased for $\text{MMD}_{r, k_\zeta}^2$, while in the third one we used Lemma 14. Now, as for the last term, we can use the fact that $0 < c \leq r(\cdot) \leq C$ to show that the first and second derivatives of $\lambda \mapsto \frac{\lambda^j}{\{n/m + \lambda r(x)\}\{n/m + \lambda r(y)\}}$ are uniformly bounded for all $\lambda > 0$, $x, y \in \mathbb{R}^d$ and $j \in \{0, 1, 2\}$. We can thus argue as we did for the second term in (39) to get

$$\mathbb{E}[\{G''(\check{\lambda})\}^2] \leq Q_1 \max_{k_1 \neq k_2} \mathbb{E}[k_\zeta^2(Z_{k_1}, Z_{k_2})] \leq Q_1 M \kappa_2(d) \zeta^d. \tag{42}$$

For the first two terms, observe that $G'(\lambda_0)$ is a two-sample second-order U-statistic with asymmetric kernel $\check{h}(x_1, x_2, y_1, y_2) = b_{XX}(x_1, x_2)k_\zeta(x_1, x_2) + b_{YY}(y_1, y_2)k_\zeta(y_1, y_2) + b_{XY}(x_1, y_2)k_\zeta(x_1, y_2) + b_{XY}(x_2, y_1)k_\zeta(x_2, y_1)$, where

$$\begin{cases} b_{XX}(x_1, x_2) := \frac{2\lambda_0(n+m)^2 r(x_1)r(x_2)}{(n+\lambda_0 m r(x_1))(n+\lambda_0 m r(x_2))} - \frac{\lambda_0^2 m(n+m)^2 r^2(x_1)r(x_2)}{(n+\lambda_0 m r(x_1))^2(n+\lambda_0 m r(x_2))} \\ \quad - \frac{\lambda_0^2 m(n+m)^2 r(x_1)r^2(x_2)}{(n+\lambda_0 m r(x_1))(n+\lambda_0 m r(x_2))^2} \\ b_{XY}(x_1, y_2) := -\frac{(n+m)^2 r(x_1)}{(n+\lambda_0 m r(x_1))(n+\lambda_0 m r(y_2))} + \frac{\lambda_0 m(n+m)^2 r^2(x_1)}{(n+\lambda_0 m r(x_1))^2(n+\lambda_0 m r(y_2))} \\ \quad + \frac{\lambda_0 m(n+m)^2 r(x_1)r(y_2)}{(n+\lambda_0 m r(x_1))(n+\lambda_0 m r(y_2))^2} \\ b_{YY}(y_1, y_2) := -\frac{m(n+m)^2 r(y_1)}{(n+\lambda_0 m r(y_1))^2(n+\lambda_0 m r(y_2))} - \frac{m(n+m)^2 r(y_2)}{(n+\lambda_0 m r(y_1))(n+\lambda_0 m r(y_2))^2}. \end{cases}$$

This is to say that

$$G'(\lambda_0) = \frac{1}{n(n-1)m(m-1)} \sum_{i_1 \neq i_2} \sum_{j_1 \neq j_2} \check{h}(X_{i_1}, X_{i_2}, Y_{j_1}, Y_{j_2}).$$

We now bound the mean and the variance of $G'(\lambda_0)$, starting from the latter. Since $\check{h}$ is not symmetric in (x_1, x_2) and (y_1, y_2) , we introduce its within-sample symmetrization

$$\bar{h}(x_1, x_2, y_1, y_2) = \frac{1}{2!2!} \sum_{\pi \in \mathcal{S}_2} \sum_{\tau \in \mathcal{S}_2} \check{h}(x_{\pi(1)}, x_{\pi(2)}, y_{\tau(1)}, y_{\tau(2)}),$$

where $\mathcal{S}_2$ denotes the symmetric group on two elements. Because $G'(\lambda_0)$ averages $\check{h}(X_{i_1}, X_{i_2}, Y_{j_1}, Y_{j_2})$ over all ordered pairs $i_1 \neq i_2$ and $j_1 \neq j_2$, replacing $\check{h}$ by $\bar{h}$ leaves the statistic unchanged. We can thus express $G'(\lambda_0)$ as a two-sample U-statistic with a kernel symmetric within each sample, so that we may apply the classical variance formula for two-sample U-statistics in [Lee \(1990, Equation 2, pag. 38\)](#). Following the simplification in [Kim et al. \(2022, Equation \(69\)\)](#), we obtain

$$\text{Var}[G'(\lambda_0)] \leq Q_2 \left\{ \frac{\bar{\sigma}_{10}^2}{n} + \frac{\bar{\sigma}_{01}^2}{m} + \left(\frac{1}{n} + \frac{1}{m} \right)^2 \bar{\sigma}_{22}^2 \right\} \leq Q_2 \left\{ \frac{\check{\sigma}_{10}^2}{n} + \frac{\check{\sigma}_{01}^2}{m} + \left(\frac{1}{n} + \frac{1}{m} \right)^2 \check{\sigma}_{22}^2 \right\}$$

for a sufficiently large universal constant $Q_2 > 0$, where $\check{\sigma}_{10}^2 = \text{Var}_{X_1}[\mathbb{E}_{X_2, Y_1, Y_2}\{\check{h}(X_1, X_2, Y_1, Y_2)\}]$, $\check{\sigma}_{01}^2 = \text{Var}_{Y_1}[\mathbb{E}_{X_1, X_2, Y_2}\{\check{h}(X_1, X_2, Y_1, Y_2)\}]$, $\check{\sigma}_{22}^2 = \text{Var}_{X_1, X_2, Y_1, Y_2}[\check{h}(X_1, X_2, Y_1, Y_2)]$, and $\bar{\sigma}_{10}^2$, $\bar{\sigma}_{01}^2$, $\bar{\sigma}_{22}^2$ are defined analogously with $\bar{h}$ in place of $\check{h}$. In particular, the final inequality uses that $\bar{h}$ is the within-sample permutation average of $\check{h}$; therefore, by Jensen's inequality, the corresponding projection variances for $\bar{h}$ are bounded up to universal constants by those for $\check{h}$, i.e. $\bar{\sigma}_{10}^2 \lesssim \check{\sigma}_{10}^2$, $\bar{\sigma}_{01}^2 \lesssim \check{\sigma}_{01}^2$, and $\bar{\sigma}_{22}^2 \leq \check{\sigma}_{22}^2$. Now, using again the boundedness assumption on $r(\cdot)$ we have that $\max\{|b_{XX}(\cdot, \cdot)|, |b_{YY}(\cdot, \cdot)|, |b_{XY}(\cdot, \cdot)|\} \leq Q_1$, which can be used to show that $\check{\sigma}_{22}^2 \leq Q_1 M \kappa_2(d) \zeta^d$ arguing as in [\(42\)](#). Furthermore,

$$\begin{aligned} \check{\sigma}_{10}^2 &= \text{Var}_{X_1}[\mathbb{E}_{X_2}\{b_{XX}(X_1, X_2)k_\zeta(X_1, X_2)\} + \mathbb{E}_{Y_2}\{b_{XY}(X_1, Y_2)k_\zeta(X_1, Y_2)\}] \\ &\leq 2\mathbb{E}_{X_1}[\mathbb{E}_{X_2}^2\{b_{XX}(X_1, X_2)k_\zeta(X_1, X_2)\} + \mathbb{E}_{Y_2}^2\{b_{XY}(X_1, Y_2)k_\zeta(X_1, Y_2)\}] \\ &\leq 2\mathbb{E}_{X_1, X_2}[b_{XX}^2(X_1, X_2)k_\zeta^2(X_1, X_2)] + 2\mathbb{E}_{X_1, Y_2}[b_{XY}^2(X_1, Y_2)k_\zeta^2(X_1, Y_2)] \leq Q_1 M \kappa_2(d) \zeta^d. \end{aligned}$$

The same holds true for $\check{\sigma}_{01}^2$, and shows that $n^{-1} \text{Var}[G'(\lambda_0)] \leq Q_1 M \kappa_2(d) n^{-2} \zeta^d$. Finally, we bound the expectation of $G'(\lambda_0)$ by $\|\varphi_\zeta * \psi_r\|_2$ up to constants, where $*$ stands for the convolution operator. Recalling $\varphi_\zeta(x - y) = k_\zeta(x, y)$ and $\psi_r = \frac{n+m}{n+\lambda_0 m r}(\lambda_0 r f - g)$, we have

$$\begin{aligned} \mathbb{E}[G'(\lambda_0)] &= \mathbb{E}[b_{XX}(X_1, X_2)k_\zeta(X_1, X_2)] + \mathbb{E}[b_{YY}(Y_1, Y_2)k_\zeta(Y_1, Y_2)] + 2\mathbb{E}[b_{XY}(X, Y)k_\zeta(X, Y)] \\ &= -\mathbb{E}\left[\frac{2m(n+m)^2\lambda_0 r^2(X_1)r(X_2)\varphi_\zeta(X_1 - X_2)}{(n+\lambda_0 m r(X_1))^2(n+\lambda_0 m r(X_2))}\right] + \mathbb{E}\left[\frac{2(n+m)^2\lambda_0 r(X_1)r(X_2)\varphi_\zeta(X_1 - X_2)}{(n+\lambda_0 m r(X_1))(n+\lambda_0 m r(X_2))}\right] \\ &\quad - \mathbb{E}\left[\frac{2m(n+m)^2r(Y_1)\varphi_\zeta(Y_1 - Y_2)}{(n+\lambda_0 m r(Y_1))^2(n+\lambda_0 m r(Y_2))}\right] - \mathbb{E}\left[\frac{2(n+m)^2r(X)\varphi_\zeta(X - Y)}{(n+\lambda_0 m r(X))(n+\lambda_0 m r(Y))}\right] \\ &\quad + \mathbb{E}\left[\frac{2m(n+m)^2\lambda_0 r^2(X)\varphi_\zeta(X - Y)}{(n+\lambda_0 m r(X))^2(n+\lambda_0 m r(Y))}\right] + \mathbb{E}\left[\frac{2m(n+m)^2\lambda_0 r(X)r(Y)\varphi_\zeta(X - Y)}{(n+\lambda_0 m r(X))(n+\lambda_0 m r(Y))^2}\right] \\ &= \int_{\mathbb{R}^d} \frac{2(n+m)r(x)f(x)}{(n+\lambda_0 m r(x))} \left\{ \int_{\mathbb{R}^d} \varphi_\zeta(x-y) \left(\frac{\lambda_0(n+m)r(y)f(y)}{(n+\lambda_0 m r(y))} - \frac{(n+m)g(y)}{(n+\lambda_0 m r(y))} \right) dy \right\} dx \\ &\quad - \int_{\mathbb{R}^d} \frac{2m(n+m)\lambda_0 r^2(x)f(x)}{(n+\lambda_0 m r(x))^2} \left\{ \int_{\mathbb{R}^d} \varphi_\zeta(x-y) \left(\frac{\lambda_0(n+m)r(y)f(y)}{(n+\lambda_0 m r(y))} - \frac{(n+m)g(y)}{(n+\lambda_0 m r(y))} \right) dy \right\} dx \\ &\quad + \int_{\mathbb{R}^d} \frac{2m(n+m)r(x)g(x)}{(n+\lambda_0 m r(x))^2} \left\{ \int_{\mathbb{R}^d} \varphi_\zeta(x-y) \left(\frac{\lambda_0(n+m)r(y)f(y)}{(n+\lambda_0 m r(y))} - \frac{(n+m)g(y)}{(n+\lambda_0 m r(y))} \right) dy \right\} dx \\ &= \int_{\mathbb{R}^d} \left\{ \frac{2(n+m)r(x)f(x)}{(n+\lambda_0 m r(x))} - \frac{2m(n+m)\lambda_0 r^2(x)f(x)}{(n+\lambda_0 m r(x))^2} + \frac{2m(n+m)r(x)g(x)}{(n+\lambda_0 m r(x))^2} \right\} (\varphi_\zeta * \psi_r)(x) dx \\ &\leq \int_{\mathbb{R}^d} \frac{2(n+m)r(x)\{nf(x) + mg(x)\}}{(n+\lambda_0 m r(x))^2} |(\varphi_\zeta * \psi_r)(x)| dx \leq Q_0 \|\varphi_\zeta * \psi_r\|_2. \end{aligned} \tag{43}$$

In particular the last step follows by combining the Cauchy–Schwartz inequality with the fact that the densities f, g satisfy $\max(\|f\|_\infty, \|g\|_\infty) \leq M$, and $n/m + \lambda_0 r$ is bounded away from zero. Combining this with (38), (41) and (42) enables to conclude the analysis of the first moment of U_{id} , showing that $|\mathbb{E}[U_{\text{id}}] - \text{MMD}_{r, k_\zeta}^2(f, g)| \leq Q_0 \sqrt{n^{-2} \zeta^d + n^{-1} \|\varphi_\zeta * \psi_r\|_2^2}$, and thus

$$|\mathbb{E}[U(Z)] - \text{MMD}_{r, k_\zeta}^2(f, g)| \leq Q_0 \sqrt{n^{-2} \zeta^d + n^{-1} \|\varphi_\zeta * \psi_r\|_2^2}. \quad (44)$$

• Variance of $\tilde{U}_{\text{id}}$

Using the second-order Taylor approximation of $G(\hat{\lambda})$ around λ_0 gives

$$\begin{aligned} \text{Var}[\tilde{U}_{\text{id}}] &\leq 3 \text{Var}[G(\lambda_0)] + 3 \text{Var}[(\hat{\lambda} - \lambda_0) G'(\lambda_0)] + \frac{3}{4} \text{Var}[(\hat{\lambda} - \lambda_0)^2 G''(\check{\lambda})] \\ &\leq Q_0(n^{-2} \zeta^d + n^{-1} \|\varphi_\zeta * \psi_r\|_2^2) + 3 \mathbb{E}[(\hat{\lambda} - \lambda_0)^2 \{G'(\lambda_0)\}^2] + \frac{3}{4} \mathbb{E}[(\hat{\lambda} - \lambda_0)^4 \{G''(\check{\lambda})\}^2] \\ &\leq Q_0(n^{-2} \zeta^d + n^{-1} \|\varphi_\zeta * \psi_r\|_2^2) + 3 \sqrt{\mathbb{E}[(\hat{\lambda} - \lambda_0)^4] \mathbb{E}[\{G'(\lambda_0)\}^4]} + \frac{3}{4} \sqrt{\mathbb{E}[(\hat{\lambda} - \lambda_0)^8] \mathbb{E}[\{G''(\check{\lambda})\}^4]} \\ &\leq Q_0(n^{-2} \zeta^d + n^{-1} \|\varphi_\zeta * \psi_r\|_2^2) + Q_1 \sqrt{n^{-2} \mathbb{E}[\{G'(\lambda_0)\}^4]} + Q_1 \sqrt{n^{-4} \mathbb{E}[\{G''(\check{\lambda})\}^4]}. \end{aligned} \quad (45)$$

Note that the second inequality can be proved following similar lines as in [Schrab et al. \(2023, Proposition 3\)](#), while the last one follows from Lemma 14. Similarly to (42) and (43), we need to control fourth-order moments of some derivatives of G . Starting from the term involving G'' , we can argue similarly to (42) to show $\mathbb{E}[\{G''(\check{\lambda})\}^4] \leq Q_1 n^{-8} \mathbb{E}[\{\sum_{k_1 \neq k_2} |k_\zeta(Z_{k_1}, Z_{k_2})|\}^4] \leq Q_1 n^{-4} \mathbb{E}[\sum_{k_1 \neq k_2} \sum_{k_3 \neq k_4} k_\zeta^2(Z_{k_1}, Z_{k_2}) k_\zeta^2(Z_{k_3}, Z_{k_4})]$. It is now just a matter of counting what is the contribution of each term in the sum, depending on how many indices are shared. In this regard, observe that we have $\mathcal{O}(n^4)$ terms like $\mathbb{E}[k_\zeta^2(Z_1, Z_2) k_\zeta^2(Z_3, Z_4)] = \mathbb{E}[k_\zeta^2(Z_1, Z_2)] \mathbb{E}[k_\zeta^2(Z_3, Z_4)] \leq M^2 \kappa_2^2(d) \zeta^{2d}$, $\mathcal{O}(n^2)$ terms like $\mathbb{E}[k_\zeta^4(Z_1, Z_2)] \leq M \kappa_4(d) \zeta^{3d}$ and $\mathcal{O}(n^3)$ terms like $\mathbb{E}[k_\zeta^2(Z_1, Z_2) k_\zeta^2(Z_1, Z_3)] = \mathbb{E}[\mathbb{E}[k_\zeta^2(Z_1, Z_2) | Z_1] \mathbb{E}[k_\zeta^2(Z_1, Z_3) | Z_1]] \leq M^2 \kappa_2^2(d) \zeta^{2d}$. Since $\zeta^d/n^2 = n^{-\frac{8s}{4s+d}} \leq 1$ for our specific choice of ζ , this suffices to show that $\sqrt{n^{-4} \mathbb{E}[\{G''(\check{\lambda})\}^4]} \leq Q_0 n^{-2} \zeta^d$.

As for the term involving the first derivative of G , we have $\mathbb{E}[\{G'(\lambda_0)\}^4] \leq 8 \mathbb{E}[\{G'(\lambda_0) - \mathbb{E}G'(\lambda_0)\}^4] + 8 \{\mathbb{E}G'(\lambda_0)\}^4 \leq 8 \mathbb{E}[\{G'(\lambda_0) - \mathbb{E}G'(\lambda_0)\}^4] + Q_1 \|\varphi_\zeta * \psi_r\|_2^4$. Hence, it just remains to bound the first term on the right-hand side. Define $h_{XX}(X_1, X_2) := b_{XX}(X_1, X_2) k_\zeta(X_1, X_2) - \mathbb{E}[b_{XX}(X_1, X_2) k_\zeta(X_1, X_2)]$ and similarly for h_{YY} and h_{XY} . We thus have

$$\begin{aligned} &\mathbb{E}[\{G'(\lambda_0) - \mathbb{E}G'(\lambda_0)\}^4] \\ &= \mathbb{E} \left[\left\{ \frac{1}{n(n-1)} \sum_{i_1 \neq i_2} h_{XX}(X_{i_1}, X_{i_2}) + \frac{1}{m(m-1)} \sum_{j_1 \neq j_2} h_{YY}(Y_{j_1}, Y_{j_2}) + \frac{2}{nm} \sum_{i,j} h_{XY}(X_i, Y_j) \right\}^4 \right] \\ &\lesssim \frac{1}{n^8} \left(\mathbb{E} \left[\left\{ \sum_{i_1 \neq i_2} h_{XX}(X_{i_1}, X_{i_2}) \right\}^4 \right] + \mathbb{E} \left[\left\{ \sum_{j_1 \neq j_2} h_{YY}(Y_{j_1}, Y_{j_2}) \right\}^4 \right] + \mathbb{E} \left[\left\{ \sum_{i,j} h_{XY}(X_i, Y_j) \right\}^4 \right] \right). \end{aligned} \quad (46)$$

These three terms can be bounded by almost identical arguments so we focus on the first. We expand it as $n^{-8} \sum_{i_1 \neq i_2} \sum_{i_3 \neq i_4} \sum_{i_5 \neq i_6} \sum_{i_7 \neq i_8} \mathbb{E}[h_{XX}(X_{i_1}, X_{i_2}) h_{XX}(X_{i_3}, X_{i_4}) h_{XX}(X_{i_5}, X_{i_6}) h_{XX}(X_{i_7}, X_{i_8})]$ and use a combinatorial argument to derive an upper bound. In this regard, we have $\mathcal{O}(n^8)$ terms with all distinct

indices and $\mathcal{O}(n^7)$ terms with seven distinct indices, but they do not contribute to the sum since their expectations are zero. This is due to the independence among the X 's and the fact that $\mathbb{E}[h_{XX}(X_1, X_2)] = 0$. Moreover, we have

$$\begin{aligned} \mathbb{E}[h_{XX}(X_{i_1}, X_{i_2})h_{XX}(X_{i_3}, X_{i_4})h_{XX}(X_{i_5}, X_{i_6})h_{XX}(X_{i_7}, X_{i_8})] &\leq \mathbb{E}[h_{XX}^4(X_1, X_2)] \\ &= \mathbb{E}[\{b_{XX}(X_1, X_2)k_\zeta(X_1, X_2) - \mathbb{E}[b_{XX}(X_1, X_2)k_\zeta(X_1, X_2)]\}^4] \leq 16\mathbb{E}[b_{XX}^4(X_1, X_2)k_\zeta^4(X_1, X_2)] \\ &\leq Q_1\mathbb{E}[k_\zeta^4(X_1, X_2)] \leq Q_1M\kappa_4(d)\zeta^{3d}, \end{aligned}$$

where in first inequality we repeatedly used the Cauchy–Schwarz inequality, while in the second inequality we used the fact that for X, X' independent and identically distributed we have $\mathbb{E}[(X - \mathbb{E}X)^4] \leq 16\mathbb{E}[X^4]$. Based on this, the contribution of the $\mathcal{O}(n^4)$ terms with four or less distinct indices is bounded above by $n^{-4}\zeta^{3d}$. It remains to bound those terms in which there are five or six different indices. As for the latter case, the only non-zero terms are those where the product decomposes into two independent blocks, each block containing two terms sharing one index; up to relabelling, a representative is of the form

$$\begin{aligned} &\mathbb{E}[h_{XX}(X_1, X_2)h_{XX}(X_1, X_3)h_{XX}(X_4, X_5)h_{XX}(X_4, X_6)] \\ &= \mathbb{E}[h_{XX}(X_1, X_2)h_{XX}(X_1, X_3)]\mathbb{E}[h_{XX}(X_4, X_5)h_{XX}(X_4, X_6)] = \mathbb{E}^2[h_{XX}(X_1, X_2)h_{XX}(X_1, X_3)] \\ &= \mathbb{E}^2[\mathbb{E}[h_{XX}(X_1, X_2) \mid X_1]\mathbb{E}[h_{XX}(X_1, X_3) \mid X_1]] \leq \mathbb{E}^2[\mathbb{E}^2[h_{XX}(X_1, X_2) \mid X_1]] \\ &= \mathbb{E}^2[\mathbb{E}^2[\{b_{XX}(X_1, X_2)k_\zeta(X_1, X_2) - \mathbb{E}[b_{XX}(X_1, X_2)k_\zeta(X_1, X_2)]\} \mid X_1]] \\ &\leq 8\mathbb{E}^2[\mathbb{E}^2[b_{XX}(X_1, X_2)k_\zeta(X_1, X_2) \mid X_1]] + 8\mathbb{E}^4[b_{XX}(X_1, X_2)k_\zeta(X_1, X_2)] \\ &\leq Q_1\mathbb{E}^2[\mathbb{E}^2[|k_\zeta(X_1, X_2)| \mid X_1]] + Q_1\mathbb{E}^4[|k_\zeta(X_1, X_2)|] \leq Q_1M^4\kappa_1^4(d), \end{aligned}$$

and since there are $\mathcal{O}(n^6)$ of these terms, their contribution is of the order n^{-2} . Finally, when there are exactly five distinct indices we can have three different typical terms:

1. $\mathbb{E}[h_{XX}(X_1, X_2)h_{XX}(X_1, X_3)h_{XX}(X_1, X_4)h_{XX}(X_1, X_5)] = \mathbb{E}[\mathbb{E}^4[h_{XX}(X_1, X_2) \mid X_1]] \leq Q_1M^4\kappa_1^4(d);$
2. $\begin{aligned} &\mathbb{E}[h_{XX}(X_1, X_2)h_{XX}(X_1, X_3)h_{XX}(X_1, X_4)h_{XX}(X_2, X_5)] \\ &= \mathbb{E}[\mathbb{E}[h_{XX}(X_1, X_2)h_{XX}(X_1, X_3)h_{XX}(X_1, X_4)h_{XX}(X_2, X_5) \mid X_1, X_2]] \\ &\leq \mathbb{E}[|h_{XX}(X_1, X_2)| \cdot \mathbb{E}^2[|h_{XX}(X_1, X_3)| \mid X_1] \cdot \mathbb{E}[|h_{XX}(X_2, X_5)| \mid X_2]] \leq Q_1M^4\kappa_1^4(d); \end{aligned}$
3. $\begin{aligned} &\mathbb{E}[h_{XX}^2(X_1, X_2)h_{XX}(X_3, X_4)h_{XX}(X_3, X_5)] = \mathbb{E}[h_{XX}^2(X_1, X_2)]\mathbb{E}[h_{XX}(X_3, X_4)h_{XX}(X_3, X_5)] \\ &= \mathbb{E}[h_{XX}^2(X_1, X_2)]\mathbb{E}[\mathbb{E}^2[h_{XX}(X_3, X_4) \mid X_3]] \leq Q_1M^3\kappa_1^2(d)\kappa_2(d)\zeta^d. \end{aligned}$

There are $\mathcal{O}(n^5)$ of these terms, hence their contribution is of the order $\zeta^d n^{-3} \leq \zeta^{2d} n^{-3} \leq \zeta^{2d} n^{-2}$. This is enough to show that $\sqrt{n^{-2}(n^{-8}\mathbb{E}[\sum_{i_1 \neq i_2} h_{XX}(X_{i_1}, X_{i_2}))^4])} \leq Q_0(n^{-2}\zeta^d + n^{-1}\|\varphi_\zeta * \psi_r\|_2^2)$ when combined with $\zeta^d/n^2 \leq 1$. This concludes the analysis for h_{XX} ; analogous bounds can be obtained for h_{YY} and h_{XY} . Overall, we thus have that $\sqrt{n^{-2}\mathbb{E}[\{G'(\lambda_0)\}^4]} \leq Q_0(n^{-2}\zeta^d + n^{-1}\|\varphi_\zeta * \psi_r\|_2^2)$, which further shows that

$$\text{Var}[U_{\text{id}}] \leq Q_0 \left\{ \frac{\zeta^d}{n^2} + \frac{\|\varphi_\zeta * \psi_r\|_2^2}{n} \right\}$$

when combined with (39), (45), and (46).

• **Relating the squared shifted maximum mean discrepancy to the L^2 distance**

Applying the previous argument with (29), (36) and (44) and using the fact that $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ for $x, y \geq 0$ shows that a uniform control of the type I and type II errors is possible whenever

$$\text{MMD}_{r,k_\zeta}^2(f, g) \geq Q_0 \left\{ \frac{\zeta^{d/2}}{n} + \frac{\|\varphi_\zeta * \psi_r\|_2}{\sqrt{n}} \right\}.$$

We now conclude the proof by relating the squared shifted maximum mean discrepancy metric to the L^2 distance defined in (13), thus providing an upper bound on ρ_r^* . We have

$$\begin{aligned} \text{MMD}_{r,k_\zeta}^2(f, g) &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \varphi_\zeta(x-y) \psi_r(x) \psi_r(y) dx dy = \int_{\mathbb{R}^d} \psi_r(x) (\varphi_\zeta * \psi_r)(x) dx \\ &= \langle \psi_r, \varphi_\zeta * \psi_r \rangle_2 = \frac{1}{2} (\|\psi_r\|_2^2 + \|\varphi_\zeta * \psi_r\|_2^2 - \|\psi_r - \varphi_\zeta * \psi_r\|_2^2), \end{aligned}$$

hence an equivalent sufficient condition to bound the total error is given by

$$\|\psi_r\|_2^2 \geq \|\psi_r - \varphi_\zeta * \psi_r\|_2^2 + Q_0 \frac{\zeta^{d/2}}{n} + Q_0 \frac{\|\varphi_\zeta * \psi_r\|_2}{\sqrt{n}} - \|\varphi_\zeta * \psi_r\|_2^2.$$

This can be further simplified to just $\|\psi_r\|_2^2 \geq \|\psi_r - \varphi_\zeta * \psi_r\|_2^2 + Q_0 n^{-1} \zeta^{d/2}$ in light of the fact that

$$\sqrt{\frac{Q_0^2 \|\varphi_\zeta * \psi_r\|_2^2}{n}} - \|\varphi_\zeta * \psi_r\|_2^2 \leq \frac{Q_0^2}{n} + \|\varphi_\zeta * \psi_r\|_2^2 - \|\varphi_\zeta * \psi_r\|_2^2 \leq \frac{Q_0^2 \zeta^{d/2}}{n},$$

as $\zeta \geq 1$ and $\sqrt{xy} \leq x+y$ for $x, y \geq 0$. Furthermore, we can argue as in [Schrab et al. \(2023, Theorem 6\)](#) and show that if $\psi_r \in \mathcal{S}_d^s(L)$ we have $\|\psi_r - \varphi_\zeta * \psi_r\|_2^2 \leq Q_3 \|\psi_r\|_2^2 + Q_4 \zeta^{-2s}$, for some constants $Q_3 \in (0, 1)$ and $Q_4 \equiv Q_4(d, s, L) > 0$. This shows that for $\zeta = n^{\frac{2}{4s+d}}$ there exists a constant $C_1 \equiv C_1(c, C, d, \tau, M, s, L, \alpha, \beta)$ such that

$$\rho_r^* \leq C_1 \sqrt{\frac{\zeta^{d/2}}{n} + \zeta^{-2s}} = C_1 n^{-\frac{2s}{4s+d}},$$

and completes the proof. $\square$

Lemma 14. Assume $m \leq n \leq \tau m$ for $\tau \geq 1$, and $0 < c \leq r(x) \leq C$ for all $x \in \mathcal{X}$. Let $\hat{\lambda}$ be such that

$$\sum_{i=1}^n \frac{\hat{\lambda} m r(X_i)}{n + \hat{\lambda} m r(X_i)} = \sum_{j=n+1}^{n+m} \frac{n}{n + \hat{\lambda} m r(Y_j)}$$

and λ_0 such that

$$\int \frac{nf + mg}{n + \lambda_0 m r} d\mu = 1.$$

There exists a constant $Q_0 \equiv Q_0(p, c, C, \tau) > 0$ such that

$$\mathbb{E}[|\hat{\lambda} - \lambda_0|^p] \leq Q_0 n^{-p/2} \text{ for all } p \in \mathbb{N}.$$

Proof. We know that $\hat{\lambda}$ is a Z-estimator, being the solution with respect to λ of

$$\Psi_{n,m}(\lambda) := \frac{1}{n+m} \sum_{k=1}^{n+m} \psi_{\lambda}(Z_k) = 0, \quad \text{with} \quad \psi_{\lambda}(x) := \frac{n+m}{n+\lambda mr(x)} - 1,$$

whereas λ_0 is a population quantity and satisfies

$$\int \frac{nf}{n+\lambda_0 mr} d\mu + \int \frac{mg}{n+\lambda_0 mr} d\mu = 1. \quad (47)$$

We can use the assumptions $m \leq n \leq \tau m$ and $0 < c \leq r(\cdot) \leq C$ to show that

$$\frac{\partial \psi_{\lambda}(x)}{\partial \lambda} = -\frac{(n+m)mr(x)}{\{n+\lambda mr(x)\}^2} \leq -\frac{c\tau^{-1}(1+\tau^{-1})n^2}{(1+C/c)^2 n^2} = -\frac{c^3(1+\tau)}{\tau^2(c+C)^2} =: -a(c, C, \tau) \equiv -a < 0,$$

which implies that for all $\epsilon > 0$ we have $\{|\hat{\lambda} - \lambda_0| < \epsilon\} \supseteq \{|\Psi_{n,m}(\hat{\lambda}) - \Psi_{n,m}(\lambda_0)| < a\epsilon\} = \{|\Psi_{n,m}(\lambda_0)| < a\epsilon\}$, using the fact that $\Psi_{n,m}(\hat{\lambda}) = 0$ by definition of $\hat{\lambda}$. As a result, the previous inclusion shows that the boundedness assumption on $r(\cdot)$ allows to relate how close $\hat{\lambda}$ is to λ_0 with $|\Psi_{n,m}(\lambda_0)|$, which is easier to analyze since it is the sum of zero mean, bounded independent and identically distributed random variables. In this regard, we have

$$\begin{aligned} \mathbb{P}\{|\hat{\lambda} - \lambda_0| \geq \epsilon\} &\leq \mathbb{P}\{|\Psi_{n,m}(\lambda_0)| \geq a\epsilon\} = \mathbb{P}\left\{\left|\frac{1}{n+m} \sum_{k=1}^{n+m} \left(\frac{n+m}{n+\lambda_0 mr(Z_k)} - 1\right)\right| \geq a\epsilon\right\} \\ &= \mathbb{P}\left\{\left|\frac{1}{n} \sum_{i=1}^n \frac{n}{n+\lambda_0 mr(X_i)} + \frac{1}{m} \sum_{j=1}^m \frac{m}{n+\lambda_0 mr(Y_j)} - 1\right| \geq a\epsilon\right\} \\ &\stackrel{(47)}{=} \mathbb{P}\left\{\left|\frac{1}{n} \sum_{i=1}^n \frac{n}{n+\lambda_0 mr(X_i)} + \frac{1}{m} \sum_{j=1}^m \frac{m}{n+\lambda_0 mr(Y_j)} - \int \frac{nf}{n+\lambda_0 mr} d\mu - \int \frac{mg}{n+\lambda_0 mr} d\mu\right| \geq a\epsilon\right\} \\ &\leq \mathbb{P}\left\{\left|\frac{1}{n} \sum_{i=1}^n \frac{n}{n+\lambda_0 mr(X_i)} - \int \frac{nf}{n+\lambda_0 mr} d\mu\right| \geq \frac{a\epsilon}{2}\right\} \\ &\quad + \mathbb{P}\left\{\left|\frac{1}{m} \sum_{j=1}^m \frac{m}{n+\lambda_0 mr(Y_j)} - \int \frac{mg}{n+\lambda_0 mr} d\mu\right| \geq \frac{a\epsilon}{2}\right\} \\ &\leq 2 \exp\left\{-\frac{na^2\epsilon^2}{2}\right\} + 2 \exp\left\{-\frac{ma^2\epsilon^2}{2}\right\} \leq 2 \exp\left\{-\frac{na^2\epsilon^2}{2}\right\} + 2 \exp\left\{-\frac{na^2\epsilon^2}{2\tau}\right\} \leq 4 \exp\left\{-\frac{na^2\epsilon^2}{2\tau}\right\}, \end{aligned} \quad (48)$$

where in the last line we applied Hoeffding's inequality for bounded random variables to $0 \leq \frac{m}{n+\lambda_0 mr(\cdot)} \leq \frac{n}{n+\lambda_0 mr(\cdot)} \leq 1$, and the assumption that $m \leq n \leq \tau m$. We can thus bound the moment of order p of $|\hat{\lambda} - \lambda_0|$ as

$$\mathbb{E}[|\hat{\lambda} - \lambda_0|^p] = \int_0^\infty \mathbb{P}\{|\hat{\lambda} - \lambda_0|^p \geq \epsilon\} d\epsilon = \int_0^\infty \mathbb{P}\{|\hat{\lambda} - \lambda_0| \geq \epsilon^{\frac{1}{p}}\} d\epsilon$$

$$\begin{aligned}
&= \int_0^\infty \mathbb{P}\{|\hat{\lambda} - \lambda_0| \geq \epsilon\} p \epsilon^{p-1} d\epsilon = \left(\frac{2\tau}{na^2}\right)^{\frac{p}{2}} \int_0^\infty \mathbb{P}\left\{|\hat{\lambda} - \lambda_0| \geq \sqrt{\frac{2\tau}{na^2}} t\right\} p t^{p-1} dt \\
&\leq 2p \left(\frac{2\tau}{na^2}\right)^{\frac{p}{2}} \int_0^\infty 2t^{p-1} e^{-t^2} dt = 2p \left(\frac{2\tau}{na^2}\right)^{\frac{p}{2}} \int_0^\infty t^{\frac{p}{2}-1} e^{-t} dt = 2p \left(\frac{2\tau}{na^2}\right)^{\frac{p}{2}} \Gamma\left(\frac{p}{2}\right),
\end{aligned}$$

where in the last equality we used the definition of the Gamma function. This completes the proof. $\square$

Lemma 15. Fix subsets $\bar{I} \subseteq [n]$ and $\bar{J} \subseteq \{n+1, \dots, n+m\}$, and modify Step 3 of Algorithm 1 to be: Sample a vector of couples $\tau_t = \{(i_1^t, j_1^t), \dots, (i_K^t, j_K^t)\}$ such that $(i_1^t, \dots, i_K^t)$ are sampled uniformly and without replacement from $[n] \setminus \bar{I}$, and $(j_1^t, \dots, j_K^t)$ are sampled uniformly and without replacement from $\{n+1, \dots, n+m\} \setminus \bar{J}$. Keep all the other steps the same. Then this algorithm still has (3) as its stationary distribution.

Proof. The key observation to make here is that all the steps that led to (19) in the proof of Proposition 2 remain valid in this other setting. The only difference lies in the fact that the Markov chain associated with this new algorithm is not irreducible, and hence it will not converge to (3) if we let it run long enough. Nonetheless, (3) is still a stationary distribution, and we will now show this by proving a detailed balance condition. Let $K = \min\{n - \#\bar{I}, m - \#\bar{J}\}$ and let $\tilde{\mathcal{P}}$ be the set of all K couples of the form $\{(i_1, j_1), \dots, (i_K, j_K)\}$ such that $(i_1, \dots, i_K)$ contains distinct elements from $[n] \setminus \bar{I}$, and $(j_1, \dots, j_K)$ contains distinct elements from $\{n+1, \dots, n+m\} \setminus \bar{J}$. For all $t \in \mathbb{N}_+$ and all permutations p, p' , we have

$$\mathbb{P}\{\sigma_t = p' \mid \sigma_{t-1} = p\} = \frac{1}{|\tilde{\mathcal{P}}|} \sum_{\tau \in \tilde{\mathcal{P}}} \mathbb{P}\{\sigma_t = p' \mid \sigma_{t-1} = p, \tau_t = \tau\},$$

since at each time t this new algorithm draws $\tau_t \in \tilde{\mathcal{P}}$ uniformly at random. Next, given $\tau_t = \tau$ and $\sigma_{t-1} = p$, it must be the case that σ_t satisfies $\sigma_t \sim_\tau p$, since this new algorithm still uses Steps 4-5 of Algorithm 1. Arguing as in (18) gives

$$\frac{\mathbb{P}\{\sigma_t = p' \mid \sigma_{t-1} = p, \tau_t = \tau\}}{\mathbb{P}\{\sigma_t = p'' \mid \sigma_{t-1} = p, \tau_t = \tau\}} = \frac{\mathbb{P}\{\sigma = p'\}}{\mathbb{P}\{\sigma = p''\}},$$

which implies that

$$\mathbb{P}\{\sigma_t = p' \mid \sigma_{t-1} = p\} = \frac{1}{|\tilde{\mathcal{P}}|} \sum_{\tau \in \tilde{\mathcal{P}}} \frac{\mathbb{1}\{p' \sim_\tau p\} \cdot \mathbb{P}\{\sigma = p'\}}{\sum_{p''} \mathbb{1}\{p'' \sim_\tau p\} \cdot \mathbb{P}\{\sigma = p''\}}.$$

This concludes the proof by analogous calculations to those in (19). $\square$

Proof of Theorem 10. For simplicity, we assume $n = m$ throughout the proof. The more general case $m \leq n \leq \tau m$ corresponds to a simpler problem, as it involves a larger sample size; thus, the lower bound derived here remains valid in that setting. For varying densities p, q on $\mathbb{R}^d$ and a sufficiently small $\rho > 0$, define the set

$$\tilde{\mathcal{S}}_\theta^r(\rho) := \mathcal{S}_\theta^r(\rho) \cap \left\{ (f \equiv f_p, g \equiv g_q) : f_p = \gamma_p \frac{p}{r} \left(1 + r \frac{\gamma_q}{\gamma_p}\right), g_q = \gamma_q q \left(1 + r \frac{\gamma_q}{\gamma_p}\right) \right\} \subseteq \mathcal{S}_\theta^r(\rho), \quad (49)$$

where $\gamma_p = \frac{\sqrt{B}}{A\sqrt{B+\sqrt{A}}}$ and $\gamma_q = \frac{\sqrt{A}}{A\sqrt{B+\sqrt{A}}}$, with $A = \int_{\mathbb{R}^d} p(x)/r(x) dx$ and $B = \int_{\mathbb{R}^d} q(x)r(x) dx$. One can

easily check that $\int_{\mathbb{R}^d} f_p(x)dx = \int_{\mathbb{R}^d} g_q(x)dx = 1$ for this specific choice of γ_p and γ_q . As a result, for any test $\varphi \in \Psi(\alpha)$, and for all prior distributions μ_0, μ_1 supported on H_0 and $\tilde{\mathcal{S}}_\theta^r(\rho)$, respectively, Equation (49) shows that we can bound the total error probability as

$$\begin{aligned} \alpha + \sup_{(f,g) \in \mathcal{S}_\theta^r(\rho)} \mathbb{E}_P(1 - \varphi) &\geq \sup_{g \propto r f} \mathbb{E}_P \varphi + \sup_{(f,g) \in \mathcal{S}_\theta^r(\rho)} \mathbb{E}_P(1 - \varphi) \\ &\geq \sup_{g \propto r f} \mathbb{E}_P \varphi + \sup_{(f,g) \in \tilde{\mathcal{S}}_\theta^r(\rho)} \mathbb{E}_P(1 - \varphi) \geq \mathbb{E}_{\mu_0} \{\mathbb{E}_P \varphi\} + \mathbb{E}_{\mu_1} \{\mathbb{E}_P(1 - \varphi)\} \geq 1 - \text{TV}(\mathbb{E}_{\mu_0} P, \mathbb{E}_{\mu_1} P), \end{aligned} \quad (50)$$

where we recall that $P = P_f^{\otimes n} \otimes P_g^{\otimes n}$. This demonstrates that controlling the total variation distance above is sufficient to obtain a lower bound on ρ_r^* . We proceed to do so for specific choices of μ_0 and μ_1 , using a classical perturbation-based method originating from Ingster (1987) and recently employed in two-sample and independence testing problems in Albert et al. (2022) and Li and Yuan (2024). Start by considering $q_0(x) = q_1(x) = p_0(x) = \mathbb{1}\{x \in [0, 1]^d\}$ and define $A_i = \int_{\mathbb{R}^d} p_i(x)/r(x)dx$ and $B_i = \int_{\mathbb{R}^d} q_i(x)r(x)dx$ for $i \in \{0, 1\}$. We can assume without loss of generality that $\int_{[0, 1]^d} r(x)dx = 1$ so that $B_0 = B_1 = 1$; otherwise, since the problem is scale invariant, we might replace r with $r/\int_{[0, 1]^d} r(x)dx$ without affecting the minimax separation. Finally, define p_1 to be a perturbation of p_0 of the following form. Let $\tilde{M}$ be as in (57), fix

$$B_n^{1/d} = \left\lceil \left(\frac{\sqrt{c}(2\pi)^{d/2}L}{2\sqrt{2(1+c)}\tilde{M}\rho} \right)^{1/s} \right\rceil \quad \text{and} \quad \delta_n = \sqrt{\frac{2(1+c)}{c}}\rho B_n^{-1/2},$$

and consider $\{\phi_1, \dots, \phi_{B_n}\}$ as in (56) in the proof of Lemma 16, i.e. an orthonormal set of functions in $L^2(\mathbb{R}^d)$ whose supports are disjoint and contained in $[0, 1]^d$, and satisfy

$$\int_{\mathbb{R}^d} \phi_k(x)dx = \int_{\mathbb{R}^d} \frac{\phi_k(x)}{r(x)}dx = 0 \quad \text{for all } k \in [B_n]. \quad (51)$$

It is convenient to recall that the ϕ_k 's are of the form $\phi_k(x) = \frac{B_n^{1/2}}{\|\phi_{0,k}\|_2} \phi_{0,k}(B_n^{1/d}\{x - x_k^0\})$, where x_k^0 is the lower-left corner of their support, and the $\phi_{0,k}$'s satisfy $\max_{k \in [B_n]} \left\{ \frac{\|\phi_{0,k}\|_{S_d^s}}{\|\phi_{0,k}\|_2} \vee \frac{\|\phi_{0,k}\|_\infty}{\|\phi_{0,k}\|_2} \right\} \leq \tilde{M}$, with $\|\cdot\|_{S_d^s}$ defined in (53). Based on this, we define

$$p_1(x) \equiv p_{1,a}(x) = p_0(x) + \delta_n \sum_{k=1}^{B_n} a_k \phi_k(x), \quad (52)$$

where $a = (a_1, \dots, a_{B_n})$ is a collection of independent and identically distributed Rademacher random variables, meaning that $\mathbb{P}\{a_k = 1\} = \mathbb{P}\{a_k = -1\} = 1/2$ for all $k \in [B_n]$.

With these definitions in mind, let $f_i := f_{p_i}$ and $g_i := g_{q_i}$ for $i \in \{0, 1\}$, with f_p, g_p as in (49). Now, it is clear that (f_0, g_0) satisfy the null, thus we just need to check that $(f_1, g_1) \in \tilde{\mathcal{S}}_\theta^r(\rho)$ for all $a \in \{\pm 1\}^{B_n}$, which is required to ensure that the distribution μ_1 that assigns equal probability to each of them is indeed supported on $\tilde{\mathcal{S}}_\theta^r(\rho)$. We may assume that $\|f_1\|_\infty \vee \|g_1\|_\infty \leq M$; otherwise, it suffices to choose $p_0(x) = \mathbb{1}\{x \in [0, u]^d\}$ for $u \geq 1$ sufficiently large and construct a perturbed version of it as in (52). This is clearly sufficient for g_1 , since the prefactor of q in the definition of g_q in (49) depends on r , which is uniformly bounded. A similar

argument applies to f_1 , taking into account that the magnitudes of the bumps are such that

$$\|\delta_n \sum_{k=1}^{B_n} a_k \phi_k\|_\infty \leq \delta_n \max_{k=1}^{B_n} \|\phi_k\|_\infty \leq \delta_n B_n^{1/2} \max_{k=1}^{B_n} \frac{\|\phi_{0,k}\|_\infty}{\|\phi_{0,k}\|_2} \leq \tilde{M} \delta_n B_n^{1/2} \leq \sqrt{\frac{2(1+c)}{c}} \tilde{M} \rho,$$

where ρ is sufficiently small. In particular, we will set $\rho = c_1 n^{-2s/(4s+d)}$, where the constant $c_1 = c_1(c, C, \theta, \alpha, \beta)$ is chosen so that $\sqrt{2c^{-1}(1+c)} \tilde{M} \rho \leq 1$; this also implies that f_1 is non-negative. As for the conditions involving ψ_r , observe that

$$\psi_r = 2 \frac{\lambda_0 r f_1 - g_1}{1 + \lambda_0 r} = 2 \frac{1 + r \gamma_{q_1} / \gamma_{p_1}}{1 + \lambda_0 r} (\lambda_0 \gamma_{p_1} p_1 - \gamma_{q_1} q_1) = 2 \gamma_{q_1} (p_1 - q_1),$$

since $\lambda_0 = \gamma_{q_1} / \gamma_{p_1}$ satisfies $\int_{\mathbb{R}^d} \psi_r(x) dx = 0$. Hence we need to verify that $2 \gamma_{q_1} \|p_1 - q_1\|_2 > \rho$ and $2 \gamma_{q_1} (p_1 - q_1) \in \mathcal{S}_d^s(L)$. Start by noticing that for all $a \in \{\pm 1\}^{B_n}$ we have

$$1 \geq \gamma_{q_1}^2 = (\sqrt{A_1} + 1)^{-2} \geq (2A_1 + 2)^{-1} \geq \frac{c}{2(1+c)},$$

as

$$A_1 = \int_{\mathbb{R}^d} \frac{p_1(x)}{r(x)} dx = \int_{\mathbb{R}^d} \frac{p_0(x) + \delta_n \sum_{k=1}^{B_n} a_k \phi_k(x)}{r(x)} dx \stackrel{(51)}{=} \int_{\mathbb{R}^d} \frac{p_0(x)}{r(x)} dx \leq \frac{1}{c}.$$

Then, as for the condition involving the L^2 -norm of ψ_r , we have

$$\begin{aligned} 4 \gamma_{q_1}^2 \|p_1 - q_1\|_2^2 &= 4 \gamma_{q_1}^2 \left\| \delta_n \sum_{k=1}^{B_n} a_k \phi_k \right\|_2^2 = 4 \gamma_{q_1}^2 \delta_n^2 \left\| \sum_{k=1}^{B_n} a_k \phi_k \right\|_2^2 = 4 \gamma_{q_1}^2 \delta_n^2 \int_{\mathbb{R}^d} \left(\sum_{k=1}^{B_n} a_k \phi_k(x) \right)^2 dx \\ &= 4 \gamma_{q_1}^2 \delta_n^2 \sum_{k=1}^{B_n} a_k^2 \int_{\mathbb{R}^d} \phi_k^2(x) dx = 4 \gamma_{q_1}^2 \delta_n^2 B_n \geq \frac{2c}{(1+c)} \delta_n^2 B_n = 4 \rho^2 > \rho^2 \end{aligned}$$

for our particular choice of δ_n and B_n . As for the smoothness condition, define for notational convenience the norm

$$\|p\|_{\mathcal{S}_d^s}^2 := \int_{\mathbb{R}^d} \|\xi\|_2^{2s} |\widehat{p}(\xi)|^2 d\xi, \quad (53)$$

so that $\mathcal{S}_d^s(L) = \{p \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d) : \|p\|_{\mathcal{S}_d^s}^2 \leq (2\pi)^d L^2\}$. Furthermore, since the iterated Laplacian $(-\Delta)^s$ of order $s \in \mathbb{N}_+$ is the Fourier multiplier with symbol $\|\xi\|_2^{2s}$, Plancherel's theorem gives

$$\int_{\mathbb{R}^d} \|\xi\|_2^{2s} \widehat{\phi_1}(\xi) \overline{\widehat{\phi_2}(\xi)} d\xi = \int_{\mathbb{R}^d} (-\Delta)^s \phi_1(x) \overline{\phi_2(x)} dx = 0,$$

since $(-\Delta)^s \phi_1$ is a combination of derivatives of ϕ_1 of order $2s$, and ϕ_1, ϕ_2 have disjoint supports. This implies that

$$4 \gamma_{q_1}^2 \|p_1 - q_1\|_{\mathcal{S}_d^s}^2 \leq$$

$$\begin{aligned}
4 \|p_1 - q_1\|_{\mathcal{S}_d^s}^2 &= 4 \left\| \delta_n \sum_{k=1}^{B_n} a_k \phi_k \right\|_{\mathcal{S}_d^s}^2 = 4\delta_n^2 \left\| \sum_{k=1}^{B_n} a_k \phi_k \right\|_{\mathcal{S}_d^s}^2 = 4\delta_n^2 \int_{\mathbb{R}^d} \|\xi\|_2^{2s} \left| \sum_{k=1}^{B_n} a_k \widehat{\phi_k}(\xi) \right|^2 d\xi \\
&\leq 4\delta_n^2 B_n \max_{k=1}^{B_n} \int_{\mathbb{R}^d} \|\xi\|_2^{2s} \left| \widehat{\phi_k}(\xi) \right|^2 d\xi = 4\delta_n^2 B_n \max_{k=1}^{B_n} \int_{\mathbb{R}^d} \|\xi\|_2^{2s} \left| \frac{B_n^{1/2}}{\|\phi_{0,k}\|_2} \widehat{\phi_{0,k}(B_n^{1/d}\{\cdot - x_k^0\})}(\xi) \right|^2 d\xi \\
&= 4\delta_n^2 B_n \max_{k=1}^{B_n} \int_{\mathbb{R}^d} \|\xi\|_2^{2s} \left| \frac{B_n^{-1/2}}{\|\phi_{0,k}\|_2} e^{-i\langle \xi, x_k^0 \rangle} \widehat{\phi_{0,k}}(\xi/B_n^{1/d}) \right|^2 d\xi = 4 \max_{k=1}^{B_n} \frac{\delta_n^2}{\|\phi_{0,k}\|_2^2} \int_{\mathbb{R}^d} \|\xi\|_2^{2s} \left| \widehat{\phi_{0,k}}(\xi/B_n^{1/d}) \right|^2 d\xi \\
&= 4 \max_{k=1}^{B_n} \frac{\delta_n^2}{\|\phi_{0,k}\|_2^2} \int_{\mathbb{R}^d} \|B_n^{1/d} \xi\|_2^{2s} \left| \widehat{\phi_{0,k}}(\xi) \right|^2 B_n d\xi = 4\delta_n^2 B_n^{\frac{2s+d}{d}} \max_{k=1}^{B_n} \frac{\|\phi_{0,k}\|_{\mathcal{S}_d^s}^2}{\|\phi_{0,k}\|_2^2} \leq 4\tilde{M}^2 \delta_n^2 B_n^{\frac{2s+d}{d}} \leq (2\pi)^d L^2
\end{aligned}$$

for our particular choice of δ_n and B_n .

It remains to control the total variation in (50) for this specific choice of μ_1, δ_n and B_n , and assess for which values of ρ we can bound it above by $1 - \alpha - \beta$. Writing $f_1 \equiv f_{1,a}$ and $g_1 \equiv g_{1,a}$ to highlight their dependence on $a \in \{\pm 1\}^d$ through p_1 , and using χ^2 for the chi-square divergence, we look at

$$\begin{aligned}
4 \text{TV}^2(P_{f_0}^{\otimes n} \otimes P_{g_0}^{\otimes n}, \mathbb{E}_a\{P_{f_1}^{\otimes n} \otimes P_{g_1}^{\otimes n}\}) &= 4 \text{TV}^2\left(P_{f_0}^{\otimes n} \otimes P_{g_0}^{\otimes n}, 2^{-B_n} \sum_{a \in \{\pm 1\}^d} P_{f_{1,a}}^{\otimes n} \otimes P_{g_{1,a}}^{\otimes n}\right) \\
&\leq \chi^2\left(P_{f_0}^{\otimes n} \otimes P_{g_0}^{\otimes n}, 2^{-B_n} \sum_{a \in \{\pm 1\}^d} P_{f_{1,a}}^{\otimes n} \otimes P_{g_{1,a}}^{\otimes n}\right) \\
&= 2^{-2B_n} \sum_{a_1, a_2 \in \{\pm 1\}^d} \int_{\mathbb{R}^{2nd}} \left(\prod_{i=1}^n \frac{f_{1,a_1}(x_i)}{f_0(x_i)} \prod_{j=1}^n \frac{g_{1,a_2}(y_j)}{g_0(y_j)} \right) \left(\prod_{i=1}^n \frac{f_{1,a_2}(x_i)}{f_0(x_i)} \prod_{j=1}^n \frac{g_{1,a_1}(y_j)}{g_0(y_j)} \right) dP_0 \\
&= 2^{-2B_n} \sum_{a_1, a_2 \in \{\pm 1\}^d} \left(\prod_{i=1}^n \int_{\mathbb{R}^d} \frac{f_{1,a_1}(x_i)}{f_0(x_i)} \frac{f_{1,a_2}(x_i)}{f_0(x_i)} f_0(x_i) dx_i \right) \left(\prod_{j=1}^n \int_{\mathbb{R}^d} \frac{g_{1,a_1}(y_j)}{g_0(y_j)} \frac{g_{1,a_2}(y_j)}{g_0(y_j)} g_0(y_j) dy_j \right) \\
&= 2^{-2B_n} \sum_{a_1, a_2 \in \{\pm 1\}^d} \left(\int_{\mathbb{R}^d} \frac{f_{1,a_1}(x_1)}{f_0(x_1)} \frac{f_{1,a_2}(x_1)}{f_0(x_1)} f_0(x_1) dx_1 \right)^n \left(\int_{\mathbb{R}^d} \frac{g_{1,a_1}(y_1)}{g_0(y_1)} \frac{g_{1,a_2}(y_1)}{g_0(y_1)} g_0(y_1) dy_1 \right)^n \quad (54)
\end{aligned}$$

where we set $dP_0 = \prod_{i=1}^n f_0(x_i) \prod_{j=1}^n g_0(y_j) dx dy$. We now focus on controlling the integrals in the last display. As for the latter, Equation (51) implies that $A_0 = A_1$, hence $\gamma_{p_1} = \gamma_{p_0}$ and $\gamma_{q_1} = \gamma_{q_0}$. This gives $\frac{g_{1,a}(y_1)}{g_0(y_1)} = \mathbb{1}\{x \in [0, 1]^d\}$ for all $a \in \{\pm 1\}^d$, and further shows that the second integral in (54) is equal to one. As for the other one, similar calculations show that

$$\begin{aligned}
\int_{\mathbb{R}^d} \frac{f_{1,a_1}(x)}{f_0(x)} \frac{f_{1,a_2}(x)}{f_0(x)} f_0(x) dx &= \int_{\mathbb{R}^d} \left(1 + \delta_n \sum_{k=1}^{B_n} a_{1,k} \frac{\phi_k(x)}{p_0(x)} \right) \left(1 + \delta_n \sum_{k=1}^{B_n} a_{2,k} \frac{\phi_k(x)}{p_0(x)} \right) f_0(x) dx \\
&= 1 + \delta_n^2 \sum_{k=1}^{B_n} a_{1,k} a_{2,k} \int_{\mathbb{R}^d} \frac{\phi_k^2(x)}{p_0^2(x)} f_0(x) dx + \delta_n \sum_{k=1}^{B_n} (a_{1,k} + a_{2,k}) \int_{\mathbb{R}^d} \frac{\phi_k(x)}{p_0(x)} f_0(x) dx \\
&= 1 + \delta_n^2 \sum_{k=1}^{B_n} a_{1,k} a_{2,k} \int_{\mathbb{R}^d} \frac{\phi_k^2(x)}{p_0^2(x)} f_0(x) dx + \delta_n \sum_{k=1}^{B_n} (a_{1,k} + a_{2,k}) \int_{\mathbb{R}^d} \frac{\phi_k(x)}{p_0(x)} \frac{p_0(x)}{r(x)} (\gamma_{p_0} + r \gamma_{q_0}) dx
\end{aligned}$$

$$\begin{aligned}
&= 1 + \delta_n^2 \sum_{k=1}^{B_n} a_{1,k} a_{2,k} \int_{\mathbb{R}^d} \frac{\phi_k^2(x)}{p_0^2(x)} f_0(x) dx + \delta_n \sum_{k=1}^{B_n} (a_{1,k} + a_{2,k}) \left\{ \gamma_{p_0} \int_{\mathbb{R}^d} \frac{\phi_k(x)}{r(x)} dx + \gamma_{q_0} \int_{\mathbb{R}^d} \phi_k(x) dx \right\} \\
&= 1 + \delta_n^2 \sum_{k=1}^{B_n} a_{1,k} a_{2,k} \int_{\mathbb{R}^d} \frac{\phi_k^2(x)}{p_0^2(x)} f_0(x) dx,
\end{aligned} \tag{55}$$

where the second equality follows from the fact that ϕ_{k_1} and ϕ_{k_2} have disjoint support when $k_1 \neq k_2$, and the final equality uses (51). Combining (54) with (55) then shows

$$\begin{aligned}
4 \text{TV}^2(P_{f_0}^{\otimes n} \otimes P_{g_0}^{\otimes n}, \mathbb{E}_a \{P_{f_1}^{\otimes n} \otimes P_{g_1}^{\otimes n}\}) &\leq 2^{-2B_n} \sum_{a_1, a_2 \in \{\pm 1\}^d} \left(1 + \delta_n^2 \sum_{k=1}^{B_n} a_{1,k} a_{2,k} \int_{\mathbb{R}^d} \frac{\phi_k^2(x)}{p_0^2(x)} f_0(x) dx \right)^n \\
&= \mathbb{E}_{a_1, a_2} \left[\left(1 + \delta_n^2 \sum_{k=1}^{B_n} a_{1,k} a_{2,k} \int_{\mathbb{R}^d} \frac{\phi_k^2(x)}{p_0^2(x)} f_0(x) dx \right)^n \right] = \mathbb{E}_a \left[\left(1 + \delta_n^2 \sum_{k=1}^{B_n} a_k \int_{\mathbb{R}^d} \frac{\phi_k^2(x)}{p_0^2(x)} f_0(x) dx \right)^n \right] \\
&\leq \mathbb{E}_a \left[\exp \left\{ n \delta_n^2 \sum_{k=1}^{B_n} a_k \int_{\mathbb{R}^d} \frac{\phi_k^2(x)}{p_0^2(x)} f_0(x) dx \right\} \right] = \prod_{k=1}^{B_n} \cosh \left(n \delta_n^2 \int_{\mathbb{R}^d} \frac{\phi_k^2(x)}{p_0^2(x)} f_0(x) dx \right) \\
&\leq \exp \left\{ \frac{1}{2} B_n n^2 \delta_n^4 \left(\max_{k=1}^{B_n} \int_{\mathbb{R}^d} \frac{\phi_k^2(x)}{p_0^2(x)} f_0(x) dx \right)^2 \right\} = \exp \left\{ \frac{1}{2} B_n n^2 \delta_n^4 \left(\max_{k=1}^{B_n} \int_{\mathbb{R}^d} \frac{\gamma_{p_0} + r(x) \gamma_{q_0}}{r(x)} \phi_k^2(x) dx \right)^2 \right\} \\
&\leq \exp \left\{ \frac{C + c^2}{c^2} B_n n^2 \delta_n^4 \right\},
\end{aligned}$$

where in the last step we used the fact that $c \leq r(\cdot) \leq C$ together with $\gamma_{q_0} = (\sqrt{A_0} + 1)^{-1} \leq 1$ and $\gamma_{p_0} = A_0^{-1/2} \gamma_{q_0} \leq A_0^{-1/2} = \{\int_{\mathbb{R}^d} p_0(x)/r(x)\}^{-1/2} \leq \sqrt{C}$. Now, being $B_n n^2 \delta_n^4$ of the order $n^2 \rho^{\frac{4s+d}{s}}$, the previous shows that there exists a constant $c_1 = c_1(c, C, \theta, \alpha, \beta)$ for which the previous display is upper bounded by $1 - \alpha - \beta$ whenever $\rho \leq c_1 n^{-2s/(4s+d)}$. This concludes the proof. $\square$

Lemma 16. *Let $r : \mathbb{R}^d \rightarrow \mathbb{R}_+$ be such that $0 < c \leq r(x) \leq C$ for all $x \in \mathbb{R}^d$. Fix an integer $B_n \geq 1$ such that $B_n^{1/d} \in \mathbb{N}_+$ and write $b_n := B_n^{-1/d}$. Partition $[0, 1]^d$ into the B_n disjoint cubes*

$$Q_k = \prod_{j=1}^d [i_j b_n, (i_j + 1) b_n), \quad \text{with } k = (i_1, \dots, i_d) \in \{0, \dots, b_n^{-1} - 1\}^d,$$

and denote the lower-left corner of Q_k by x_k^0 . There exist functions $\{\phi_{0,1}, \dots, \phi_{0,B_n}\}$ supported on $[0, 1]^d$ satisfying the following conditions:

(i) For all $k \in [B_n]$

$$\phi_k(x) = \frac{B_n^{1/2}}{\|\phi_{0,k}\|_2} \phi_{0,k}(B_n^{1/d} \{x - x_k^0\})$$

is $C^\infty(\mathbb{R}^d)$ and satisfies

$$\int_{\mathbb{R}^d} \phi_k(x) dx = \int_{\mathbb{R}^d} \frac{\phi_k(x)}{r(x)} dx = 0, \quad \|\phi_k\|_2 = 1, \quad \text{supp } \phi_k \subseteq Q_k;$$

(ii) There exist a constant $\tilde{M} > 0$ such that

$$\max_{k=1}^{B_n} \left\{ \frac{\|\phi_{0,k}\|_{S_d^s}}{\|\phi_{0,k}\|_2} \vee \frac{\|\phi_{0,k}\|_\infty}{\|\phi_{0,k}\|_2} \right\} \leq \tilde{M}.$$

Proof. We will start by proving part (i). Consider the standard $C^\infty(\mathbb{R})$ bump

$$\eta(t) := \begin{cases} \exp\left(-\frac{1}{1-t^2}\right), & |t| < 1, \\ 0, & |t| \geq 1, \end{cases}$$

and define $f_+(y_1) = \eta(6y_1 - 1)$, $f_-(y_1) = \eta(6y_1 - 3)$ and $f_0(y_1) = \eta(6y_1 - 5)$. It is immediate to see that f_+ , f_- and f_0 are supported on $[0, \frac{1}{3}]$, $[\frac{1}{3}, \frac{2}{3}]$ and $[\frac{2}{3}, 1]$, respectively. Based on this, define three bumps in $\mathbb{R}^d$ by

$$\phi_+(y) = f_+(y_1) \prod_{l=2}^d \eta(y_l), \quad \phi_-(y) = f_-(y_1) \prod_{l=2}^d \eta(y_l), \quad \phi_0(y) = f_0(y_1) \prod_{l=2}^d \eta(y_l)$$

for all $y = (y_1, \dots, y_d) \in \mathbb{R}^d$, and observe that all three functions are in $C^\infty(\mathbb{R}^d)$ and supported in $[0, \frac{1}{3}] \times [0, 1]^{d-1}$, $[\frac{1}{3}, \frac{2}{3}] \times [0, 1]^{d-1}$ and $[\frac{2}{3}, 1] \times [0, 1]^{d-1}$, respectively. Fix a cube Q_k and write points in it as $x = x_k^0 + b_n y$ with $y \in [0, 1]^d$. Because r^{-1} is bounded on Q_k , each of the numbers

$$a_k = \int_{\mathbb{R}^d} \frac{\phi_+(y)}{r(x_k^0 + b_n y)} dy, \quad b_k = \int_{\mathbb{R}^d} \frac{\phi_-(y)}{r(x_k^0 + b_n y)} dy, \quad c_k = \int_{\mathbb{R}^d} \frac{\phi_0(y)}{r(x_k^0 + b_n y)} dy$$

is finite and strictly positive. If $a_k \neq b_k$, set

$$u_k = \frac{b_k - c_k}{b_k - a_k}, \quad v_k = \frac{a_k - c_k}{b_k - a_k}, \quad w_k = 1 \quad \text{and} \quad \phi_{0,k}(y) = u_k \phi_+(y) - v_k \phi_-(y) - w_k \phi_0(y).$$

If instead $a_k = b_k$ but $a_k \neq c_k$, we switch the roles of ϕ_- and ϕ_0 and apply the same formula. Finally, if $a_k = b_k = c_k$, we simply set $u_k = w_k = 1$ and $v_k = 0$, so that $\phi_{0,k}(y) = \phi_+(y) - \phi_0(y)$. A straightforward calculation shows that $\phi_{0,k}$ has zero average both with respect to Lebesgue measure and with respect to the weight r^{-1} evaluated at $x_k^0 + b_n y$:

$$\int_{\mathbb{R}^d} \phi_{0,k}(y) dy = \left(\int_{\mathbb{R}^d} \phi_+(y) dy \right) (u_k - v_k - w_k) = 0, \quad \int_{\mathbb{R}^d} \frac{\phi_{0,k}(y)}{r(x_k^0 + b_n y)} dy = u_k a_k - v_k b_k - w_k c_k = 0.$$

We now verify that the function

$$\phi_k(x) := \frac{B_n^{1/2}}{\|\phi_{0,k}\|_2} \phi_{0,k}(B_n^{1/d} \{x - x_k^0\}) \quad (56)$$

satisfies the desired properties. Because $dx = b_n^d dy = B_n^{-1} dy$, the two zero-average identities above translate to the x -scale, yielding

$$\int_{\mathbb{R}^d} \phi_k(x) dx = \int_{\mathbb{R}^d} \frac{\phi_k(x)}{r(x)} dx = 0.$$

Moreover $\|\phi_k\|_2^2 = B_n \|\phi_{0,k}\|_2^{-2} B_n^{-1} \|\phi_{0,k}\|_2^2 = 1$. Finally, the support of ϕ_k is contained in Q_k , and different

cubes do not intersect, hence for $k_1 \neq k_2$ we also have $\langle \phi_{k_1}, \phi_{k_2} \rangle_2 = 0$. This completes the proof of the first part of the statement.

As for part (ii), set $J_\ell := \|\phi_+\|_\ell = \|\phi_-\|_\ell = \|\phi_0\|_\ell$ for $\ell \in \{\infty, 2, \mathcal{S}_d^s\}$, and observe that these constants depend only on the shape of the function η . The disjoint structure of the supports of ϕ_+, ϕ_-, ϕ_0 gives

$$\begin{cases} \|\phi_{0,k}\|_\infty = J_\infty \max\{|u_k|, |v_k|, |w_k|\} \\ \|\phi_{0,k}\|_2 = J_2 \sqrt{u_k^2 + v_k^2 + w_k^2} \\ \|\phi_{0,k}\|_{\mathcal{S}_d^s} = J_{\mathcal{S}_d^s} \sqrt{u_k^2 + v_k^2 + w_k^2} \end{cases}$$

for all $k \in [B_k]$, and implies that

$$\frac{\|\phi_{0,k}\|_{\mathcal{S}_d^s}}{\|\phi_{0,k}\|_2} = \frac{J_{\mathcal{S}_d^s}}{J_2} \quad \text{and} \quad \frac{\|\phi_{0,k}\|_\infty}{\|\phi_{0,k}\|_2} = \frac{J_\infty \max\{|u_k|, |v_k|, |w_k|\}}{J_2 \sqrt{u_k^2 + v_k^2 + w_k^2}} \leq \frac{J_\infty (|u_k| + |v_k| + |w_k|)}{J_2 \sqrt{u_k^2 + v_k^2 + w_k^2}} \leq \frac{\sqrt{3} J_\infty}{J_2}.$$

Note the denominator is always well-defined for our choices of (u_k, v_k, w_k) as $u_k^2 + v_k^2 + w_k^2 \geq 1$. This shows that the claim in part (ii) holds with

$$\tilde{M} := \frac{J_{\mathcal{S}_d^s}}{J_2} \vee \frac{\sqrt{3} J_\infty}{J_2} \quad (57)$$

and concludes the proof of the lemma. $\square$

A.4 Proofs for Section 5

Proof of Proposition 11. The reason we cannot directly apply Theorem 3 is that, although $g \propto r_\star f$ for a certain $r \in \mathcal{R}$, Algorithm 2 now generates $Z^{(1)}, \dots, Z^{(H)}$ using an approximation $\hat{r}$ to $r_\star$. To address this mismatch, let $\tilde{X} = (\tilde{X}_1, \dots, \tilde{X}_n)$, $\tilde{Y} = (\tilde{Y}_1, \dots, \tilde{Y}_m)$ be such that $\tilde{X} \perp\!\!\!\perp \tilde{Y}$, $\tilde{X}_i \stackrel{\text{i.i.d.}}{\sim} f$ and $\tilde{Y}_i \stackrel{\text{i.i.d.}}{\sim} \tilde{r} f$. Define $\tilde{Z} = (\tilde{X}, \tilde{Y})$ and let $\tilde{Z}^{(1)}, \dots, \tilde{Z}^{(H)}$ be draws of Algorithm 2 based on $\hat{r}$ when we sample from the values of $\tilde{Z}$ instead of Z . That is, for every $h \in [H]$ independently we have

$$\tilde{Z}^{(h)} = \tilde{Z}_{\tilde{\sigma}^{(h)}} \quad \text{where} \quad \mathbb{P}\left\{\tilde{\sigma}^{(h)} = \sigma \mid \tilde{Z}\right\} \propto \prod_{i \in \{n+1, \dots, n+m\}} \hat{r}(\tilde{Z}_{\sigma(i)}),$$

where $\tilde{Z}_\sigma$ is defined analogously to Z_σ . Next, by comparing to the sampling mechanism outlined in Algorithm 2, we observe that the $\tilde{Z}^{(h)}$'s, conditional on $\tilde{Z}$, are generated with the same mechanism as the $Z^{(h)}$'s conditional on Z . That is, for every $z \in \mathcal{X}^{n+m}$ we have

$$\left((\tilde{Z}^{(1)}, \dots, \tilde{Z}^{(H)}) \mid \tilde{Z} = z\right) \stackrel{d}{=} \left((Z^{(1)}, \dots, Z^{(H)}) \mid Z = z\right).$$

We can now use the fact that, if $(V \mid U = u) \stackrel{d}{=} (V' \mid U' = u)$ for all u , then $\text{TV}((U, V), (U', V')) = \text{TV}(U, U')$. It follows that

$$\begin{aligned} \text{TV}\left((\tilde{Z}, \tilde{Z}^{(1)}, \dots, \tilde{Z}^{(H)}), (Z, Z^{(1)}, \dots, Z^{(H)})\right) &= \text{TV}(\tilde{Z}, Z) = \text{TV}\left((\tilde{X}, \tilde{Y}), (X, Y)\right) \\ &= \text{TV}(\tilde{Y}, Y) = \text{TV}\left(\{\tilde{r} \cdot f\}^{\otimes m}, \{\bar{r}_\star \cdot f\}^{\otimes m}\right). \end{aligned} \quad (58)$$

Hence, if we define

$$A_\alpha := \left\{ (z, z^{(1)}, \dots, z^{(H)}) : \frac{1 + \sum_{h=1}^H \mathbb{1}\{T(z^{(h)}) \geq T(z)\}}{1 + H} \leq \alpha \right\},$$

we can bound the type I error as

$$\begin{aligned} \mathbb{P}\{p_{\hat{r}} \leq \alpha\} &= \mathbb{P}\{(Z, Z^{(1)}, \dots, Z^{(H)}) \in A_\alpha\} \\ &\leq \mathbb{P}\{(\tilde{Z}, \tilde{Z}^{(1)}, \dots, \tilde{Z}^{(H)}) \in A_\alpha\} + \text{TV}\left((\tilde{Z}, \tilde{Z}^{(1)}, \dots, \tilde{Z}^{(H)}), (Z, Z^{(1)}, \dots, Z^{(H)})\right) \\ &= \mathbb{P}\{(\tilde{Z}, \tilde{Z}^{(1)}, \dots, \tilde{Z}^{(H)}) \in A_\alpha\} + \text{TV}\left(\{\tilde{r} \cdot f\}^{\otimes m}, \{\tilde{r}_* \cdot f\}^{\otimes m}\right) \leq \alpha + \text{TV}\left(\{\tilde{r} \cdot f\}^{\otimes m}, \{\tilde{r}_* \cdot f\}^{\otimes m}\right), \end{aligned}$$

where the first inequality follows from the definition of total variation distance, the second equality from (58), and the final inequality from the exchangeability of $(\tilde{Z}, \tilde{Z}^{(1)}, \dots, \tilde{Z}^{(H)})$. This arises from the fact that the $\tilde{Y}_i$'s are independent and identically distributed from a distribution proportional to $\hat{r} f$, and Algorithm 2 copies $\tilde{Z}^{(1)}, \dots, \tilde{Z}^{(H)}$ are generated using the same approximation $\hat{r}$. $\square$

Proof of Proposition 13. Arguing as in the proof of Proposition 11, if we define

$$A_\alpha := \left\{ (z, z^{(1)}, \dots, z^{(H)}) : \frac{1 + \sum_{h=1}^H \mathbb{1}\{T(z^{(h)}) \geq T(z)\}}{1 + H} \leq \alpha \right\},$$

we can bound

$$\begin{aligned} \mathbb{P}\{p_{\hat{r}} > \alpha\} &= \mathbb{P}\{(Z, Z_{\hat{r}}^{(1)}, \dots, Z_{\hat{r}}^{(H)}) \in A_\alpha^c\} \\ &\leq \mathbb{P}\{(Z, Z_{r_*}^{(1)}, \dots, Z_{r_*}^{(H)}) \in A_\alpha^c\} + \text{TV}\{(Z, Z_{\hat{r}}^{(1)}, \dots, Z_{\hat{r}}^{(H)}), (Z, Z_{r_*}^{(1)}, \dots, Z_{r_*}^{(H)})\} \\ &= \mathbb{P}\{p_{r_*} > \alpha\} + \text{TV}\{(Z, Z_{\hat{r}}^{(1)}, \dots, Z_{\hat{r}}^{(H)}), (Z, Z_{r_*}^{(1)}, \dots, Z_{r_*}^{(H)})\}, \end{aligned}$$

which shows that it is sufficient to control the total variation in the last display to conclude the proof. In this regard, let $Q_r(\cdot | z)$ be the law of $Z_r^{(1)}$ conditionally on $Z = z$, and $\pi_r(\cdot | z)$ be the law of $\sigma^{(1)}$ conditionally on $Z = z$ in the case where (3) is based on r , meaning that for all $C \subseteq \mathcal{S}_{n+m}$ we have

$$\pi_r(C | z) = \frac{\sum_{\sigma \in C} \prod_{i \notin [n]} r(z_{\sigma(i)})}{\sum_{\sigma \in \mathcal{S}_{n+m}} \prod_{i \notin [n]} r(z_{\sigma(i)})}.$$

Based on this, we can write

$$\begin{aligned} \text{TV}\{(Z, Z_{\hat{r}}^{(1)}, \dots, Z_{\hat{r}}^{(H)}), (Z, Z_{r_*}^{(1)}, \dots, Z_{r_*}^{(H)})\} &\leq \mathbb{E} \left[\text{TV}\{(Z_{\hat{r}}^{(1)}, \dots, Z_{\hat{r}}^{(H)} | Z), (Z_{r_*}^{(1)}, \dots, Z_{r_*}^{(H)} | Z)\} \right] \\ &= \mathbb{E} \left[\text{TV}\{Q_{\hat{r}}(\cdot | Z)^{\otimes H}, Q_{r_*}(\cdot | Z)^{\otimes H}\} \right] \leq H \mathbb{E} [\text{TV}\{Q_{\hat{r}}(\cdot | Z), Q_{r_*}(\cdot | Z)\}], \end{aligned}$$

where in the penultimate step we used the fact that $(Z_r^{(1)}, \dots, Z_r^{(H)})$ are i.i.d. conditionally on Z . Now, let

$$\text{KL}(P, Q) := \begin{cases} \int \log\left(\frac{dP}{dQ}\right) dP & \text{if } P \ll Q, \\ +\infty & \text{otherwise,} \end{cases}$$

denote the Kullback-Leibler divergence between two distributions defined on the same measurable space, and for all measurable $B \subseteq \mathcal{X}^{\setminus \uparrow + \uparrow}$ define $C_B := \{\sigma \in \mathcal{S}_{n+m} : Z_\sigma \in B\}$. By the sampling rule (3), for all measurable $B \subseteq \mathcal{X}^{n+m}$ we have

$$Q_r(B \mid Z) = \mathbb{P}\{Z_r^{(1)} \in B \mid Z\} = \sum_{\sigma: z_\sigma \in B} \mathbb{P}\{\sigma^{(1)} = \sigma \mid Z; r\} = \frac{\sum_{\sigma: z_\sigma \in B} \prod_{i \notin [n]} r(Z_{\sigma(i)})}{\sum_{\sigma \in \mathcal{S}_{n+m}} \prod_{i \notin [n]} r(Z_{\sigma(i)})} = \pi_r(C_B \mid Z),$$

hence

$$\begin{aligned} \text{TV}\{Q_{\hat{r}}(\cdot \mid Z), Q_{r_*}(\cdot \mid Z)\} &= \sup_{B \subseteq \mathcal{X}^{n+m}} |Q_{\hat{r}}(B \mid Z) - Q_{r_*}(B \mid Z)| \\ &= \sup_{B \subseteq \mathcal{X}^{n+m}} |\pi_{\hat{r}}(C_B \mid Z) - \pi_{r_*}(C_B \mid Z)| \leq \sup_{C \subseteq \mathcal{S}_{n+m}} |\pi_{\hat{r}}(C \mid Z) - \pi_{r_*}(C \mid Z)| = \text{TV}\{\pi_{\hat{r}}(\cdot \mid Z), \pi_{r_*}(\cdot \mid Z)\} \\ &\leq \sqrt{1 - \exp\{-\text{KL}(\pi_{\hat{r}}(\cdot \mid Z), \pi_{r_*}(\cdot \mid Z))\}} \leq \sqrt{1 - \exp\{-2m\eta\}}. \end{aligned}$$

In particular, the first inequality holds since $\{C_B : \text{measurable } B \subseteq \mathcal{X}^{n+m}\} \subseteq \{C : C \subseteq \mathcal{S}_{n+m}\}$, the second one follows from the Bretagnolle–Huber inequality (Bretagnolle and Huber, 1978, Lemma 2.1), while the third one uses

$$\frac{\prod_{i \notin [n]} \hat{r}(Z_{\sigma(i)})}{\prod_{i \notin [n]} r_*(Z_{\sigma(i)})} \quad , \quad \frac{\sum_{\sigma \in \mathcal{S}_{n+m}} \prod_{i \notin [n]} r_*(Z_{\sigma(i)})}{\sum_{\sigma \in \mathcal{S}_{n+m}} \prod_{i \notin [n]} \hat{r}(Z_{\sigma(i)})} \in [e^{-m\eta}, e^{m\eta}],$$

which follows from the definition of $\eta := \max_{i \in [n+m]} |\log \hat{r}(Z_i) - \log r_*(Z_i)|$. This concludes the proof. $\square$

Appendix B Additional theoretical results

We first derive an explicit representation of the population discrepancy measure $T_{\mathcal{F},r}(f, g)$, introduced in Section 3.1, in the regime of highly unbalanced sample sizes. Since λ_0 is a continuous function of n/m , the fact that $\lim_{t \rightarrow 0^+} \lambda_0 = \int g/r \, d\mu$ and $\lim_{t \rightarrow \infty} \lambda_0^{-1} = \int r f \, d\mu$ also ensures that λ_0 is bounded away from zero and infinity in every regime of n and m , provided that $\int g/r \, d\mu < \infty$ and $\int r f \, d\mu < \infty$. This applies also to $T_{\mathcal{F},r}(f, g)$ when f, g are fixed alternatives.

Proposition 17. *Recall $T_{\mathcal{F},r}(f, g) := \sup_{\varphi \in \mathcal{F}} \left| \int \frac{n/m+1}{n/m+\lambda_0 r} (\lambda_0 r f - g) \varphi \, d\mu \right|$, and assume $\int g/r \, d\mu < \infty$, $\int r f \, d\mu < \infty$ and $\|\varphi\|_\infty \leq \gamma$ for all $\varphi \in \mathcal{F}$. Then, we have*

$$\lambda_0 \rightarrow \int \frac{g}{r} \, d\mu, \quad T_{\mathcal{F},r}(f, g) \rightarrow \sup_{\varphi \in \mathcal{F}} \left| \int \left\{ f - \left(\frac{g}{r} \right) \left(\int \frac{g}{r} \, d\mu \right)^{-1} \right\} \varphi \, d\mu \right|$$

when $n/m \rightarrow 0^+$. Similarly,

$$\lambda_0^{-1} \rightarrow \int r f \, d\mu, \quad T_{\mathcal{F},r}(f, g) \rightarrow \sup_{\varphi \in \mathcal{F}} \left| \int \left\{ (r f) \left(\int r f \, d\mu \right)^{-1} - g \right\} \varphi \, d\mu \right|$$

when $n/m \rightarrow \infty$.

Proof. We focus on the case $t := n/m \rightarrow 0^+$. For each fixed $t > 0$, define

$$F_t : \begin{cases} \mathbb{R}_+ \rightarrow [-1, \infty) \\ \lambda \mapsto \int \frac{tf+g}{t+\lambda r} d\mu - 1, \end{cases} \quad \text{and} \quad F_0 : \begin{cases} \mathbb{R}_+ \rightarrow [-1, \infty) \\ \lambda \mapsto \int \frac{g}{\lambda r} d\mu - 1. \end{cases}$$

Arguing as in Lemma 4, there exists a unique $\lambda_0(t) > 0$ such that $F_t(\lambda_0(t)) = 0$. Similarly, F_0 is continuous, strictly decreasing, with $\lim_{\lambda \rightarrow 0^+} F_0(\lambda) = \infty$, $\lim_{\lambda \rightarrow \infty} F_0(\lambda) = -1$, so there is a unique $\lambda_\star \in (0, \infty)$ such that $F_0(\lambda_\star) = 0$, namely $\lambda_\star = \int g/r d\mu$. We will now show that $\lambda_0(t) \rightarrow \lambda_\star$ as $t \rightarrow 0^+$. In this regard, for $0 < a < \lambda_\star < b < \infty$ and for fixed $x \in \mathcal{X}$, we have

$$\begin{aligned} \sup_{\lambda \in [a, b]} \left| \frac{tf(x) + g(x)}{t + \lambda r(x)} - \frac{g(x)}{\lambda r(x)} \right| &= \sup_{\lambda \in [a, b]} \left| \frac{t(\lambda r(x)f(x) - g(x))}{(t + \lambda r(x))\lambda r(x)} \right| \\ &\leq \min \left(1, \frac{t}{ar(x)} \right) \sup_{\lambda \in [a, b]} \frac{\lambda r(x)f(x) + g(x)}{\lambda r(x)} \leq f(x) + \frac{g(x)}{ar(x)}. \end{aligned}$$

This both shows that the left-hand side converges to zero as $t \rightarrow 0^+$, and that

$$\sup_{\lambda \in [a, b]} |F_t(\lambda) - F_0(\lambda)| \leq \int \sup_{\lambda \in [a, b]} \left| \frac{tf(x) + g(x)}{t + \lambda r(x)} - \frac{g(x)}{\lambda r(x)} \right| d\mu \leq 1 + \frac{1}{a} \int \frac{g}{r} d\mu.$$

Taken together, the previous claims show that $F_t \rightarrow F_0$ uniformly on $[a, b]$ by the dominated convergence theorem. This is enough to show that $\lambda_0(t) \rightarrow \lambda_\star$ as $t \rightarrow 0^+$. Indeed, for an arbitrary $\epsilon > 0$, define $\alpha_\epsilon := F_0(\lambda_\star - \epsilon)$ and $\beta_\epsilon := F_0(\lambda_\star + \epsilon)$ with $[\lambda_\star - \epsilon, \lambda_\star + \epsilon] \subset (a, b)$. Further set $\eta_\epsilon = \frac{1}{2} \min(\alpha_\epsilon, \beta_\epsilon)$, so that $\alpha_\epsilon \geq 2\eta_\epsilon$ and $\beta_\epsilon \leq -2\eta_\epsilon$. By the uniform convergence of F_t to F_0 on $[a, b]$, there exists $t_\epsilon > 0$ such that for all $0 < t < t_\epsilon$ we have $\sup_{\lambda \in [a, b]} |F_t(\lambda) - F_0(\lambda)| \leq \eta_\epsilon$. In particular, we have

$$F_t(\lambda_\star - \epsilon) \geq F_0(\lambda_\star - \epsilon) - |F_t(\lambda_\star - \epsilon) - F_0(\lambda_\star - \epsilon)| \geq 2\eta_\epsilon - \eta_\epsilon = \eta_\epsilon > 0,$$

and, similarly, $F_t(\lambda_\star + \epsilon) \leq -\eta_\epsilon < 0$. Combined with the fact that F_t is continuous and monotonically decreasing, this implies by the intermediate value theorem that $\lambda_0(t) \in (\lambda_\star - \epsilon, \lambda_\star + \epsilon)$, which yields $\lambda_0(t) \rightarrow \lambda_\star$ as $t \rightarrow 0^+$ by the arbitrariness of ϵ .

As for the second part of the claim, for $t > 0$ define

$$k_t(x) := \frac{t+1}{t + \lambda_0(t)r(x)} (\lambda_0(t)r(x)f(x) - g(x)), \quad \text{and} \quad k(x) := f(x) - \frac{g(x)}{\lambda_\star r(x)}.$$

In light of the previous result, $k_t(x) \rightarrow k(x)$ for all $x \in \mathcal{X}$ as $t \rightarrow 0^+$. Furthermore, we proved there exist $\epsilon > 0$ and $t_\epsilon > 0$ such that $\lambda_0(t) \in (\lambda_\star - \epsilon, \lambda_\star + \epsilon)$ for all $0 < t < t_\epsilon$; set $\lambda_{\min} := \lambda_\star - \epsilon > 0$ and $\lambda_{\max} := \lambda_\star + \epsilon < \infty$. For fixed $x \in \mathcal{X}$ and $0 < t < t_\epsilon$, we thus have

$$\left| \frac{t+1}{t + \lambda_0(t)r(x)} \right| \leq \frac{t_\epsilon + 1}{\lambda_{\min}r(x)}, \quad \text{and} \quad |\lambda_0(t)r(x)f(x) - g(x)| \leq \lambda_{\max}r(x)f(x) + g(x),$$

so

$$|k_t(x) - k(x)| \leq |k_t(x)| + |k(x)| \leq \left\{ (t_\epsilon + 1) \frac{\lambda_{\max}}{\lambda_{\min}} + 1 \right\} f(x) + \frac{t_\epsilon + 2}{\lambda_{\min}r(x)} g(x).$$

Since the right-hand side is integrable, the dominated convergence implies that $\int |k_t - k| d\mu \rightarrow 0$. This, together with the uniform boundedness assumption on the elements of $\mathcal{F}$, immediately yields

$$\begin{aligned} \left| \sup_{\varphi \in \mathcal{F}} \left| \int k_t \varphi d\mu \right| - \sup_{\varphi \in \mathcal{F}} \left| \int k \varphi d\mu \right| \right| &\leq \sup_{\varphi \in \mathcal{F}} \int |k_t - k| |\varphi| d\mu \\ &\leq \left(\sup_{\varphi \in \mathcal{F}} \|\varphi\|_\infty \right) \int |k_t - k| d\mu \leq \gamma \int |k_t - k| d\mu, \end{aligned}$$

and concludes the proof for the regime $n/m \rightarrow 0^+$.

We omit the proof for the regime $t \rightarrow \infty$, as it follows by analogous steps to the previous case. $\square$

We also formally justify the claim made in Section 4.2 that, on compact domains, the minimax testing rate with separation measured by $\|\psi_r\|_2$ is equivalent, up to multiplicative constants, to the minimax testing rate when the separation is measured with $\|g - rf(\int r f d\mu)^{-1}\|_2$. By Proposition 19 below, the same conclusion applies to $\|f - g/r(\int g/r d\mu)^{-1}\|_2$.

Proposition 18. *Assume that $m \leq n \leq \tau m$ for a fixed $\tau \geq 1$, $\text{supp}(f) = \text{supp}(g) = \text{supp}(r) = [0, 1]^d$, $\|f\|_\infty \vee \|g\|_\infty \leq M$, and $c \leq r(x) \leq C$ for all $x \in [0, 1]^d$. There exist positive constants $C_0 \equiv C_0(\tau, c, C, M)$ and $c_0 \equiv c_0(\tau, c, C, M)$ such that $c_0 \|\psi_r\|_2 \leq \|g - \bar{r}f\|_2 \leq C_0 \|\psi_r\|_2$, where $\bar{r} = r / \int_{[0, 1]^d} r f d\mu$.*

Proof. We can assume without loss of generality that $\int_{[0, 1]^d} r f d\mu = 1$. Otherwise, it is sufficient to replace r by $r / \int_{[0, 1]^d} r f d\mu = 1$; this normalization only rescales the relevant quantities by multiplicative constants depending on c and C via bounds on $\int_{[0, 1]^d} r f d\mu$, which we absorb into c_0 and C_0 . Recall that $h = \frac{nf + mg}{n + \lambda_0 mr}$, and $\lambda_0 = (\int r h d\mu)^{-1}$, which was shown in the proof of Lemma 4. Define also $\tilde{\psi}_r := f - h = \frac{\lambda_0 mr f - mg}{n + \lambda_0 mr}$, so that $(n + m) \tilde{\psi}_r = m \psi_r$. We have

$$1 - \lambda_0^{-1} = \int r f d\mu - \int r h d\mu = \int r \tilde{\psi}_r d\mu,$$

by which $\lambda_0 - 1 = \int r \tilde{\psi}_r d\mu / \int r h d\mu$. Based on this, we have

$$g - rf = g - \lambda_0 rf + (\lambda_0 - 1)rf = -\frac{n + \lambda_0 mr}{m} \tilde{\psi}_r + \frac{\int r \tilde{\psi}_r d\mu}{\int r h d\mu} rf,$$

implying that

$$\begin{aligned} \|g - rf\|_2 &\leq \|(n/m + \lambda_0 r) \tilde{\psi}_r\|_2 + \left| \frac{\int r \tilde{\psi}_r d\mu}{\int r h d\mu} \right| \|rf\|_2 \\ &\leq \left\{ \tau + C/c + \frac{\|r\|_2 \|rf\|_2}{\int r h d\mu} \right\} \|\tilde{\psi}_r\|_2 \leq \{\tau + C/c + C^2 M^{1/2}/c\} \|\tilde{\psi}_r\|_2, \end{aligned}$$

where in the last step we used $\int_{[0, 1]^d} r^2 d\mu \leq C^2$, $\int_{[0, 1]^d} r^2 f^2 d\mu \leq C^2 M$, $\int_{[0, 1]^d} r h d\mu \geq c$; in particular this follows from the fact that λ_0 is such that $\int h d\mu = 1$.

Similarly, we can use the fact that $\lambda_0 - 1 = \int r \tilde{\psi}_r d\mu / \int r h d\mu$ to write

$$1 - \lambda_0^{-1} = (1 - \lambda_0^{-1}) \int \frac{\lambda_0 mr}{n + \lambda_0 mr} rf d\mu + \int \frac{mr(rf - g)}{n + \lambda_0 mr} d\mu,$$

which further implies $1 - \lambda_0^{-1} = \int \frac{mr(rf-g)}{n+\lambda_0 mr} d\mu / \int \frac{nr f}{n+\lambda_0 mr} d\mu$ upon reordering. This, together with

$$\tilde{\psi}_r = \frac{(\lambda_0 - 1)mr f}{n + \lambda_0 mr} + \frac{m(rf - g)}{n + \lambda_0 mr},$$

allows bounding

$$\begin{aligned} \|\tilde{\psi}_r\|_2 &\leq \left\| \frac{m}{n + \lambda_0 mr} (g - rf) \right\|_2 + \left\| \frac{(\lambda_0 - 1)mr}{n + \lambda_0 mr} f \right\|_2 \leq \|g - rf\|_2 + |1 - \lambda_0^{-1}| \|f\|_2 \\ &\leq \|g - rf\|_2 + \left| \frac{\int \frac{mr(rf-g)}{n+\lambda_0 mr} d\mu}{\int \frac{nr f}{n+\lambda_0 mr} d\mu} \right| \|f\|_2 \leq \{1 + CM^{1/2}(c+C)/c\} \|g - rf\|_2. \end{aligned}$$

Notice that in the third inequality we used the fact that $\int \frac{mr}{n+\lambda_0 mr} d\mu \leq \lambda_0^{-1} \leq C$ and $\int \frac{nr f}{n+\lambda_0 mr} d\mu \geq \int \frac{rf}{1+\lambda_0 r} d\mu \geq \int \frac{rf}{1+C/c} d\mu \geq c/(c+C)$. This yields the reverse inequality and shows that $\|\tilde{\psi}_r\|_2 \asymp_{\tau, C, c, M} \|g - rf\|_2$. Finally, since $m \leq n \leq \tau m$, we obtain $\|\tilde{\psi}_r\|_2 = \|\frac{m}{n+m} \psi_r\|_2 \asymp_{\tau} \|\psi_r\|_2$, which completes the proof. $\square$

A key quantity that appears throughout the paper is the function $\psi_r := (1+n/m)(f-h)$, which intuitively captures the discrepancy between g and rf , as formalized in Proposition 18 above. We now justify the claim that ψ_r is a more natural notion of separation by showing that it is invariant, up to a sign, to taking reciprocal transformations of r and relabelling the samples.

Proposition 19. Define $\psi(n, f, m, g, r) := (n+m)m^{-1}(f-h)$, where $h \equiv h(n, f, m, g, r) = (nf+mg)/(n+\lambda_0 mr)$ for a suitable constant $\lambda_0 > 0$ such that $\int h d\mu = 1$. We have $\psi(n, f, m, g, r) = -\psi(m, g, n, f, r^{-1})$.

Proof. Arguing as in Remark 2, we can show that the constant $\theta_0 > 0$ that makes $h' \equiv h(m, g, n, f, r^{-1}) = (mg+nf)/(m+\theta_0 nr^{-1})$ a density satisfies $\theta_0 = \lambda_0^{-1}$, where λ_0 is such that $\int h(n, f, m, g, r) d\mu = 1$. This implies that

$$\begin{aligned} \psi(m, g, n, f, r^{-1}) &= (1+m/n)(g-h') = (1+m/n) \left(g - \frac{mg+nf}{m+n\lambda_0^{-1}r^{-1}} \right) = \frac{n+m}{n} \frac{ng\lambda_0^{-1}r^{-1} - nf}{m+n\lambda_0^{-1}r^{-1}} \\ &= \frac{n+m}{n} \frac{ng - \lambda_0 nr f}{n + \lambda_0 mr} = \frac{n+m}{m} \frac{mg - \lambda_0 mr f}{n + \lambda_0 mr} = -\psi(n, f, m, g, r), \end{aligned}$$

thus concluding the proof. $\square$

Finally, we further discuss Proposition 11 in Section 5 and show, in the conditional two-sample setting, how to implement sample splitting so that $\text{TV}(\{\tilde{r}f\}^{\otimes(n-N)}, \{\tilde{r}_*f\}^{\otimes(n-N)}) \xrightarrow{\mathbb{P}} 0$. Assume for simplicity that $n = m$ and write $Z = (X_1, \dots, X_n, Y_1, \dots, Y_n)$. Split Z into an estimation and a testing part, $Z^{\text{estim}} = (X_1, \dots, X_N, Y_1, \dots, Y_N)$ and $Z^{\text{test}} = (X_{N+1}, \dots, X_n, Y_{N+1}, \dots, Y_n)$ with $1 \leq N < n$; use Z^{estim} to fit $\hat{r}$ as an estimator of r_* , then run Algorithm 2 based on $\hat{r}$ on Z^{test} . For Proposition 11 to be practically informative, we must identify regimes in which N can be chosen so that r_* is estimated with sufficient accuracy and the resulting excess type I error is small. As an illustrative example, we use Theorem 2 of Nguyen et al. (2008) to control estimation of the marginal density ratio. Alternative density-ratio estimators, such as those analyzed in Theorem 1 and Example 1 of Sugiyama et al. (2008) or Theorem 2 of Kanamori et al. (2012), can yield analogous guarantees under comparable regularity conditions.

Example 1. Consider the setting of the conditional two-sample testing problem with $\mathcal{X} \times \mathcal{Y} \subseteq \mathbb{R}^{d_X} \times \mathbb{R}^{d_Y}$, i.e. we identify the X -values with $S := [d_X] \subset [d_X + d_Y]$ and the Y -values with $\{d_X + 1, \dots, d_X + d_Y\}$. Here, the density ratio corresponds to $f_X^{(2)}(x)/f_X^{(1)}(x)$, and we can write $r(x, y) = r_S(x)$ for a function r_S acting on $\mathbb{R}^{d_X}$. We will now show that the estimation error is small provided that r_S is sufficiently smooth. In particular, consider $\mathcal{R} = \mathcal{S}_{d_X}^s(L)$, the Sobolev ball introduced in (14), with integer s satisfying $s > d_X/2$, and assume that r_S lies in $\mathcal{R}$. Let $\hat{r}_S$ represent the first estimator introduced in Nguyen et al. (2008) for r_S . Define also $\bar{r}_S := r_S / \int r_S f d\mu_X$ and $\tilde{r}_S := \hat{r}_S / \int \hat{r}_S f d\mu_X$. If $0 < c \leq r(\cdot) \leq C$ for all $r \in \mathcal{R}$, Theorem 2 in Nguyen et al. (2008) and the remarks thereafter ensure that $\int (\sqrt{\bar{r}_S} f - \sqrt{\tilde{r}_S} f)^2 d\mu = \mathcal{O}_{\mathbb{P}}(N^{-\frac{2s}{2s+d}})$. As a result, writing $H^2(p, q) = 2(1 - \int \sqrt{pq} d\mu)$ for the squared Hellinger distance between densities p and q , the Cauchy–Schwarz inequality gives

$$H^2(\{\tilde{r}_S f\}, \{\bar{r}_S f\}) = \int (\sqrt{\tilde{r}_S} f - \sqrt{\bar{r}_S} f)^2 d\mu_X \leq 4 \int (\sqrt{\hat{r}_S} f - \sqrt{\bar{r}_S} f)^2 d\mu_X = \mathcal{O}_{\mathbb{P}}(N^{-\frac{2s}{2s+d}}).$$

This, combined with standard inequalities, yields $\text{TV}(\{\tilde{r}_S f\}^{\otimes(n-N)}, \{\bar{r}_S f\}^{\otimes(n-N)}) \lesssim \sqrt{(n-N)N^{-\frac{2s}{2s+d}}}$ with high probability, which implies that the total variation vanishes for sufficiently large N , i.e. $n \ll N^{\frac{2s}{2s+d}}$.

Related results in the literature can be leveraged to derive analogous guarantees for alternative estimators of r_* across different settings. Broadly speaking, theoretical insights emphasize the importance of choosing the estimation sample size N to be significantly larger than the testing sample size $n - N$ in order to keep the excess type I error low. More broadly, Proposition 11 together with Example 1 shows that our method is a viable approach to conditional two-sample testing. Its key feature is permutation-based calibration, so any excess type I error arises solely from density ratio estimation. Although our non-uniform permutation calibration is novel, the use of marginal density ratios parallels Lee et al. (2024); their classifier-based and linear-time maximum mean discrepancy procedures, reweighted by an estimate of marginal importance, are asymptotically valid, and align with the spirit of Section 5. The problem is well studied, with numerous alternatives (Hu and Lei, 2020; Chatterjee et al., 2024; Yan et al., 2024; Huang et al., 2024; Zheng et al., 2025; Chen and Lei, 2025); we compare Algorithm 2 with Lee et al. (2024) and several of these methods in Section 6 and Appendix C.1.

Appendix C Supplementary experiments and setup

C.1 Additional simulation studies

We present additional simulation studies beyond those in Section 6. First, we consider a univariate setting and compare it to the bivariate one shown in Fig. 1(a). We set $P_f = \mathcal{N}(0, 1)$ and define $P_g = (1 + \eta)^{-1} \mathcal{N}(0, 1/9) + \eta(1 + \eta)^{-1} \text{Exp}(1)$ for $\eta \in \{0, \dots, 0.40\}$. For $\eta = 0$, g satisfies the null hypothesis with $r(x) = e^{-4x^2}$, while larger values of η correspond to greater departures from the null. The results are presented in Figure 4. The purple curve corresponds to (E1), and we compare it against (E2), shown in green. Furthermore, we also consider a similar setting (shown by the orange and blue lines) in which $P_f = \mathcal{N}(0, 1)$ and $P_g = (1 + \eta)^{-1} \mathcal{N}(0, 1/3) + \eta(1 + \eta)^{-1} \text{Exp}(1)$ for $\eta \in \{0, \dots, 0.40\}$, which means that the null hypothesis is satisfied for $r'(x) = e^{-x^2}$. Across all four simulations we fix $n = m = 150$. For each setting, each test is run 500 times, and the mean average decision is reported as the estimated power

function. Compared with Fig. 1(a), the four curves have approximately the same shape. Note, however, that here we consider $\eta \in \{0, \dots, 0.40\}$ rather than $\{0, \dots, 0.80\}$. Although this is not a general rule, it is interesting that with twice the dimensionality we recover the same qualitative behaviours by considering alternatives that are twice as far apart.

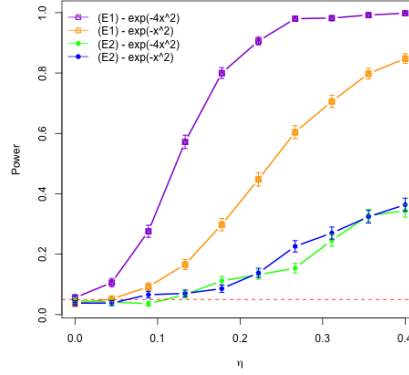


Figure 4: Purple: simulation of (E1); here $r(x) = e^{-4x^2}$. Orange: same setting but with $r'(x) = e^{-x^2}$. Green and blue: alternative approaches based on (E2). Error bars show ± 1 standard errors.

Furthermore, in Table 2 we include a more comprehensive version of Table 1. Using the same experimental design as in Section 6.2.3 for the diamonds dataset, we compare our method against a broader set of alternatives. As before, we report the performance of the single-split classifier-based test and the linear-time maximum mean discrepancy test from Lee et al. (2024), but also include their cross-fit counterparts. Additionally, we include the conformal prediction test based on conformity scores (Hu and Lei, 2020), and the debiased conformal prediction test, which enhances the former through Neyman orthogonality and cross-fitting (Chen and Lei, 2025). The results are shown in Table 2 and agree with Fig. 4 in Lee et al. (2024). The cross-fitted variants exhibit similar size and higher power if compared to their non-cross-fitted counterparts presented in the main body. The conformal prediction test attains the highest power in the table but shows slightly inflated type I error. This likely reflects the added sensitivity of conformal-prediction-based ranks together with error from estimating density ratios and other nuisance functions when no orthogonalization is used. By contrast, the debiased conformal prediction test achieves very high power, often exceeding our method, while maintaining conservative type I error. This pattern aligns with its Neyman-orthogonal, cross-fitted construction, which mitigates first-order estimation bias.

C.2 Training setup for neural ratio estimators

We briefly describe the training of the neural density-ratio estimator $\hat{r}_{\text{NRE}}$ and its balanced variant $\hat{r}_{\text{BNRE}}$ used in Section 6.2.2. We use the respective losses as implemented in the Python package LAMPE. The estimators take the concatenated input $(\theta_1, \theta_2, x_1, x_2) \in \mathbb{R}^4$ and output an estimate of $\log r(\theta_1, \theta_2, x_1, x_2)$ via a multilayer perceptron with five hidden layers, each of width 128, and exponential linear unit activations. For each training budget $N_{\text{train}} \in \{2^4, 2^7, 2^9, 2^{12}, 2^{15}\}$, we draw a fixed set of N_{train} joint pairs once, shuffle each epoch, and optimize with AdamW, with learning rate 10^{-3} and gradient clipping set to 1, for 400 epochs

Table 2: Simulation results for the conditional two-sample testing problem on the diamonds dataset. LL stands for linear logistic regression, KLR for kernel logistic regression. (E1) is used, and compared against several alternative methods. CLF stands for the single-split classifier test, MMD- l for the linear-time maximum mean discrepancy test. $\dagger$ CLF and $\dagger$ MMD- l indicate their cross-fitted counterpart. CP indicates the conformal prediction test, and DCP its debiased version. N denotes the total sampling budget.

Estimator	Hypothesis	Test	$N = 200$	$N = 400$	$N = 800$	$N = 1200$	$N = 1600$	$N = 2000$
LL	Null	CLF	0.0900	0.0650	0.0750	0.0450	0.0675	0.0425
LL	Null	CP	0.0650	0.0875	0.0925	0.0575	0.1100	0.0925
LL	Null	$\dagger$ CLF	0.0950	0.0675	0.0850	0.0750	0.0450	0.0675
LL	Null	$\dagger$ MMD- l	0.0700	0.0700	0.0675	0.0550	0.0750	0.0625
LL	Null	MMD- l	0.0750	0.0650	0.0750	0.0500	0.0575	0.0575
LL	Null	DCP	0.0375	0.0400	0.0350	0.0425	0.0325	0.0400
LL	Null	DRPT (E1)	0.0600	0.0550	0.0850	0.0600	0.0500	0.0350
LL	Alternative	CLF	0.1575	0.2050	0.2650	0.3600	0.3925	0.4800
LL	Alternative	CP	0.2950	0.5275	0.6900	0.8700	0.9050	0.9300
LL	Alternative	$\dagger$ CLF	0.2425	0.3675	0.4700	0.6225	0.6575	0.7575
LL	Alternative	$\dagger$ MMD- l	0.0975	0.1100	0.0900	0.1075	0.1125	0.1275
LL	Alternative	MMD- l	0.0650	0.0675	0.0850	0.0825	0.0975	0.0850
LL	Alternative	DCP	0.1750	0.4150	0.6750	0.7925	0.8250	0.7950
LL	Alternative	DRPT (E1)	0.1500	0.1800	0.4150	0.5450	0.6350	0.7150
KLR	Null	CLF	0.0750	0.0575	0.0675	0.0350	0.0475	0.0450
KLR	Null	CP	0.0450	0.0675	0.0575	0.0400	0.0675	0.0500
KLR	Null	$\dagger$ CLF	0.0825	0.0375	0.0475	0.0400	0.0425	0.0500
KLR	Null	$\dagger$ MMD- l	0.0200	0.0450	0.0675	0.0650	0.0550	0.0550
KLR	Null	MMD- l	0.0600	0.0625	0.0725	0.0575	0.0550	0.0600
KLR	Null	DCP	0.0275	0.0150	0.0275	0.0175	0.0275	0.0200
KLR	Null	DRPT (E1)	0.0400	0.0350	0.0600	0.0500	0.0600	0.0300
KLR	Alternative	CLF	0.0975	0.1525	0.2600	0.3550	0.3675	0.4450
KLR	Alternative	CP	0.3100	0.5450	0.7450	0.9025	0.9600	0.9725
KLR	Alternative	$\dagger$ CLF	0.1675	0.2650	0.3900	0.5750	0.6275	0.7150
KLR	Alternative	$\dagger$ MMD- l	0.0700	0.0625	0.0950	0.1000	0.1225	0.1425
KLR	Alternative	MMD- l	0.0725	0.0675	0.0800	0.0900	0.0950	0.1050
KLR	Alternative	DCP	0.2425	0.4325	0.6850	0.8175	0.9200	0.9700
KLR	Alternative	DRPT (E1)	0.1250	0.1750	0.3750	0.5000	0.6300	0.6650

using batch size $\min(256, N_{\text{train}})$.

Appendix D Algorithm 2 for discrete data

D.1 Methodology

One significant drawback of permutation tests is their computational cost, which in our setting stems from choosing large values for H in (2) and S in Algorithm 2. In this subsection, we introduce an alternative method for discrete data with finite support that is computationally more efficient, since it enables direct sampling from (3) without relying on the Markov Chain Monte Carlo sampler. Let $\mathcal{X} = \{0, \dots, J\} =: \mathcal{J}$ with $J \geq 1$, and let $X_i \stackrel{\text{i.i.d.}}{\sim} f$ and $Y_i \stackrel{\text{i.i.d.}}{\sim} g$, where $\mathbb{P}_f\{X = j\} = f_j$ and $\mathbb{P}_g\{Y = j\} = g_j$, for all $j \in \mathcal{J}$, with $\sum_{j=0}^J f_j = \sum_{j=0}^J g_j = 1$. Given a sequence of positive numbers $(r_0, \dots, r_J)$, we can write

$r(x) = \sum_{j \in \mathcal{J}} r_j \mathbb{1}\{x = j\}$, so that the testing problem becomes

$$H_0 : g_j \propto r_j f_j \quad \text{for all } j \in \mathcal{J}.$$

At the sample level, the methodology relies on generating a permutation σ given Z according to the distribution (3). Now, conditioned on the data, each permutation preserves the total number of units in each of the $J + 1$ categories, where the count for category $j \in \mathcal{J}$ is denoted by tot_j . What changes is how these values are split between the first n and the last m data points. We can represent this using the following table:

	$Z_{\sigma(1:n)}$	$Z_{\sigma(n+1:n+m)}$	+
0	$\text{tot}_0 - N_{Y,0}^\sigma$	$N_{Y,0}^\sigma$	tot_0
$\vdots$	$\vdots$	$\vdots$	$\vdots$
J	$\text{tot}_J - N_{Y,J}^\sigma$	$N_{Y,J}^\sigma$	tot_J
+	n	m	$m + n$

where $N_{Y,j}^\sigma = \#\{i \notin [n] : Z_{\sigma(i)} = j\} = \mathbb{1}\{Z_{\sigma(n+1)} = j\} + \dots + \mathbb{1}\{Z_{\sigma(n+m)} = j\}$ for all $j \in \mathcal{J}$. If we restrict attention to test statistics that are functions of this frequency table, which is natural given that it is a sufficient statistic in our model, we can characterize (3) through the distribution of $(N_{Y,0}^\sigma, \dots, N_{Y,J}^\sigma) \mid Z$. In this regard, given the data and for $w = (w_0, \dots, w_J)$ such that $\sum_{j=0}^J w_j = m$ and $\max(0, \text{tot}_j - n) \leq w_j \leq \min(m, \text{tot}_j)$ for all $j \in \mathcal{J}$, we have

$$\begin{aligned} \mathbb{P}\{N_{Y,0}^\sigma = w_0, \dots, N_{Y,J}^\sigma = w_J \mid Z\} &= \sum_{\{\sigma \in \mathcal{S}_{n+m} : (N_{Y,0}^\sigma, \dots, N_{Y,J}^\sigma) = w\}} \mathbb{P}\{\sigma^{(1)} = \sigma \mid Z\} \\ &\propto \sum_{\{\sigma \in \mathcal{S}_{n+m} : (N_{Y,0}^\sigma, \dots, N_{Y,J}^\sigma) = w\}} \prod_{i \notin [n]} r(Z_{\sigma(i)}) = \sum_{\{\sigma \in \mathcal{S}_{n+m} : (N_{Y,0}^\sigma, \dots, N_{Y,J}^\sigma) = w\}} \prod_{j \in \mathcal{J}} r_j^{N_{Y,j}^\sigma} \\ &= \left(\prod_{j \in \mathcal{J}} r_j^{w_j} \right) \#\{\sigma \in \mathcal{S}_{n+m} : (N_{Y,0}^\sigma, \dots, N_{Y,J}^\sigma) = w\} = n!m! \prod_{j \in \mathcal{J}} r_j^{w_j} \binom{\text{tot}_j}{w_j} \propto \prod_{j \in \mathcal{J}} r_j^{w_j} \binom{\text{tot}_j}{w_j}. \end{aligned} \quad (59)$$

Notice that in the penultimate step we used the fact that $\#\{\sigma \in \mathcal{S}_{n+m} : (N_{Y,0}^\sigma, \dots, N_{Y,J}^\sigma) = w\} = n!m! \prod_{j \in \mathcal{J}} \binom{\text{tot}_j}{w_j}$, as there are $\binom{\text{tot}_j}{w_j}$ ways of choosing w_j many j 's for the last m data points for all $j \in \mathcal{J}$. Considering all the possible ways in which we can further permute the first n and last m values gives the extra factor of $n!m!$. This shows that $(N_{Y,0}^\sigma, \dots, N_{Y,J}^\sigma) \mid Z$ is distributed according to Fisher's multivariate noncentral hypergeometric distribution, which is a generalization of the hypergeometric distribution where sampling probabilities are adjusted by weight factors (see, e.g. McCullagh and Nelder, 1989, Section 7). Coming back to the testing problem (1), the previous argument shows that in the case of discrete data with finite support we can avoid sampling permutations from (3), as it is sufficient to sample $(N_{Y,0}^\sigma, \dots, N_{Y,J}^\sigma) \mid Z$ from (59). This can be done efficiently using for example the R-function `rMFNCHypergeo` from R-package `BiasedUrn`.

Compared to the Markov Chain Monte Carlo-based approach, aside from reducing computational runtime, Algorithm 2 copies of the table above generated through independent and identically distributed draws from (59) are conditionally independent, given the row totals, across different $h \in [H]$, while the $Z^{(h)}$'s generated by Algorithm 2 exhibit non-zero correlation due to their shared initialization. Nonetheless, this

dependence decreases for larger and larger values of S . Regarding its theoretical guarantees, finite-sample validity is ensured by Theorem 1.

D.2 Power analysis

We now provide some further insights for this version of Algorithm 2, henceforth referred to as the discrete density ratio permutation test. For the case $\mathcal{X} = \{0, 1\}$, i.e. $J = 1$, we recall that the testing problem (1) is equivalent to

$$H_0 : \frac{g_1}{g_0} = \frac{r_1 f_1}{r_0 f_0},$$

for $r_0, r_1 > 0$. As a result, we may assume without loss of generality that $r_0 = 1$, since our interest lies solely in the ratio r_1/r_0 , and $r_1 \equiv r \geq 1$; if this condition does not hold, we can simply switch the roles of f and g and consider $1/r$ instead. Now, it is instructive to analyze the behaviour of the sample mean of the permuted data. In this regard, Lemma 7 implies the following unconditional result. In what follows, with a slight abuse of notation, whenever we write an expression such as $(\tau f_1 + g_1)/(\tau + \lambda r)$ with $\tau = \infty$, it is to be interpreted as its limit as $\tau \rightarrow \infty$; for example, $(\tau f_1 + g_1)/(\tau + \lambda r)|_{\tau=\infty} := \lim_{\tau \rightarrow \infty} (\tau f_1 + g_1)/(\tau + \lambda r) = f_1$.

Corollary 20. *Let $N_{Y,1}^\sigma = \sum_{j=n+1}^{n+m} Z_{\sigma(j)}$, where σ is sampled according to (3). Calling $\tau := \lim_{n,m \rightarrow \infty} n/m \in \mathbb{R}_+ \cup \{0, \infty\}$, for all $k \in \mathbb{N}$ we have $\mathbb{E}[(m^{-1} N_{Y,1}^\sigma)^k] \rightarrow \gamma_1^k$ as $n, m \rightarrow \infty$, where*

$$\gamma_1(f_1, g_1, r, \tau) \equiv \gamma_1 = \begin{cases} \frac{\tau f_1 + g_1}{\tau + 1} + \frac{(r-1)^{\frac{\tau-1}{\tau+1}} (\tau f_1 + g_1) + \tau + r - \sqrt{(\tau + r + (r-1)(\tau f_1 + g_1))^2 - 4(r-1)r(\tau f_1 + g_1)}}{2(r-1)} & \text{if } r > 1, \\ \frac{\tau f_1 + g_1}{\tau + 1} & \text{if } r = 1. \end{cases} \quad (60)$$

Proof. Let μ be the counting measure and equip the set $\{0, 1\}$ with the discrete topology. We can show as a corollary of Lemma 7 in Subsection 3.2 that

$$\frac{1}{m} \sum_{j=n+1}^{n+m} \varphi(Z_{\sigma(j)}) \xrightarrow{\mathbb{P}} \int \varphi \lambda_\infty r_{\text{func}} \frac{\tau f + g}{\tau + \lambda_\infty r_{\text{func}}} d\mu$$

for every bounded and continuous function φ , where $r_{\text{func}}(x) = r \mathbb{1}\{x = 1\} + \mathbb{1}\{x = 0\}$ and $\lambda_\infty > 0$ is the positive solution of

$$1 = \int \frac{\tau f + g}{\tau + \lambda_\infty r_{\text{func}}} d\mu = \frac{\tau f_1 + g_1}{\tau + \lambda_\infty r} + \frac{\tau f_0 + g_0}{\tau + \lambda_\infty} = \frac{\tau f_1 + g_1}{\tau + \lambda_\infty r} - \frac{\tau f_1 + g_1}{\tau + \lambda_\infty} + \frac{\tau + 1}{\tau + \lambda_\infty}.$$

As $\mathcal{X} = \{0, 1\}$, we can choose $\varphi = \text{id}$ and obtain that $m^{-1} N_{Y,1}^\sigma \xrightarrow{\mathbb{P}} \lambda_\infty r (\tau f_1 + g_1)/(\tau + \lambda_\infty r) = \gamma_1$, where in the last step we simply plugged in the expression for λ_∞ , which can be found explicitly by solving the equation above. We have thus established that $m^{-1} N_{Y,1}^\sigma \xrightarrow{\mathbb{P}} \gamma_1$, which implies the existence of a subsequence along which convergence holds almost surely; applying the dominated convergence theorem yields $\mathbb{E}[(m^{-1} N_{Y,1}^\sigma)^k] \rightarrow \gamma_1^k$ for all $k \in \mathbb{N}$ along this subsequence, and by the uniqueness of limits, the convergence extends to the entire sequence. This completes the proof. $\square$

This result shows that $m^{-1} N_{Y,1}^\sigma \xrightarrow{\mathbb{P}} \gamma_1$, which offers a more explicit interpretation of the result in Lemma 7 and emphasizes the intricate dependence of the limiting distribution of the permuted data on the initial pa-

rameters (f_1, g_1, r, τ) , even in simple cases. Interestingly, under the null hypothesis, $\gamma_1 = g_1$, as expected. Furthermore, if $r = 1$, we find that $\gamma_1 = (\tau f_1 + g_1)/(\tau + 1)$, reflecting the fact that Algorithm 2 selects permutations uniformly at random when $r = 1$, consistent with (6). Similarly, we can establish an analogous result for the convergence of $n^{-1} \sum_{i=1}^n Z_{\sigma(i)}$ to ν_1 , where ν_1 satisfies $\frac{\tau}{1+\tau} f_1 + \frac{1}{1+\tau} g_1 = \frac{\tau}{1+\tau} \nu_1 + \frac{1}{1+\tau} \gamma_1$, by leveraging the constraint that the total number of ones must remain conserved.

As for the analysis of power, Theorem 6 already guarantees that the discrete density ratio permutation test is consistent when the test statistic is an integral probability metric that depends on $(N_{Y,0}^\sigma, \dots, N_{Y,J}^\sigma)$. We now establish a further consistency result for an alternative choice of test statistic. In this setting, the null hypothesis (1) is equivalent to

$$H_0 : \frac{g_j}{g_0} = \frac{r_j f_j}{r_0 f_0} \quad \text{for all } j \in [J],$$

for fixed $r = (r_0, r_1, \dots, r_J) \in R_+^{J+1}$. As before, we can assume $r_0 = 1$ without loss of generality. This motivates the introduction of

$$T(Z_\sigma) = \frac{1}{nm} \sum_{j=1}^J \left| r_j^{-1/2} N_{Y,j}^\sigma(\text{tot}_0 - N_{Y,0}^\sigma) - r_j^{1/2} N_{Y,0}^\sigma(\text{tot}_j - N_{Y,j}^\sigma) \right|, \quad (61)$$

which serves as an estimator of $D(f, g) \equiv D^r(f, g) := \sum_{j \in [J]} \left| r_j^{-1/2} g_j f_0 - r_j^{1/2} f_j g_0 \right|$, which is a population measure of discrepancy that characterizes the null. Similarly to $T_{\mathcal{F},r}$ introduced in the main body, this quantity is invariant under taking the reciprocal transformations of r and relabelling the sample. We can prove the following result.

Proposition 21. *Let $\mathcal{X} = \{0, \dots, J\} =: \mathcal{J}$ with $J \geq 1$, $r = (r_0 = 1, r_1, \dots, r_J) \in \mathbb{R}_+^{J+1}$, and define $H_0 : g_j \propto r_j f_j$ for all $j \in \mathcal{J}$. Provided that $H > \lceil 1/\alpha - 1 \rceil$ for fixed $\alpha \in (0, 1)$, the discrete density ratio permutation test using (61) as its test statistic is consistent for H_0 .*

Proof. Let μ be the counting measure, and equip the set $\mathcal{J}$ with the discrete topology. Then, arguing as in the proof of Corollary 20, we can use Lemma 7 to show that $m^{-1} N_{Y,j}^\sigma = m^{-1} \sum_{i=n+1}^{n+m} \mathbb{1}\{Z_{\sigma(i)} = j\} \xrightarrow{\mathbb{P}} \lambda_\infty r_j (\tau f_j + g_j)/(\tau + \lambda_\infty r_j)$, where λ_∞ is a positive solution of $\sum_{k=0}^J (\tau f_k + g_k)/(\tau + \lambda_\infty r_k) = 1$. As a result, the generic j -th term of the test statistic (61) satisfies

$$\begin{aligned} & \frac{1}{nm} \left\{ r_j^{-1/2} N_{Y,j}^\sigma(\text{tot}_0 - N_{Y,0}^\sigma) - r_j^{1/2} N_{Y,0}^\sigma(\text{tot}_j - N_{Y,j}^\sigma) \right\} \\ & \xrightarrow{\mathbb{P}} r_j^{-1/2} \lambda_\infty r_j \frac{\tau f_j + g_j}{\tau + \lambda_\infty r_j} \frac{\tau f_0 + g_0}{\tau + \lambda_\infty r_0} - r_j^{1/2} \lambda_\infty r_0 \frac{\tau f_0 + g_0}{\tau + \lambda_\infty r_0} \frac{\tau f_j + g_j}{\tau + \lambda_\infty r_j} = 0, \end{aligned}$$

since $r_0 = 1$. This shows that $T(Z_\sigma) \xrightarrow{\mathbb{P}} 0$.

As for the behaviour of our test statistic under for the unpermuted sample, we can use the weak law of large numbers to show that $T(Z) \xrightarrow{\mathbb{P}} D(f, g) > 0$ under the alternative. Since we assumed $H > \lceil 1/\alpha - 1 \rceil$,

we have $\alpha(1+H) - 1 > 0$ and we can therefore apply Markov's inequality to see that

$$\begin{aligned}\mathbb{P}\{p > \alpha\} &= \mathbb{P}\left\{1 + \sum_{h=1}^H \mathbb{1}\{T(Z_{\sigma^{(h)}}) \geq T(Z)\} > \alpha(1+H)\right\} \\ &\leq \frac{\mathbb{E}\left[\sum_{h=1}^H \mathbb{1}\{T(Z_{\sigma^{(h)}}) \geq T(Z)\}\right]}{\alpha(1+H) - 1} = \frac{H}{\alpha(1+H) - 1} \mathbb{P}\{T(Z_{\sigma^{(1)}}) \geq T(Z)\} \\ &\leq \frac{H}{\alpha(1+H) - 1} (\mathbb{P}\{T(Z_{\sigma^{(1)}}) \geq \tfrac{1}{2}D(f, g)\} + \mathbb{P}\{T(Z) \leq \tfrac{1}{2}D(f, g)\}) \rightarrow 0,\end{aligned}$$

where the penultimate step follows from exchangeability. This implies $\mathbb{P}\{p \leq \alpha\} \rightarrow 1$ and concludes the proof. $\square$

Finally, coming back again to the case of binary data, the dependence of the minimax separation on r can be analyzed more effectively compared to Theorem 10. In this regard, let $\mathcal{X} = \{0, 1\}$ and $r \geq 1$, and consider the measure of discrepancy defined above, i.e. $D(f, g) = |r^{-1/2}g_1f_0 - r^{1/2}f_1g_0|$. For fixed $r \geq 1$ and $\rho > 0$, consider

$$H_0 : \frac{g_1}{g_0} = r \frac{f_1}{f_0} \quad \text{vs.} \quad H_1^r(\rho) : D(f, g) \geq \rho.$$

Write Ψ for the set of all tests, that is randomized functions of $(X_1, \dots, X_n, Y_1, \dots, Y_m)$, and $\Psi(\alpha)$ for the set of tests of size α , with $\alpha \in (0, 1)$. For $\beta \in (0, 1 - \alpha)$, we may define the minimax separation as $\rho_r^* \equiv \rho_r^*(n, m, \alpha, \beta) := \inf \left\{ \rho > 0 : \inf_{\varphi \in \Psi(\alpha)} \sup_{(f, g) \in H_1^r(\rho)} \mathbb{E}_P(1 - \varphi) \leq \beta \right\}$, where $P = P_f^{\otimes n} \otimes P_g^{\otimes m}$. We now prove a lower bound on ρ_r^* .

Proposition 22. *Let ρ_r^* the minimax separation defined above and suppose that $\alpha + \beta < 1/2$. We have that $\rho_r^*(n, m, \alpha, \beta) \geq \sqrt{\frac{r}{(n \wedge m)(1+r)^2} \{1 - 2(\alpha + \beta)\}}$.*

Proof. For $0 < \rho^2 \leq \frac{r}{(1+r)^2}$, consider

$$(f^{(0)}, g^{(0)}) = \left(\left(\frac{1}{2}, \frac{1}{2} \right), \left(\frac{1}{1+r}, \frac{r}{1+r} \right) \right) \quad \text{and} \quad (f^{(1)}, g^{(1)}) = \left(\left(\frac{1-\gamma}{2}, \frac{1+\gamma}{2} \right), \left(\frac{1}{1+r}, \frac{r}{1+r} \right) \right),$$

with $\gamma = \frac{1+r}{\sqrt{r}}\rho$. Observe that $D(f^{(0)}, g^{(0)}) = 0$ and $D(f^{(1)}, g^{(1)}) = \rho$. We will use the well-known fact that the squared Hellinger distance between two discrete probability distributions p, q supported on $[J]$ is given by $H^2(p, q) = \sum_{i \in [J]} (\sqrt{p_i} - \sqrt{q_i})^2$. Writing $P_{f^{(i)}}$ for the Bernoulli distribution with parameter $f^{(i)}$ for $i \in \{0, 1\}$, we thus have

$$H^2(P_{f^{(0)}}, P_{f^{(1)}}) = \frac{1}{2} \left\{ \left(1 - \sqrt{1-\gamma} \right)^2 + \left(1 - \sqrt{1+\gamma} \right)^2 \right\} \leq \gamma^2 = \frac{(1+r)^2}{r} \rho^2,$$

where the last inequality relies on $(1 - \sqrt{1 \pm \gamma})^2 \leq \gamma^2$, and further shows that $H^2(P_{f^{(0)}}, P_{f^{(1)}}) \leq 1$ since $\rho^2 \leq \frac{r}{(1+r)^2}$. We can then bound the minimax risk using a standard Le-Cam two-point argument as

$$\begin{aligned}\alpha + \sup_{(f, g) \in H_1^r(\rho)} \mathbb{E}_P(1 - \varphi) &\geq \sup_{(f, g) \in H_0} \mathbb{E}_P \varphi + \sup_{(f, g) \in H_1^r(\rho)} \mathbb{E}_P(1 - \varphi) \\ &\geq 1 - \text{TV} \left(P_{f^{(0)}}^{\otimes n} \otimes P_{g^{(0)}}^{\otimes m}, P_{f^{(1)}}^{\otimes n} \otimes P_{g^{(1)}}^{\otimes m} \right) \geq 1 - \left\{ \text{TV} \left(P_{f^{(0)}}^{\otimes n}, P_{f^{(1)}}^{\otimes n} \right) + \text{TV} \left(P_{g^{(0)}}^{\otimes m}, P_{g^{(1)}}^{\otimes m} \right) \right\}\end{aligned}$$

$$\begin{aligned}
&= 1 - \text{TV} \left(P_{f^{(0)}}^{\otimes n}, P_{f^{(1)}}^{\otimes n} \right) \geq \frac{1}{2} \left(1 - \frac{1}{2} \text{H}^2(P_{f^{(0)}}, P_{f^{(1)}}) \right)^{2n} \geq \frac{1}{2} (1 - n \text{H}^2(P_{f^{(0)}}, P_{f^{(1)}})) \\
&\geq \frac{1}{2} \left(1 - \frac{n(1+r)^2}{r} \rho^2 \right),
\end{aligned}$$

where in the fifth inequality we used the fact that $(1-x)^n \geq 1-nx$ for $n \in \mathbb{N}$ and $x \leq 1$. The last display is lower bounded by $\alpha + \beta$ if and only if $\rho^2 \leq \frac{r}{n(1+r)^2} \{1 - 2(\alpha + \beta)\}$. Note that $0 < \rho^2 \leq \frac{r}{(1+r)^2}$ is necessarily satisfied since $n \geq 1$ and $0 \leq \alpha + \beta < 1/2$. A similar construction, where $f^{(0)} = f^{(1)}$ and $g^{(1)}$ is a perturbation of $g^{(0)}$, gives an analogous lower bound with m in place of n , and concludes the proof. $\square$

Proposition 22 suggests that the testing problem is the hardest when $r = 1$. As already mentioned in Subsection 4.2, this is in accordance with the goodness-of-fit testing problem, where the goal is to test the null hypothesis $f = f_0$ for a fixed density f_0 , based on independent and identically distributed samples $X_1, \dots, X_n \sim f$. In this setting, the minimax separation rate depends on the choice of f_0 , and it has been shown that the problem is hardest when f_0 is the uniform distribution (e.g. Balakrishnan and Wasserman, 2019). In complete analogy, and according to the simulation results in Section 6.1, Proposition 22 seems to indicate that $r = 1$ corresponds to the harder testing problem. In other words, more extreme shifts should be easier to detect. We validate this conjecture through simulations on synthetic data in Appendix D.3.

Establishing the optimality of the rate $1/\sqrt{r(n \wedge m)}$ is more delicate. While the two-moment method used in the proof of Theorem 9 suffices to derive an upper bound on ρ_r^* , it yields a loose dependence on r , even though it accurately captures the scaling with n and m . Specifically, we can show that $\text{Var}(T(Z)) \lesssim r/n$ and $\text{Var}(T(Z_\sigma)) \lesssim r^2/n$. The first inequality follows from the independence between the X 's and the Y 's, while the second relies on the following proposition, which we include for completeness.

Proposition 23. *Assume $r \geq 1$ and $\mathcal{X} = \{0, 1\}$. Define $N_{Y,1}^\sigma = \sum_{i=n+1}^{n+m} Z_{\sigma(i)}$, where σ is sampled according to (3), and let γ_1 be as in (60). Then*

$$\mathbb{E}[(N_{Y,1}^\sigma - m\gamma_1)^2] \leq (1+r)n \wedge m + 2f_1(1-f_1)n + 2g_1(1-g_1)m.$$

Proof. All the expectations are to be intended conditionally to Z . We will assume $n = \tau m$ throughout the proof to simplify the computations. Write $\mathcal{T} = \text{tot}_1$, and define

$$p_1^x = \frac{(n - \mathcal{T} + x)x}{(1+r)nm} \quad \text{and} \quad p_2^x = \frac{r(\mathcal{T} - x)(m - x)}{(1+r)nm}.$$

Let $\gamma(\mathcal{T})$ be the solution to $p_1^x = p_2^x$, i.e.

$$\gamma(\mathcal{T}) = \frac{1}{2(r-1)} \left\{ (r-1)\mathcal{T} + n + rm - \sqrt{[(r-1)\mathcal{T} + n - rm]^2 + 4rmn} \right\}.$$

We will show that $\mathbb{E}[(N_{Y,1}^\sigma - \gamma(\mathcal{T}))^2] \leq \frac{1+r}{2} n \wedge m$ and $\mathbb{E}[(m\gamma_1 - \gamma(\mathcal{T}))^2] \leq nf_1(1-f_1) + mg_1(1-g_1)$, which imply the desired result since $\mathbb{E}[(N_{Y,1}^\sigma - m\gamma_1)^2] \leq 2\mathbb{E}[(N_{Y,1}^\sigma - \gamma(\mathcal{T}))^2] + 2\mathbb{E}[(m\gamma_1 - \gamma(\mathcal{T}))^2]$. Now, as for the

latter, observe that $m\gamma_1 = \gamma(nf_1 + mg_1)$, and that $\gamma(\cdot)$ is 1-Lipschitz. This is due to fact that

$$|\gamma'(x)| \leq \frac{1}{2} \left(1 + \frac{|(r-1)x + n - rm|}{\sqrt{[(r-1)x + n - rm]^2 + 4rnm}} \right) \leq 1.$$

Hence,

$$\begin{aligned} \mathbb{E}[(m\gamma_1 - \gamma(\mathcal{T}))^2] &= \mathbb{E}[\{\gamma(nf_1 + mg_1) - \gamma(\mathcal{T})\}^2] \leq \mathbb{E}[\{(nf_1 + mg_1) - \mathcal{T}\}^2] \\ &= \mathbb{E} \left[\left\{ (nf_1 + mg_1) - \sum_{i=1}^n X_i - \sum_{i=1}^m Y_i \right\}^2 \right] = \text{Var} \left[\sum_{i=1}^n X_i \right] + \text{Var} \left[\sum_{i=1}^m Y_i \right] = nf_1(1 - f_1) + mg_1(1 - g_1). \end{aligned}$$

As for the other term, start by noticing that Algorithm 1 works the same if at every time step $t \in \mathbb{N}$ we just choose a single couple (i, j) with $i \in [n]$ and $j \in \{n+1, \dots, n+m\}$ at random, and then switch $Z_{\sigma_t(i)}$ with $Z_{\sigma_t(j)}$ with probability equal to $r^{Z_{\sigma_t(i)}} / (r^{Z_{\sigma_t(i)}} + r^{Z_{\sigma_t(j)}})$. This is not efficient from a computational point of view, but simplifies the proof, since every time step t corresponds at most to one switch. Now, let K_t be the sum of the last m observations after t steps of this simplified algorithm, and observe that it has the same distribution of $N_{Y,1}^\sigma$ for every $t \in \mathbb{N}$ when the procedure is initialized at stationarity. Thus, for $\gamma \equiv \gamma(\mathcal{T})$ to ease notation, it follows that

$$\begin{aligned} \mathbb{E}[(K_{t+1} - \gamma)^2 | K_t] &= (K_t - \gamma)^2 = p_1^{K_t} + p_2^{K_t} + 2\{(p_2^{K_t} - p_2^\gamma) - (p_1^{K_t} - p_1^\gamma)\}(K_t - \gamma) \\ &= p_1^{K_t} + p_2^{K_t} - \frac{2(K_t - \gamma)^2}{(1+r)nm} \sqrt{[(r-1)\mathcal{T} + n - rm]^2 + 4rnm} + \frac{2(r-1)}{(1+r)nm} (K_t - \gamma)^3 \\ &\leq 1 - \frac{2(K_t - \gamma)^2}{(1+r)nm} \sqrt{[(r-1)\mathcal{T} + n - rm]^2 + 4rnm} + \frac{2(r-1)(m-\gamma)}{(1+r)nm} (K_t - \gamma)^2 \\ &= 1 - \frac{2(K_t - \gamma)^2}{(1+r)nm} \left\{ \sqrt{[(r-1)\mathcal{T} + n - rm]^2 + 4rnm} - (r-1)(m-\gamma) \right\} \\ &= 1 - \frac{(K_t - \gamma)^2}{(1+r)nm} \left\{ \sqrt{[(r-1)\mathcal{T} + n - rm]^2 + 4rnm} + (r-1)\mathcal{T} + n - (r-2)m \right\} \\ &\leq 1 - \frac{(K_t - \gamma)^2}{(1+r)nm} \{ |(r-1)\mathcal{T} + n - rm| + (r-1)\mathcal{T} + n - rm + 2m \} \\ &\leq 1 - \frac{2(K_t - \gamma)^2}{(1+r)n} \leq 1 - \frac{2(K_t - \gamma)^2}{(1+r)n}. \end{aligned}$$

Taking expectation with respect to K_t under stationarity yields

$$0 \leq 1 - \frac{2}{(1+r)n} \mathbb{E}[\{K_t - \gamma(\mathcal{T})\}^2] = 1 - \frac{2}{(1+r)n} \mathbb{E}[\{N_{Y,1}^\sigma - \gamma(\mathcal{T})\}^2],$$

implying $\mathbb{E}[\{N_{Y,1}^\sigma - \gamma(\mathcal{T})\}^2] \leq (1+r)n/2$. By symmetry, we can repeat the same computations for $N_{X,1}^\sigma := \mathcal{T} - N_{Y,1}^\sigma$ and $\nu(\mathcal{T}) := \mathcal{T} - \gamma(\mathcal{T})$ and get $\mathbb{E}[\{N_{Y,1}^\sigma - \gamma(\mathcal{T})\}^2] = \mathbb{E}[\{N_{X,1}^\sigma - \nu(\mathcal{T})\}^2] \leq (1+r)m/2$, which gives $\mathbb{E}[\{N_{Y,1}^\sigma - \gamma(\mathcal{T})\}^2] \leq \frac{1+r}{2}n \wedge m$ and concludes the proof. $\square$

D.3 Simulation studies

We introduce some variants of the discrete density ratio permutation test, each with $S = 50$ and $H = 99$. (E4) stands for the discrete density ratio permutation test with V-statistic (11) and collision kernel $k(x, y) = \sum_{j=0}^J \mathbb{1}\{x = j\} \mathbb{1}\{y = j\}$. (E5) corresponds to the discrete density ratio permutation test with U-statistic (12) and the same kernel. Finally, (E6) identifies the discrete density ratio permutation test with test statistic (61).

We start by replicating the setup used in the proof of Proposition 22, selecting

$$(f_r, g_r) = \left(\left(\frac{1 - \gamma_r}{2}, \frac{1 + \gamma_r}{2} \right), \left(\frac{1}{1 + r}, \frac{r}{1 + r} \right) \right) \quad \text{with } \gamma_r = \frac{1 + r}{\sqrt{r}} \eta.$$

In Fig. 5(a), we assess the performance of the discrete (E6) as its test statistic over 3000 repetitions with $n = m = 500$, $r \in \{0.1, 0.5, 1, 2, 10\}$ and plotting an estimate of the power function for varying $\eta \in \{0, \dots, 0.10\}$. The results empirically support the tightness of the lower bound of Proposition 22, demonstrating that the power of the discrete density ratio permutation test increases when r moves further away from 1.

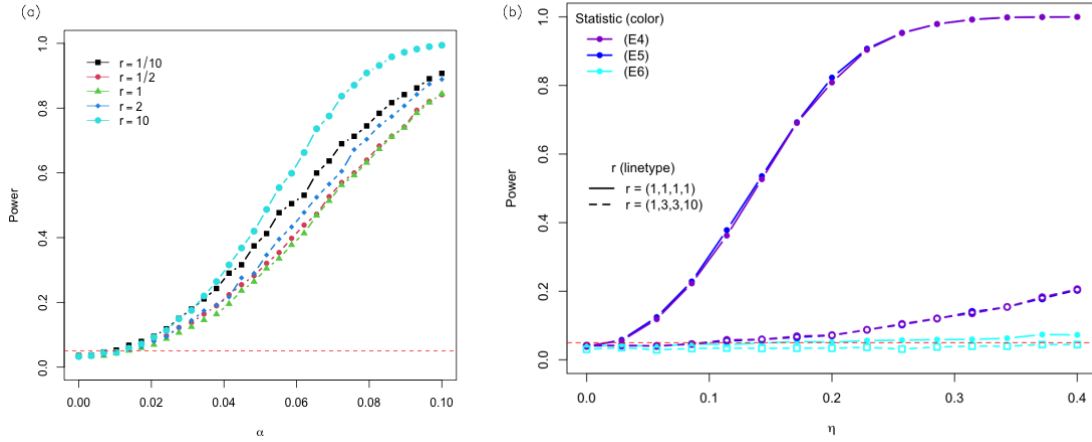


Figure 5: (a) Empirical power of (E6) in a synthetic binary data setting for varying $r \in \{0.1, 0.5, 1, 2, 10\}$. (b) Comparison of (E4), (E5), (E6) in two four-dimensional settings.

We now evaluate the performance of our reproducing kernel Hilbert space-based methodology. In this setting, when we use the kernel $k(x, y) := \sum_{j=0}^J \mathbb{1}\{x = j\} \mathbb{1}\{y = j\}$, we can express the V-statistic (11) and the U-statistic (12) as

$$V(x_1, \dots, x_n, y_1, \dots, y_m) = \sum_{j=0}^J \frac{1}{(n/m + \hat{\lambda}r_j)^2} \left\{ \frac{\hat{\lambda}r_j}{n} \sum_{i=1}^n \mathbb{1}\{x_i = j\} - \frac{1}{m} \sum_{i=1}^m \mathbb{1}\{y_i = j\} \right\}^2,$$

and

$$U(x_1, \dots, x_n, y_1, \dots, y_m) = V - \sum_{j=0}^J \frac{\hat{\lambda}^2 r_j^2}{(n/m + \hat{\lambda}r_j)^2} \left\{ \frac{1}{n^2} \sum_{i=1}^n \mathbb{1}\{x_i = j\} \right\} - \sum_{j=0}^J \frac{1}{(n/m + \hat{\lambda}r_j)^2} \left\{ \frac{1}{m^2} \sum_{i=1}^m \mathbb{1}\{y_i = j\} \right\},$$

which shows that both U and V are functions of $(N_{Y,0}^\sigma, \dots, N_{Y,J}^p)$, in analogy with (61). In particular, for (E4)

and (E5) it is sufficient to sample $(N_{Y,0}^\sigma, \dots, N_{Y,J}^\sigma) \mid Z$ according to (59) via the R function `rMFNCHypergeo`, without resorting to Algorithm 1, which substantially reduces the computational cost of Algorithm 2.

We apply this approach in two settings, described next. Before detailing the data-generating processes, note that our empirical findings here are consistent with Section 6. The only practical difference is that the discrete density ratio permutation test is much faster to run, allowing substantially larger Monte Carlo sample sizes in each simulation. For the first setting, we set $J = 3$ and $n = m = 250$, where the X_i 's were independently drawn from a multinomial distribution over $\{0, 1, 2, 3\}$ with probabilities $p_X = (\frac{1}{8}, \frac{1}{8}, \frac{3}{8}, \frac{3}{8})$, and the Y_j 's were sampled from a multinomial distribution with probabilities $p_Y = (\frac{1}{43}, \frac{3}{43}, \frac{9+25\eta}{43}, \frac{30-25\eta}{43})$, with $\eta \in \{0, \dots, 0.4\}$. When $\eta = 0$, the null hypothesis holds with $r = (1, 3, 3, 10)$; increasing η corresponds to larger deviations from the null. We repeated each experiment 5000 times, comparing (E4), (E5) and (E6) in this setting. Fig. 5(b) presents the results in the dashed lines. Overall, the results show a notable drop in power when (E6) based on rejection sampling is used. Furthermore, the (E4) and (E5) are nearly indistinguishable, as expected, since the difference $V - U$ vanishes asymptotically as the sample size increases, while (E6) seems significantly less powerful than the reproducing kernel Hilbert space-based approach in this particular setting. For the second simulation setting, we consider a less extreme shift. Specifically, the shift vector is $r = (1, 1, 1, 1)$, so the problem reduces to the classical two-sample testing setting. The X_i 's are drawn independently and identically distributed from the uniform distribution on $\{0, 1, 2, 3\}$, that is, $p'_X = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, while the Y_j 's are sampled from a multinomial distribution with $p'_Y = (\frac{1}{4}, \frac{1}{4}, \frac{1+\gamma}{4}, \frac{1-\gamma}{4})$, where $\gamma = \frac{25\eta}{43}(\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{10}})$. This specific choice of γ ensures that $D(p_X, p_Y) = D(p'_X, p'_Y)$ for all values of η , where the separation $D(f, g)$ is defined right after (61). The solid curve in Fig. 5(b) supports our earlier conjecture that near-uniform shifts are generally harder to detect.